

אוניברסיטת תל אביב

הפקולטה להנדסה ע"ש איבי ואלדר פליישמן

בית הספר לתארים מתקדמים ע"ש זנדמן-סליינר

חימום מיקרוגל ממוקד של בזלת –

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בהנדסת חשמל ואלקטרוניקה על ידי

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**Localized microwave-heating of basalts –
Melting, lava-like eruption, and dusty-plasma ejection**

A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering

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This research was carried out in The School of Electrical
Engineering- Department of Physical Electronics,

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Prof. Eli Jerby

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תקציר

חימום מיקרוגל ממוקד (Localized microwave-heating LMH) והיווצרות של נקודה חמה בתהליך של בריחה תרמית עלולים לגרום להיתוך מקומי, ואף לפליטה של פלזמה על ידי חומרים שונים. תופעה זו, הנוצרת על ידי אינטראקציה מקומית של קרינת מיקרוגל עם חומר, מאפשרת חימום מהיר בתוך אזור מוגבל (נקודה חמה, Hotspot) שממדיו קטנים בהרבה מאורך הגל. תהליך חימום מקומי זה מתפתח עד להופעת מעבר פאזה של החומר המוצק, לנוזל, גז או פלזמה. התנאים הדרושים להיווצרות נקודה חמה תלויים בהספק מקור המיקרוגל, וכן בגיאומטריה של המוליך ובמאפייני החומר המוקרנים.

עבודה זו מציגה מחקר ניסיוני ותיאורטי של אינטרקציה החימום הממוקד בין בזלת למיקרוגל. בעבודה זו אנו חוקרים לראשונה תופעה של פליטת פלזמה מאובקת מאבן בזלת מותכת, וכן היווצרות עמודות אש ופלסמוידים (plasmoids) דמויי כדור אש. בפרט, אנו מציגים את הדמיון בין השפעות LMH שונות באבני הבזלת (שנצפו בקנה מידה מעבדתי) לבין תופעות וולקניות ענקיות בטבע, כגון התפרצות וזרימה של לבה, פליטה של ברק-כדורי (Ball lightning) ויצירת אבק וולקני.

מערך הניסוי שלנו כולל מהוד מיקרוגל בנפח של $\sim 10^3 \text{ cm}^3$ המעורר בהספק של 1kW. בתוך המהוד מונחת אבן בזלת בנפח של $\sim 30 \text{ cm}^3$. בתהליך LMH, הליבה הפנימית של אבן הבזלת מתחממת ומותכת תחילה. הקרנה נוספת מובילה להתפרצות בזלת מותכת מליבת האבן אל פני השטח (בדומה להתפרצות הר געש ופליטה של לבה מתוכו). הבזלת המותכת מתקררת ומתמצקת לזכוכית וולקנית שחורה (Obsidian). במחקר זה מצאנו כי הקרנה נוספת גורמת לבזלת לפלוט סילון של פלזמה מאובקת, המתפתחת לעמוד פלסמה, ולאחר מכן לפלסמויד הדומה לברק-כדורי כפי שנצפה לעתים רחוקות בטבע.

עמודי-אש אלו והפלסמוידים בצורת כדורי-אש מאופיינים בניסויים באמצעים שונים, תוך כדי התהליך ולאחר סיומו. טמפרטורות העירור והסיבוב של הפלסמה בהם ($T_{\text{exc}}, T_{\text{rot}}$) מוערכות בטווח של 0.3-0.6 eV, עם צפיפות אלקטרונים ואבק של $n_e \sim 10^{17} - 10^{18} \text{ m}^{-3}$ ו- $n_d \sim 10^{16} \text{ m}^{-3}$, בהתאמה. תכולת היסודות המרכיבים את הפלזמה המאובקת, המתגלים על ידי פליטת הספקטרום האטומי שלה, נמצאה דומה לתכולתה של אבן הבזלת ממנה נפלטת הפלזמה, וכן דומה לתצפיות ספקטרליות של ברק-כדור טבעי. שכבת האבק שנצברה על גבי המשטח המאסף (בקצב צבירה של $\sim 0.2 \text{ mm}^3/\text{s}$) מכילה מיקרו-ננו-חלקיקים במבנים שונים. אבקה זו, שהופקה על ידי הפלזמה המאובקת, מאופיינת בהרכב יסודות הדומה לאבן הבזלת הפולטת, וכן לאפר וולקני.

תוצאות הניסויים נידונות בהיבטים מדעיים ומעשיים שונים, כגון הרלוונטיות שלהן לתופעת הברק הכדורי ותופעות וולקניות נוספות וכן ביחס ליישומים שונים, כגון ייצור אבקה, ספקטרוסקופיה באמצעות מיקרוגל (Microwave-induced breakdown spectroscopy MIBS), ומיצוי של מינרלים מאבני בזלת.

Abstract

Localized microwave-heating (LMH) and the consequent formation of hotspots by thermal-runaway instability may cause local melting, and even plasma ejection by various materials. This phenomenon, generated by the microwave-material interaction, enables a rapid heating within a localized zone (a hotspot, much smaller than the microwave's wavelength). This local heating process evolves up to the occurrence of a phase transition of the solid material into liquid, gas or plasma. The necessary conditions for the hotspot formation depend on the microwave power source, the geometry of the applicator, and the irradiated material properties. This work presents experimental and numerical investigations of the LMH interaction of basalts and microwaves.

In this work we first study a phenomenon of dusty-plasma ejection from molten basalt, forming fire-column and fireball-like plasmoids. We also indicate the resemblance of various LMH effects in basalts (observed in laboratory scales) to the gigantic volcanic phenomena in nature, such as lava eruption and flow, ball-lightning emission, and creation of volcanic dust.

Our experimental setup employs a microwave cavity of a $\sim 10^3\text{-cm}^3$ volume, powered by a 1-kW magnetron. A basalt stone of a $\sim 30\text{-cm}^3$ volume is situated in the cavity. By applying LMH, the inner core of the basalt is first melted. Further irradiation leads to the eruption of the molten basalt, from the core to the surface (similarly to lava eruption and flow in volcanoes). The molten basalt cools down and solidifies to a black glass (obsidian). Here we show that further irradiation causes the basalt to emit a plume of dusty plasma, which evolves to a confined plasma column, and further to a plasmoid similar to a natural ball-lightning. These fire-column and fireball-like plasmoids are characterized by various *in-* and *ex-situ* means. Their excitation and rotational temperatures are estimated in the range $\sim 0.3\text{-}0.6$ eV, with electron and dust densities of $n_e \sim 10^{17} - 10^{18} \text{ m}^{-3}$ and $n_d \sim 10^{16} \text{ m}^{-3}$, respectively.

The elemental content of the dusty-plasma, revealed by its atomic spectral emission, is found similar to the content of the originating basalt, and to spectral observations of natural ball-lightning as well. The dust layer accumulated on the cavity's upper collector (in a deposition rate of $\sim 0.2 \text{ mm}^3/\text{s}$) contains micro- and nano-

particles in various structures. This powder, deposited by the dusty plasma generated in the cavity, possess an elemental composition similar to the emitting basalt stone, and to natural volcanic ash as well.

The experimental results are discussed in various scientific and practical aspects. These include their potential relevance to natural ball-lightning and volcanic effects, as well as their significance for several applications, such as powder generation, microwave-induced breakdown spectroscopy (MIBS), and mineral extraction.

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The measurements of material properties by means of SEM, EDS and XRD in Chapter 3 were initiated by Dr. Zahava Barkay and Dr. David Levi from Tel-Aviv University. This research is supported by The Israel Science Foundation (Grant No. 1896/16

Finally, I would like to thank my family for their love and support that made this work possible.

Dedication

I am dedicating this thesis to the memory of my grandmother, Rivka Segal R.I.P, who was always curious about science and research, and showed great interest in this work progress and publications.

This thesis is also dedicated to the memory of my grandfather, Ezra Shoshani R.I.P, who always showed great respect and admiration for academic achievements.

Both passed away during the last months.

Your memory will light my way and will ever be with me.

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1. Introduction

1.1. About basalts

Basalts, the most common volcanic rocks on Earth, appear in nature in a range of physical and chemical properties, and in a variety of outer shapes [1]. Basalt underlies more of Earth's surface than any other type of rock, and is the key component of oceanic crust. Above sea level, basalt is common in hotspot islands and around volcanic arcs, especially those on thin crust. The largest volumes of basalt on land correspond to continental flood basalt, which are known to exist in India, Canada, Brazil, Russia, South Africa, USA, etc. It is also common on the moon and on other rocky planets in the solar system. The dark areas visible on Earth's moon, the *Lunar Maria* are plains of flood basaltic lava flows. Basalt commonly erupts on Io (Jupiter's third largest moon), and has also formed on Mars, Venus and the asteroid Vesta [2].

Basalt is used for a wide variety of purposes. It is most commonly crushed for use as an aggregate in construction projects, such as road base, aggregates for concrete and for asphalt pavement, railroad ballast, filter stone in drain fields, and many other purposes. Basalt is also cut into dimension stone. Thin slabs are cut and sometimes polished for use as floor tiles, building veneer, monuments, and other stone objects. Additional use for basalt is fiber reinforcement, in which thousands of fibers are compressed, and wound spirally into a structural reinforcement rod [3]. The basalt rebar is corrosion and rust-resistant, it was originally viewed as an ideal material for use in marine construction and on overpasses and bridges which were often exposed to salt.

Basalts are common aphanitic igneous extrusive (volcanic) fine grained rocks. It is composed of minute grains of plagioclase feldspar (generally labradorite), pyroxene, olivine, biotite and hornblende [4]. Basalts are formed by the rapid cooling of basaltic lava from the interior of the Earth mantle and exposed at, or very close to, the surface of the Earth. Significant amounts of basalt are produced above oceanic hotspots. These are locations where a small plume of hot rock rises up through the mantle from a hotspot on Earth's core. The Hawaiian Islands are an example of where basaltic volcanoes have been built above an oceanic hotspot, Kilauea (had a series of eruptions this year, 2018) is the most active of these volcanoes. It has been in an almost continuous eruption since January 1983. Basalt flows from Kilauea have extruded over one cubic mile of lava,

which currently covers about 48 square miles of land. These flows have travelled over seven miles to reach the ocean, covering highways, homes and entire country side areas that were in their path.

More than 500 million people live within the potential exposure range of a volcano, according to Doocy *et al* [5], there were a total of 91,789 deaths, 14,068 injuries, and 4.72 million people affected by volcanic events between 1900 and 2008. Volcanic eruptions not only produce direct material losses; they can also affect global climate thus affecting society as a whole, even a small eruption, as occurred in Iceland of mount Eyjafjallajökull in 2010, caused major air traffic disruptions over northern Europe causing disastrous socio-economic impacts [6]. According to forecasts by scientists at the *Geological Society of London* [7], agriculture would be impossible in many volcanic-eruption affected regions. Furthermore, crops will even be destroyed by a layer of ash just 1-cm thick. Etna, the biggest volcano in Europe, caused enormous damage when it erupted in 2001 and again in 2002/03. In 2002, the ash rain alone caused economic losses of around 800 m€ [8].

In the past year (2018) we have experienced several hazardous volcanic eruptions, leading to massive number of casualties, evacuation of populations and damage to houses, property and infrastructure. The violent eruption of Guatemala's Fuego volcano [9] on June 3, 2018 [10,11], Indonesia's Mount Merapi volcano in a series of eruptions from mid-May for several weeks, Telica volcano in Nicaragua on June 18, 2018 [11], and Hawaii's Kilauea volcano eruption on May 4 [12], all accompanied by earthquakes, fissures, ash plumes, flying rocks and pyroclastic flows, fast-moving currents of extremely hot gas and volcanic matter that pour down volcanic slopes at incredibly high speeds [13]. As seen in the latest Kilauea eruption, pyroclastic flow is the most dangerous volcanic activity, ongoing for several weeks, and forcing thousands of people to evacuate their homes.

Volcanoes directly or indirectly produce, or host, deposits of aluminium, diamonds, gold, nickel, lead, zinc, and copper. Intensely altered and fractured basalt is a common host rock for gold. Diamonds are brought to the surface from the mantle in a rare type of magma called Kimberlite, and erupted at a rare type of volcanic vent called a diatreme or pipe. Some South African mines operate at 25 carats of diamond per 100 cubic meters of rock, or about 2 grams of diamonds per 100 tons of rock.

Basalt drilling is used for geo-thermal wells, tunnelling and mining as basalt being an abundant rock in the Earth crust. Mechanical drilling of basalt and other hard rock has been proven to be the most economical and hence most commonly used one. During basalt breakage in mining and civil applications, the performance of mechanical equipment such as tunnel boring machines (TBM's) is limited by high levels of bit wear and maintenance. Therefore, methods of preconditioning, to weaken natural rock prior to mechanical breakage (for example, flame torch-assisted TBM's [14]) has been researched in the last decades.

1.2. Microwave processing of basalts

The concept of using microwave energy to generate heat for rock fracture and drilling was briefly examined in the 1960's [15], but due to technical issues at the time, it was not deemed economical and was not further investigated, till the 2000's [16].

Heating, cracking, braking and melting of basalt rock by means of microwave irradiation have been studied in the last decades as microwave technology has advanced. Basalt microwave-heating experiments were presented in various aspects. Melting of basalts in a microwave cavity was introduced in 2004 [17]. More recently, microwave heating of basalt at lower temperatures was studied in various conditions [18]. The formation of cracks due to temperature gradients and thermal stresses after microwave heating in a cavity was analysed by a thermo-mechanical model [19], point load and strength measurements were taken [20]. Rock breakage and cracking by means of horn antenna microwave irradiation was conducted, testing the temperature and strength of the basalt rocks after different irradiation powers and distances from the antenna [21]. All studies showed melting of the basalt rock after a period of time depending on the irradiation power. Larger basalt stones, of $\sim 400 \text{ cm}^3$, were heated by microwaves to $\sim 500^\circ\text{C}$ in their core [22]. In 2013, Jerby *et al* showed that basalt melting by localized microwave heating (LMH) involves a thermal-runaway instability leading to a hotspot formation in the core of the basalt stone causing a volcano like effect of core melting, cracks and lava flow from the core towards the surface [23].

1.3. Microwave heating

Microwave heating of materials is a unique technology for industrial processing, widely studied and implemented [24-26]. The absorption of microwave energy allows volumetric heating of materials thus it leads to time saving and reduces energy consumption comparing to conventional heating. The microwave heating may comprise of several distinctive mechanisms such as ohmic, dielectric and magnetic losses. In conventional heating, the temperature gradient transfers heat from the surface towards the bulk via heat conduction. In contrast, in microwave volumetric heating the temperature gradient plays an opposite role, it directs heat outward from the bulk thus tends to cool it down through heat convection from the surface. Since heat transfer is a diffusive process by nature, the typical heating rate of a thick sample by means of conventional heating might be slower than by microwave volumetric heating. Microwave heating of materials is often preformed volumetrically in a resonant cavity with transverse dimensions comparable to the microwave wavelength λ_0 employed. The electromagnetic (EM) energy is then dissipated into heat within the matter [27].

Microwave volumetric heating is limited to non-metallic materials and to metal powders. In these cases, the more lossy the material is (in the microwave range), the more drastically the energy decays within it, hence the less it penetrates into the bulk's core.

Microwave power applications are nowadays wide-spread throughout industry, science, and domestic households. The best known is thermal processing of foods, but one should also note other industrial, scientific, and medical (ISM) applications, such as drying of textiles, fibers, paper and board, material processing and sintering, and pre-heating and curing of plastics and rubber.

1.4. Localized microwave heating (LMH)

Microwave-heating techniques used in industrial and domestic applications have a unique feature of inner-body heating [28]. The hotspot effect (undesired in most microwave applications) may evolve due to thermal-runaway instability [29]. This tends to occur in materials characterized by properties with certain temperature dependencies (e.g. if the dielectric losses increase and/or the thermal conductivity decreases as the

temperature rises). Then, the hotter part of the material (in the core) tends to absorb relatively more microwave energy than the cooler part (on the surface). This positive-feedback effect induces localized microwave-heating (LMH) and thermal-energy concentration, which creates a molten hotspot inside the material core [30].

While most of the known microwave-processing techniques are volumetric in nature, the microwave-drill is an example for an intentionally localized applicator [16, 31]. The underlying physical effect that enables its sub-wavelength focusing of the microwave energy is the induced thermal-runaway mechanism [32]. This effect enables for instance the focusing of 2.45-GHz microwaves (12-cm wavelength) within the material into a $\sim 1\text{-mm}^{\varnothing}$ spot ($\sim 10^{-2}$ smaller than the wavelength). This unique capability of LMH enables for instance melting, drilling and other localized material-processing applications usually performed by lasers. On the other hand, unlike the remote laser operation, the localized microwave concentration is a near-field effect which requires a closed proximity (or even a physical contact) of the microwave applicator and the processed object. LMH effects were demonstrated by microwave-drill type applicators in various materials, including concrete, glass, ceramics, basalts, and bones [17,33-35].

1.5. Microwave-induced plasma ejection from solid materials

In the 1950's, Kapitza [36, 37] produced fireballs using high-power radio waves at atmospheric pressures. He suggested accordingly an external electromagnetic-energy mechanism for the production of ball-lightning in nature. The excitations of laboratory ball-lightning-like effects have also been demonstrated experimentally using high-voltage discharges on silicon [38-40] and in water [41]. Though various characteristics of ball lightning have been demonstrated experimentally, also by microwave-generated fireballs, a definitive experimental demonstration of a long lived, floating ball-lightning-like object has yet to be achieved.

Plasma-columns, plasmoids and fireballs, directly excited by localized microwaves from solid substrates to air atmosphere, were first presented in 2006 by Dikhtyar and Jerby [42] and have been studied since in various experiments [43-48]. A molten hotspot is created on the substrate's surface in a thermal-runaway instability caused by the localized microwave heating (LMH). This leads to a plasma-column ejection from the hotspot, and to oxidation or combustion processes within the plasma.

The plasmoid may be further evolved to form a self-sustained fireball while its feeding plasma-column is detached from the molten hotspot. Measurements of the optical emission spectra of such plasmoids show the dominance of elements imported from their originating solid substrates (either dielectric or metallic). These microwave-generated fireballs float in the air, hence demonstrating to some extent also the buoyancy feature of ball lightning.

Using synchrotron radiation, the small-angle X-ray scattering (SAXS) method was applied as well to study fireballs of this type, and to find their particle size distribution, density and decay rate. SAXS results of fireballs ejected from borosilicate glass [43] show that these fireballs contain particles with diameters of ~ 50 nm and average number densities of $\sim 5 \times 10^9 \text{ cm}^{-3}$. Hence, these fireballs are considered a form of dusty plasma [49-51] consisting of ensembles of charged nanoparticles in the plasma volume.

The creation of artificial and natural fireballs was studied also by Bychkov [52] as originating from a reservoir made up of an oxide cover filled with metal powder. As in natural ball lightning, it is electrically charged due to the transfer of charges from the linear lightning. A plasmon resonance theory [53] proposes that an object irradiated by EM waves may visually manifest itself as ball lightning. The plasmon resonance effect may cause also a strong concentration of EM-energy near particles, as observed experimentally in optical wavelengths.

A dusty (or complex) plasma is a normal electron-ion plasma with an additional charged component of micron-sized particulates. Complex (dusty) plasmas occur for instance in candle flame, the zodiacal light, in lightning in thunderstorms containing smoke-contaminated air, in volcanic eruptions, and in ball lightning. The dust particles in the plasma are charged by thermionic emission from the heated dust grains. Thermionic emission involving plasma plumes are seen in the process of laser ablation when laser pulses are emitted on a metal surface [54], the ablation process is dominated by heat conduction, melting, evaporation and plasma formation [55]. The ablation of metals involves a complex sequence of different processes. One of the first is the thermionic emission process, followed by plasma plume ejection from the heated area. Surface electron emission due to thermionic and photoelectric effect contribute to plasma initiation [56].

1.6. The ball-lightning enigma

The ball-lightning enigma has attracted the interest of both the public and scientists [57-59]. It is yet a major motivation for studies of microwave-excited fireballs, which mimic ball lightning. This rare natural phenomenon has been accidentally observed by eyewitnesses, mostly during thunderstorms, earthquakes, volcanic activity, and tornadoes. These sightings describe ball lightning as glowing luminescent objects floating in the air and lasting for up to several seconds. Ball-lightning effects were observed in various shapes, bouncing on the ground, rotating as tornados, moving against winds, and even passing through glass windows.

A model for ball-lightning generation in nature, proposed by Abrahamson and Dinniss [60,61] suggests that normal lightning striking the ground evaporates silica and carbon, and ejects a plume of nanoparticles into the atmosphere. The reduction of silicon dioxide by carbon to silicon nanoparticles, in reactions such as



and their oxidation again, generate the heat energy that sustains the reaction. The ball-lightning luminosity is maintained by a slow combustion process in which the silicon nanoparticles are gradually fused together to form larger silicon spheres with oxidized outer surfaces. The oxide layer slows the further oxidation of the interior due to the higher melting point of the oxide, hence delaying the process by reducing further air oxygen diffusion to the silicon cores of the particles. The ball lightning is then evolved as an elastic aerogel structure because of mutual electrostatic aggregation of bipolar charged nanoparticles. Therefore, according to this model, the ball lightning seems to be a hybrid of plasma and combustion effects.

Ball-lightning and fireballs effects have been long associated with volcanic eruptions and volcanic air pollution. A large number of fireball sightings has been recorded in the last centuries [62], many occurring near volcanos or at times of heavy air pollution caused by volcanic eruptions. Spectrograph measurements of natural ball-lightning reveals the presence of Si, Fe, Ca and Al [63], as in soil, in line with the theory suggesting that ball lightning is generated by lightning striking the soil [64].

1.7. Research objectives and methodology

In this thesis, we have investigated various LMH-basalt interactions using a commercial 1kW magnetron at 2.45 GHz. This work advances previous studies performed by our group, including thermal-runaway effects in basalt-microwave interactions (such as core melting, lava eruption and microwave-drilling of basalt) [17, 23]. This work is also related to other studies of LMH interactions with various materials and metals, such as silicon, titanium and copper [42-48]. These studies presented dusty-plasma ejections from a molten hotspot in the irradiated material in the form of fire-column and fire-ball.

The thesis objectives were also to validate the basalt-LMH theoretical model by experiments and numerical simulations. A hypothesis of plasma ejection from basalt hotspots was proposed, and means for hotspot formation and plasma ejection were studied. The integration of an electrode and later an array of electrodes in the experimental setup was conducted in order to examine ways to expedite and intensify the LMH process in basalts. The effect of the basalt rock shape on the melting process and plasma ejection was studied as well. Another objective was to characterize the dusty-plasma ejected from basalt.

The thesis consists of an experimental work and theoretical analyses of the different setups, including diagnostics of the heating and melting processes (and comparison to the theoretical model), and characterization of the dusty-plasma ejected from the basalt samples. The plasma temperature, the electromagnetic properties, and the elemental composition, were measured by several *in-* and *ex-situ* means. Also, characterization of the dust accumulated from the dusty-plasma is presented in this study.

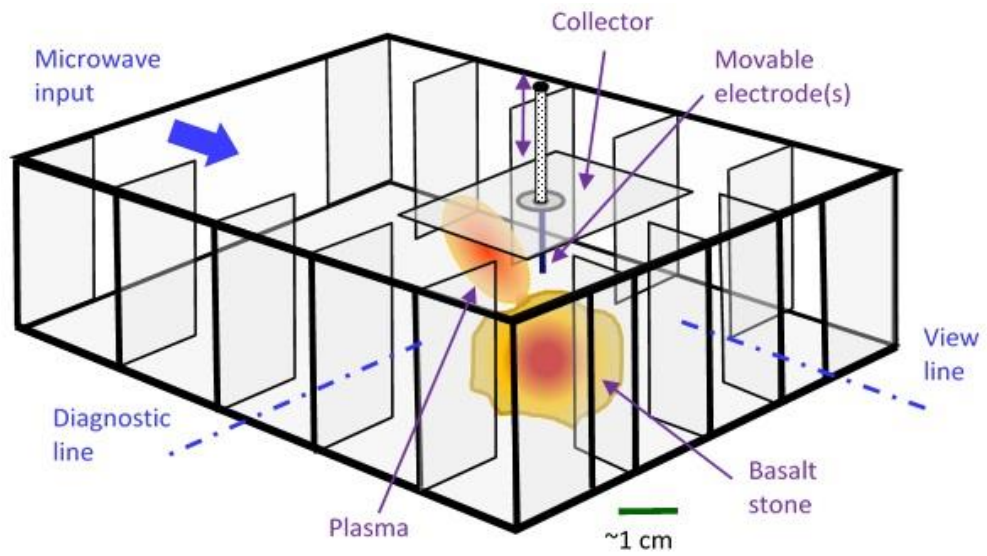
2. Experimental setup and procedures

2.1. The LMH furnace

The experimental setup shown in Fig. 2.1 consists of a microwave cavity fed by a 2.45-GHz, 1-kW magnetron. The cavity is made of a 10x5 cm² cross-section waveguide, shorted at the end by a mirror made of metallic vanes under cutoff. This arrangement enables a direct view into the waveguide, with no disturbance to the various diagnostic means (e.g. the thermal camera which requires a direct line of sight). A similar array of vanes under cutoff is also installed as the side wall of the cavity, enabling a broader view (from both perpendicular directions) and more in-situ diagnostic means simultaneously. As shown in the block diagram (Fig. 2.1b), the magnetron's switched-mode power supply (MagDrive-1000, Dipolar Ltd., Sweden) provides an adjustable input microwave power up to 1 kW. The microwave power is delivered to the cavity via an isolator and an impedance auto-tuner (Homer, S-Team Ltd., Bratislava, Slovak republic). The incident and reflected waves are recorded by the impedance-analyser mode of the auto-tuner. This enables adaptive impedance matching and optimal transmission of the microwave power into the cavity. The diagnostic means include video recording (up to 200 fps) and thermal imaging by FLIR-SC300 camera. The optical spectra emitted from the lava and plasma eruptions are captured by an optical spectrometer (Avaspec-3648) with 0.3-nm resolution in the wavelength range of 200-1000 nm.

A LabView script was written for each component of the setup to synchronize operation, acquire data and write time-stamps. The control screen offers: (a) Magnetron output power control via MagDrive programming, (b) HOMER definition control, measured and de-embedded reflection coefficient (magnitude and phase) data acquisition, (c) video-camera recordings control, and (d) voltage measurements operation and acquisition for the Langmuir probe measurements (Appendix A).

(a)



(b)

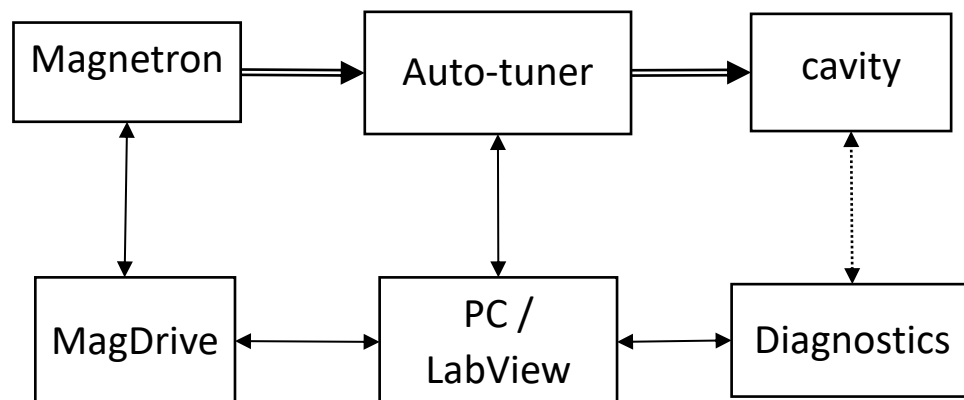


Figure 2.1: (a) A schematic of the experimental setup: chamber, movable electrode and basalt sample.(b) A block diagram of the experimental instrumentation, with the diagnostics including an optical spectrometer, thermal and video cameras, and an I - V probe.

2.2. Electrodes

In this experiment we use basalt stones (from the Golan heights) either in their original natural shapes or cut to (3-cm^3) cubic bricks. The basalt-rock-load is placed inside the cavity in an optimized position (discussed in Section 4.3) whereas the movable electrode directs the microwave energy locally into its substrate. Various levels of focusing the microwave energy are optionally available in this setup, either by moving a single electrode shown in Fig. 2.1a, or by manipulating an array of 9 or 18 electrodes (Fig. 2.2a and b respectively) installed in the cavity's ceiling via the collector, and in the cavity floor. Each electrode in this array expedites the intentional excitation of the hotspot in its vicinity, when brought into contact with the basalt-stone surface, and hence enables the stimulation of the fire-column (and consequently of a fireball) in a more controlled fashion.

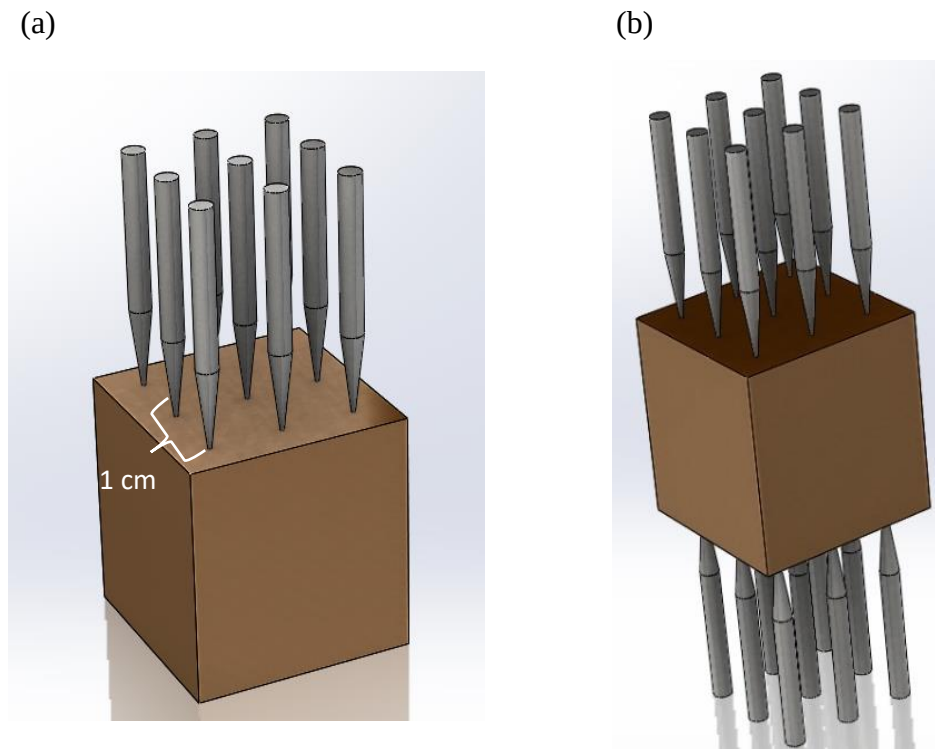


Figure 2.2: Electrode arrays. (a) An array of 9 electrodes pushed against the basalt surface.

(b) An array of 18 electrodes, with 9 electrodes on each face.

2.3. Diagnostic means and procedures

The video camera recording is used to monitor, measure times, and synchronize between the various effects observed (e.g. inner hotspot formation, cracks, breakage, lava eruption and plasma ejection) and the other diagnostic means used in the experiment. Once ejected, the plasma column size, shape and position can be found from the video recording. Also, the evolution of fireballs, as well as their shapes and movements, are monitored by the video camera.

The FLIR-SC300 thermal camera monitors the spatial temperature evolution of the sample, hotspot formation, temperature and size. Recordings and analyses are made by FLIR-Tools+™ software. Profile temperatures, temperature evolution in time, the hotspot size and temperature evolution, are all monitored and later analysed and synchronized with other experimental measurements. The thermal imaging data is later compared to simulation in order to validate the physical model and the simulation results.

2.3.1. Microwave scattering, reflection and transmission analyses

The HOMER impedance tuner (by S-Team Ltd.) is controlled as part of the LabView™ system. Adaptive tuning is applied to intensify the hotspot heating in order to eject the plasma. The synchronized de-embedded reflection coefficient (S_{11}) measurements are presented and stored for offline analyses. In order to get more accurate measurements, once the plasma is ejected, one can cancel the adaptive tuning action in order to measure the actual reflection coefficient instead of the de-embedded one. This is made possible due to the self-tuning effect of the plasmoid (later discussed in Section 3.2.1), which provides an alternative to the adaptive impedance tuning by the impedance tuner.

The plasma transmission coefficient (S_{21}) measurements were carried out by a Tektronix-TTR500 series vector network analyser (VNA). The two-port system was constructed of two excitation ports with the plasma column in-between (Fig. 2.3a). For this purpose, two electrodes are placed alongside the plasma column and serve as the excitation ports. The plasma can be characterized by measuring changes in the coupling between the two electrodes using a VNA for the S-parameter measurements. At frequencies higher than 3 GHz, the S-parameter measurements are too noisy and thus it is difficult to notice changes in the coupling. A different approach for high frequency

measurement was then implemented. A two-port system for higher frequencies was constructed of two WR-187 waveguides with coaxial SMA input ports (Fig. 2.3b).

The measurement setup needs to be protected from the relative high-power radiation inside the cavity, therefore filtering measures were used. For the low frequency setup, two band-stop filters for 2.45-GHz were placed between the VNA ports and the electrode ports. For the high frequency setup, the cutoff frequency of the WR-187 waveguide ports satisfies the filtering requirements.

In order to measure S -parameters and to later analyse the results with respect to other measurements, a computer written script (Appendix B) was used to record the continuous VNA measurements and synchronize it with the entire experimental setup.

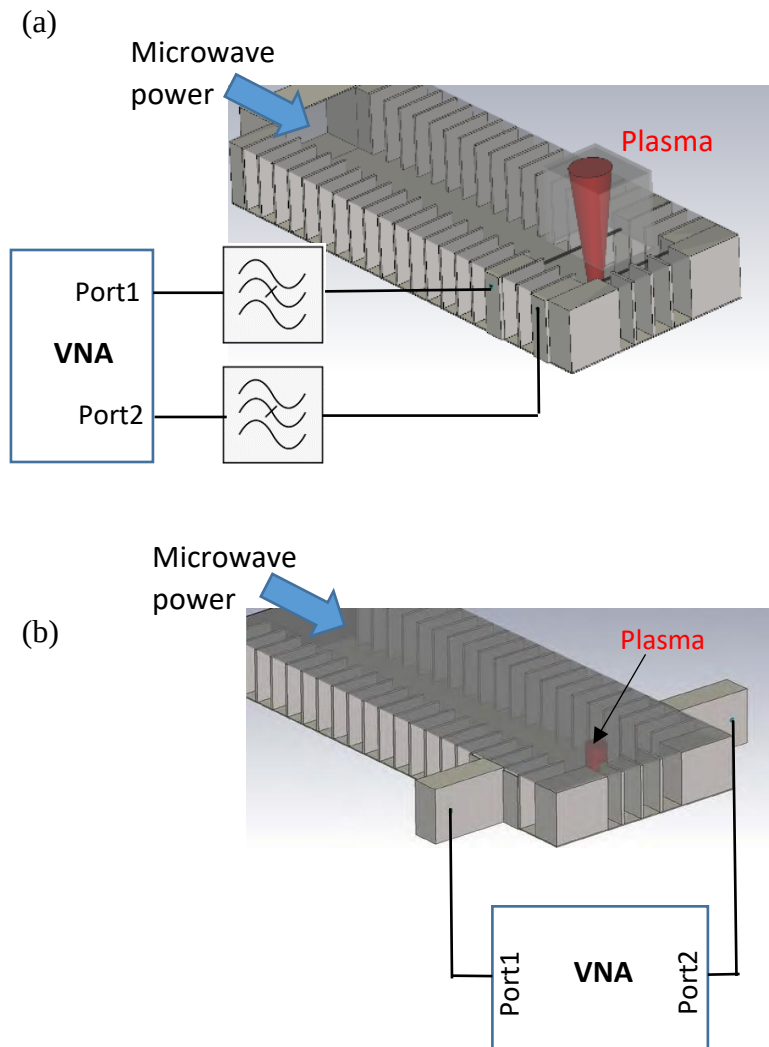


Figure 2.3: Transmission scattering parameters measurement setup schematics:

- (a) Two electrodes as coupling ports, and stop-band filters for VNA protection.
- (b) Two WR-187 wave guides as coupling ports.

2.3.2. Optical spectrum analyses

In this experiment we use an AvaSpecTM-3648 spectrometer with a resolution of 0.3 nm, controlled by the Avasoft.8TM computer software. The continuous measurements are recorded and later synchronized with the video recordings, and analyzed. The spectral lines are used for identifying the elements contained in the plasma by comparing the wavelength of the emitted spectral lines to the NIST [65] library. Regarding the spectrometer resolution (0.3 nm) and the higher resolution of the wavelengths in the library (0.001 Å), the lines were compared to elements which are presumed to compose the dusty plasma. Spectrometer measurements were also used to evaluate the rotational and excitation temperatures of the plasma. Several elements with strong and differentiable spectral lines, such as Fe and Ti where later used for the excitation temperature approximation by Boltzmann plot. A fitting tool for the radical emission curve (LifBaseTM) was used to fit the measured spectrum of OH radicals to the theoretical plots, thus evaluating the rotational temperature.

2.3.3. Boltzmann plot

Using the Boltzmann-plot technique, the estimated temperature of the plasma can be depicted through the relative intensity of the spectral lines emitted by excited atoms. Finding the excitation temperature (T_{exc}) can give an estimate for the electron temperature in the emitted plasma [66].

The Boltzmann plot method is a simple and widely used method for measuring the electron temperature of plasma by using the relative intensity measurements of two or more spectral lines having a relatively large difference of energy. However, in order to practically apply the Boltzmann plot method for the measurement of electron temperature, the excitation level needs to be reached under a local thermodynamic equilibrium (LTE) condition (discussed later). Let us consider two energy levels, E_i and E_j (where $E_i < E_j$), of two different line intensities with respective atomic densities of N_i and N_j . Under the thermal equilibrium condition, the relation between N_i and N_j can be established using the Boltzmann distribution:

$$\frac{N_j}{N_i} = \frac{g_j}{g_i} \exp\left[-\frac{E_j - E_i}{kT}\right], \quad (2.1)$$

where g_i, g_j are the statistical weights of the respective states, k is the Boltzmann constant (1.38×10^{-23} J/K), and T is the temperature in K. If the total population density is N , the Boltzmann relation related to the population distribution over a given atomic state is given by

$$\frac{N_j}{N} = \frac{g_j}{Z(T)} \exp\left[-\frac{E_j - E_i}{kT}\right], \quad (2.2)$$

where $Z(T)$ is the sum of the weighted Boltzmann function of all the discrete energy levels, called the partition function and given by

$$Z(T) = \sum g_m \exp\left[-\frac{E_m}{kT}\right]. \quad (2.3)$$

When an atom in the plasma is de-excited from the upper energy level, E_j , to the lower energy level, E_i , the emission coefficient of spectral line can be expressed as

$$\varepsilon_{ji} = \frac{hc_0}{4\pi\lambda_{ji}} A_{ji} N_j, \quad (2.4)$$

where λ_{ji} is the wavelength of the emitted light, h is the Planck's constant (6.626×10^{-34} J×s), c_0 is the speed of light in vacuum (3×10^8 m/s), and A_{ji} is the transition probability, which is the probability per second that an atom in state j spontaneously emits in a random direction and de-excited to state i . From Equations 2.2, and 2.4 we get:

$$\frac{\varepsilon_{ji} \lambda_{ji}}{A_{ji} g_j} = \frac{hc_0 N}{4\pi Z(T)} \exp\left[-\frac{E_j}{kT}\right]. \quad (2.5)$$

Taking the logarithm in both sides of Eq. (2.5), one gets

$$\ln\left(\frac{\varepsilon_{ji} \lambda_{ji}}{A_{ji} g_j}\right) = -\frac{E_j}{kT} + C, \quad (2.6)$$

where C is a constant. The emission coefficient can be replaced by the relative intensity I_{ji} (in arbitrary units) of the emission line between energy levels i and j . leading to

$$\ln \left(\frac{I_{ji} \lambda_{ji}}{A_{ji} g_j} \right) = -\frac{E_j}{kT} + C . \quad (2.7)$$

Finally, for a given spectrum, a plot of the logarithmic term in the left-hand side of Eq. (2.7) versus E_j yields a straight line whose slope is equal to $1/kT$. Thus, we can obtain the excitation temperature from the slope a Using $T = 1/ka$. To increase the precision, many spectral lines must be used to determine the excitation temperature. The spectroscopic data corresponding to the Fe lines studied here are presented in Table 2.1, as taken from the NIST atomic line database [65].

Table 2.1: Fe atomic spectroscopy data for measured Fe spectral lines.

λ [Å]	A_{ji} [$10^8 s^{-1}$]	g_j	E_j [eV]	I_{ji} [a.u.]
2714.4129	0.57	6	5.552602	9432.67
2813.2864	0.342	11	5.320399	9364.67
2825.5557	0.132	9	5.344824	11511.33
2947.8760	0.183	7	4.256222	17853
3586.9849	0.166	5	4.445627	17000
3679.9133	0.0138	9	3.368256	18415
3878.5732	0.0617	3	3.283026	31695

2.3.4. Local thermodynamic equilibrium (LTE) in plasma

A thermodynamic equilibrium (TE) in a plasma located in a closed system means that each process is in equilibrium with its reverse process (detailed balance). Such a plasma can be characterized by several parameters, such as the temperature, pressure, and number densities of each single species (electrons, ions, neutrals, etc.). The black-body radiation is described by Planck's law. The population of the excited states obeys a Boltzmann distribution, while the velocity has a Maxwellian distribution.

In a real plasma, deviations from equilibrium occur. For example, radiation escapes out of the plasma, disturbing the detailed balance. If the equilibrium still holds

in a local sense, the plasma is considered in local thermodynamic equilibrium (LTE). The plasma can still be described (locally) by the above-mentioned relations, with the exception of Planck's law. LTE can only exist if the radiation processes can be neglected and the plasma is determined by collisions. Therefore, a sufficiently high electron density is required.

The electrons are more susceptible to the electric field than the heavy particles, due to their significantly lower mass. Electrons therefore gain more energy than the heavy particles per unit time. If the pressure is low or the electron density is small, the heavy particles' temperature remains lower than the electron temperature. The plasma can no longer be described by a single temperature.

The energy difference between the ground state and the first excited state is normally large, whereas the energy difference between the excited states is small, especially in noble gases. If the electron density is sufficiently high, the excited states can be in equilibrium among each other (excluding the ground state). The velocity distribution within the different kind of species can still be Maxwellian (in spite of different temperatures). In such a case, the plasma is considered as in partial local thermodynamic equilibrium (PLTE).

2.3.5. Langmuir probe

Langmuir probes are used to measure plasma densities, electron temperatures and plasma potentials [67,68]. The probe consists of a bare wire which is inserted to the plasma and is electrically biased with respect to a reference electrode to collect the electron and/or ion currents. In a typical plasma, the electrons have significantly higher thermal speeds (due to their smaller mass) than positive ions, even if the electrons and ions are at the same temperature. Although, a plasma is electrically neutral, and the electron and ion densities are nearly equal, a floating probe will tend initially to draw a higher electron current because the electrons reach the probe faster than the heavier ions.

A floating probe inserted to the plasma will measure the floating potential V_f of the probe which is generally not the same as the plasma potential. Because the net current to the floating probe must be zero, the probe floats to a negative potential

relative to the plasma so that further collection of electrons is retarded, and ion collection is enhanced. Thus, the floating potential is less than the plasma potential. The plasma potential V_p is the potential of the plasma with respect to the walls of the device (the microwave cavity in our case) at a given location in the plasma.

The Langmuir's method obtains the current-voltage ($I - V$) characteristic of the probe as the applied bias voltage, V_B , is swept from a negative to a positive potential. When $V_B \gg V_p$, an electron current I_e is collected and the probe current is negative. When $V_B \ll V_p$ an ion current I_i is collected. A typical I-V curve can be seen in Fig. 2.4. The plasma potential V_p is near the “knee” of the curve. At the far left, where all the electrons have been repelled, we have the Ion Saturation current, I_{sat} . The Floating Potential V_f , is where the ion and electron currents are equal, and the net current is zero. In the Transition Region, the ion current is negligible, and the electrons are partially repelled by the negative potential $V_B - V_p$. In a Maxwellian plasma, this part of the curve is exponential and is expressed by

$$I_e = I_{es} \exp \left[\frac{e(V_B - V_p)}{kT_e} \right]. \quad (2.8)$$

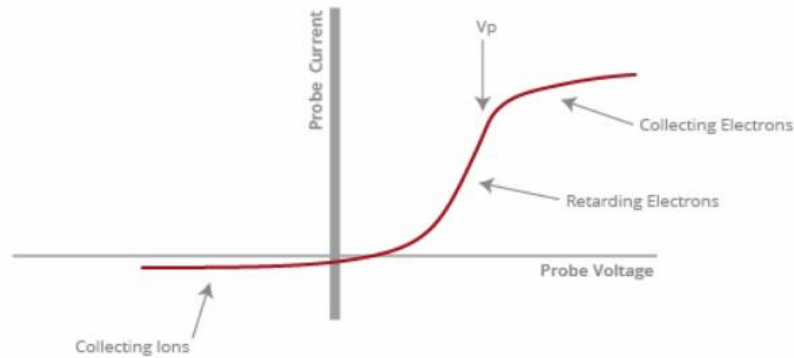


Figure 2.4: Typical I - V measurement by a Langmuir probe [69].

We can see from Equation (2.8) that the slope of the $\ln(I) - V_p$ curve is exactly $1/T_{eV}$ and is a good estimate of the electron temperature.

In our experimental setup a Langmuir-like probe is inserted into the plasma in order to measure its $I - V$ characteristics. As shown in Fig. 2.5, the 1.6-mm diameter,

50-mm long probe is fed by a 50-Hz, ~100-V alternating voltage. Valid results are only observed when the melt exists, and closes the electric circuits to enable the current flow (otherwise much smaller currents are erroneously measured).

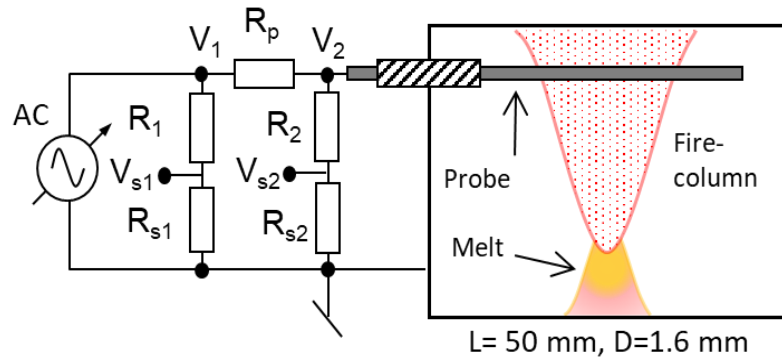


Figure 2.5: The electrical setup of the probe fed by a 50-Hz, 100-V alternating voltage.

The measurements are sampled in a 100 samples per second rate by National-Instruments™ USB-6008 Multifunction I/O Device operated by LabVIEW™. The samples are stored with a time-stamp attached, and the measurements are then analysed to extract the relevant I - V curves.

2.3.6. Radical emission

Discrete emission spectra in the UV and visible range correspond to electronically excited species produced in the chemical reactions in the plasma flame [70]:



This chemically excited radicals or molecule D^* may be destroyed by spontaneous emission ($D^* \rightarrow D + h\nu$) or collisional quenching ($D^* + M \rightarrow D + M$). Chemiluminescence corresponds to the spontaneous emission of photons (such as $h\nu = E_2 - E_1$ with E_2 the energy of the excited state and E_1 the energy of the final state). This emission generally takes place in the UV or visible range. Each molecule or radical produced in an excited state is responsible for a particular spectrum, which is related to its quantum properties and can be identified. The $OH^-(A^2\Sigma^+ - X^2\Pi)$ band emission in the range 300–330 nm reveals the rotational temperature of the radicals, estimated by fitting the line shape to the LifBase™ simulation [71].

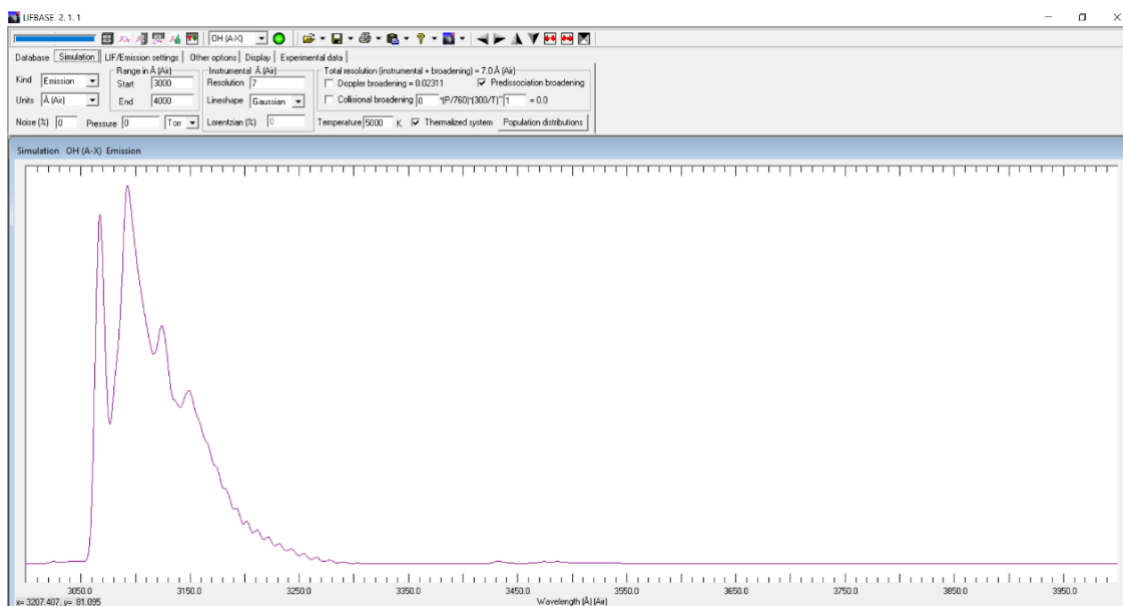


Figure 2.6: An example of a LifBase™ software screen with $OH^-(A^2\Sigma^+ - X^2\Pi)$ radical spectral emission simulated for a temperature of 5000 K and resolution of 0.3 nm [71].

The LifBase™ software simulates the radical emission spectrum for a certain temperature. The temperature, wavelength range, and resolution can be controlled (as in Fig. 2.6). The measured data is then loaded to the software from a file. The two plots (measured and simulated) are shown on the same graph and can then be fitted. Another option is to export the simulated graphs for different temperatures, and compare and fit it to the measured spectrum.

3. Experimental results

A conceptual scheme of the experiment process is presented in Fig. 3.1. The sample is irradiated with microwave energy inside the cavity. The microwave energy irradiated into the core of the basalt stone is converted to heat inside. It melts the core by the LMH mechanism (the molten core is seen in Fig. 3.1a as an inner magma via the porous solid surface). A further irradiation of the basalt stone by a regulated microwave power supply can stably maintain the molten hotspot inside, in a thermal equilibrium. However, a slight increase in the power may elevate the temperature and pressure inside the stone. Cracks may appear then on the rock surface, and the melt may erupt outwards and flow as a lava stream over the rock surface (Fig. 3.1b). This sight looks as a miniature volcano.

In the present work, we have observed that by a further LMH irradiation of the basalt, the hotspot continues to heat up to a point in which plasma is ejected from it, as shown in Fig. 3.1c. Starting from a molten hotspot, the plasma is first emitted and evolves as a stable fire-column. After a few seconds it may also emit a fireball, buoyant in the air atmosphere within the cavity, as shown in Fig. 3.1d.

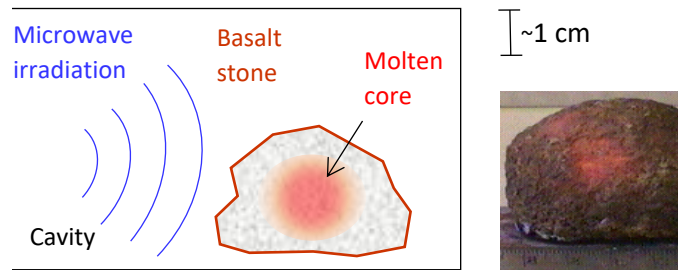
3.1. Basalt melting

3.1.1. Cracks, lava eruption and vitrification (Obsidian)

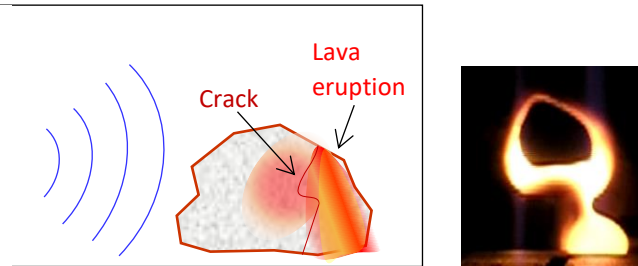
Effects of core melting, followed by lava eruption and flows, have been observed in LMH of natural-shaped basalt stones, as shown in Fig. 3.2. Alternatively, cubic bricks of 3-cm side length (cut from natural basalts) were irradiated instead of the naturally-shaped stones (in order to enable comparison with numerical simulations).

The experiments with basalt cubes resulted in similar effects of lava eruption and flows, as shown in Fig. 3.3. The thermal image shows the molten lava, reaching a $\sim 1200^{\circ}\text{C}$ temperature (Fig. 3.3e). It flows from the core outwards and down by gravity (in an average front-flow velocity of $\sim 1\text{cm/s}$), while its color varies from luminous white to orange and red. It eventually cools down and vitrifies to black glass, namely obsidian. Various volcano-like phenomena are observed, including formation of bubbles and domes, fissures and craters, vents, blasts, bursts of lava, and lava tunneling.

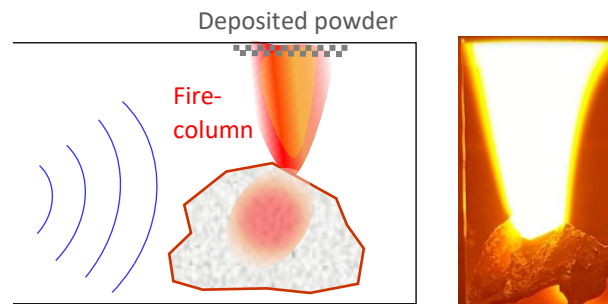
(a) **Core melting**



(b) **Lava eruption and flow**



(c) **Plasma ejection**



(d) **Floating fireball**

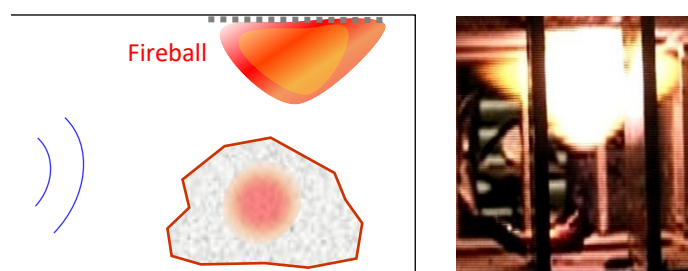


Figure 3.1: A conceptual scheme of the experimental process: (a) Core melting, (b) lava eruption, (c) dusty-plasma ejection and (d) fireball evolution.

The use of electrode (or several electrodes) expedites the heating process (Fig 3.3), reaching melting temperature more rapidly. It generates hotspots on the surface, in addition to the inner hotspot, from which lava erupts more rapidly with no need for cracks to be formed (producing lava-tunneling effect). After a cooling process, most of the partially hollow rock interior vitrifies to black glass, making the rock fragile and crisp.

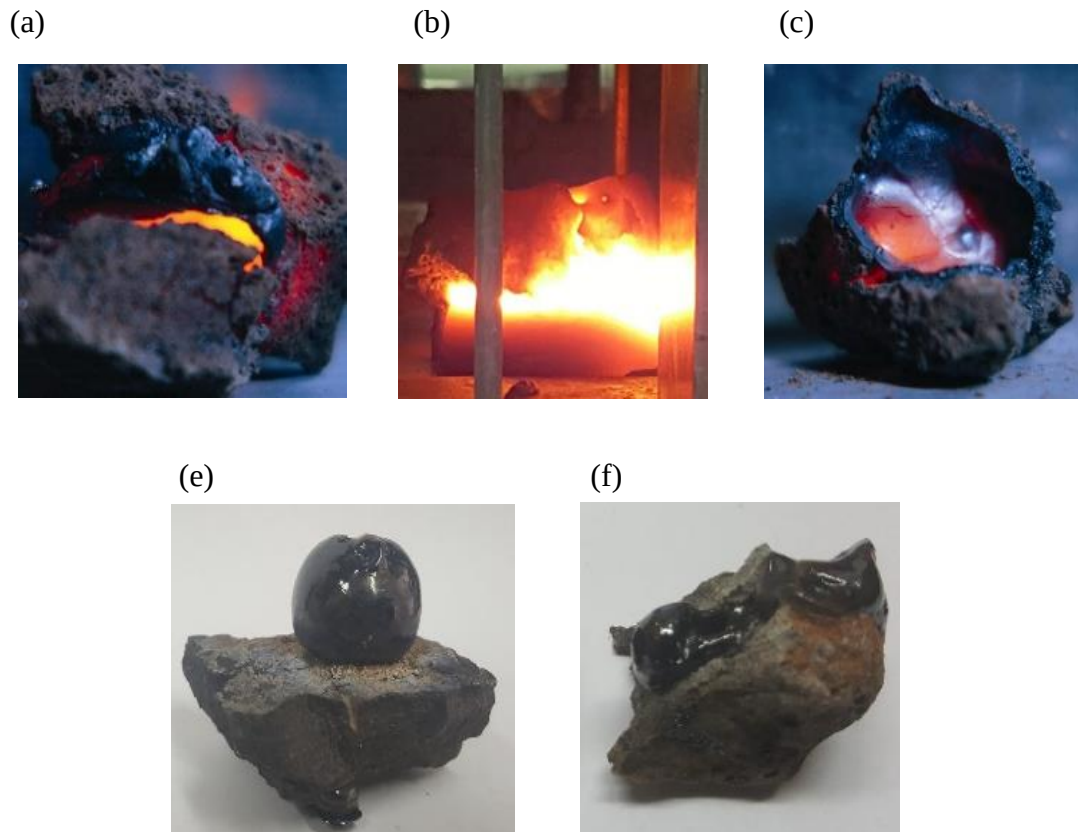


Figure 3.2: Basalt heating and melting by LMH, lava flow, and vitrification to obsidian. The images show naturally-shaped basalt stones (of ~5-cm length) under LMH in various phases: (a) Inner core melting [23]; (b) lava eruption and flow; (c) formation of a crater [23]; (d) a dome of a thin layer of volcanic glass (obsidian) created by lava bubbling; and (e) a trace of lava stream solidified to obsidian.

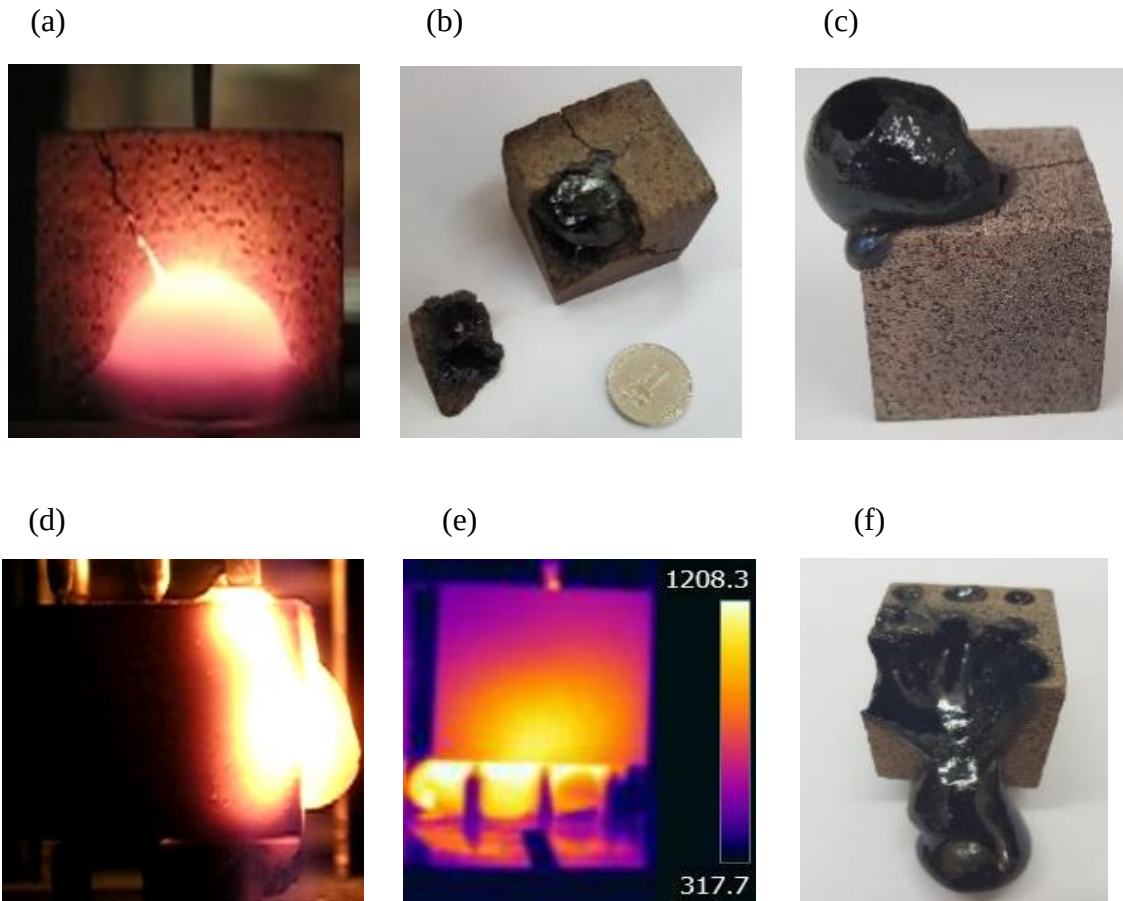


Figure 3.3: Electrode-enhanced LMH interactions with basalt stones cut to 3-cm side cubes: (a) Lava flow from molten core; (b) a basalt cube after a partial LMH interaction (stopped before eruption) revealing the molten core inside which has been solidified to obsidian; (c) a bubble dome; (d) electrode array expedites lava flow from molten core; (e) a thermal image of lava eruption from the core through multiple hotspots in the bottom surface induced by an electrode array; and (f) a hollow core remains after the lava has erupted and streamed outside, leaving a void inside and a solidified traces of obsidian.

3.1.2. Hotspot formation

Video and thermal images can imply hotspot formation in the rock core, as shown in Fig 3.4. Only the rock surface is visible thus only its surface temperature can be measured. Later, numerical simulation can help estimating the core temperature by measuring its surface temperature. Near-surface hotspots can be achieved by using electrodes (as in a microwave-drill device [16]), thus hotspots size and temperature can be measured by a thermal camera (Fig 3.4 d). Another method to obtain near-surface hotspots is by shaping the basalt stone in a sharp fin form, or using natural sharp shaped stones.

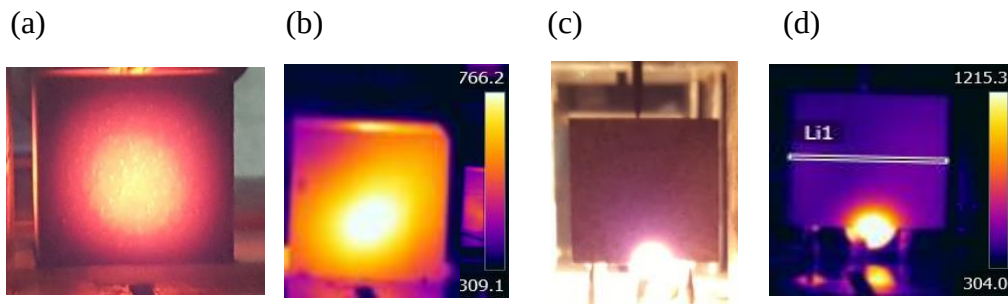


Figure 3.4: Hotspot formation in the basalt sample: (a) An image of the inner hotspot captured via the solid outer side of the cube; (b) a thermal image of the molten core captured via the solid outer sides of the cube (note also the upper side); (c) an image of hotspot induced in the bottom surface of the cube by a multiple electrode array; and (d) a thermal image of the induced bottom surface hotspot.

3.1.3. Temperature profiles

The temperature was measured by the FLIR-SC300TM thermal camera, thus only surface temperature can be measured. The temperature in the centre of the rock face was measured during the heating process, while irradiating the rock either with a single electrode, or by a 1+9 electrode array (one on the top and nine at the bottom), or by a 9+9 electrode arrays. A comparison of the temperature evolution versus time is presented in Fig 3.5. It can be seen that adding electrodes reduces the heating process in the centre face, implying a cooler core. This can be explained by the multiple surficial hotspots attracting the electromagnetic energy from the basalt core, also generating rapid lava flow, in a lava-tunnelling effect, leaving the core hollow and thus not attracting heating energy.

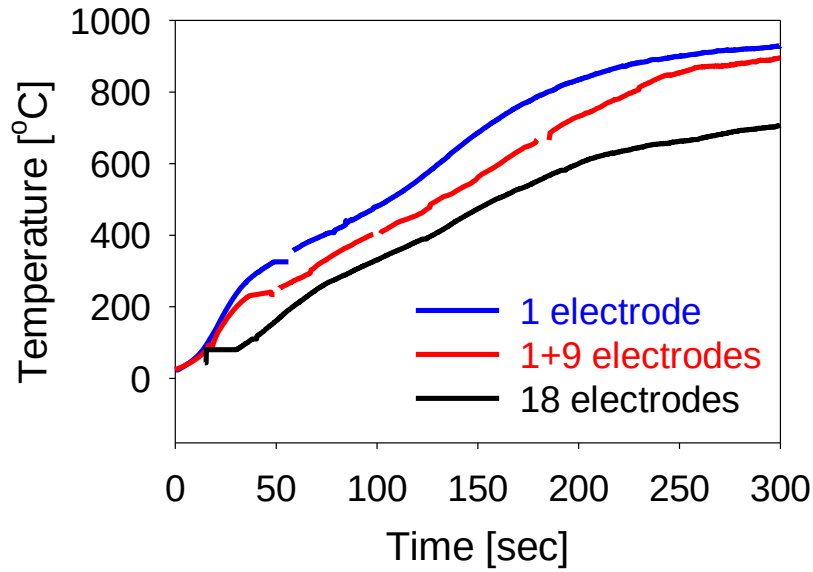


Figure 3.5: Temperature evolution on the surface center of the basalt cube for different irradiation setups.

The temporal evolution of the surface line-profile temperature was also examined. Figure 3.6 presents the centre face line-profile temperature (bottom to top) of: (a) No-electrode heating, (b), one-electrode heating, and (c) 1+9 electrode heating. In the no-electrode case (Fig. 3.6a), the spatial profile satisfies the core heating mechanism. For the one electrode case, placed on the top surface (Fig. 3.6b), it can be seen that the heat is concentrated on the rock upper zone, where the electrode is concentrating the EM-energy. In the 1+9 electrodes case (Fig. 3.6c), at first, heat is attracted to the top and bottom parts (where the LMH process starts) and after ~ 130 seconds concentrates back inside the rock core.

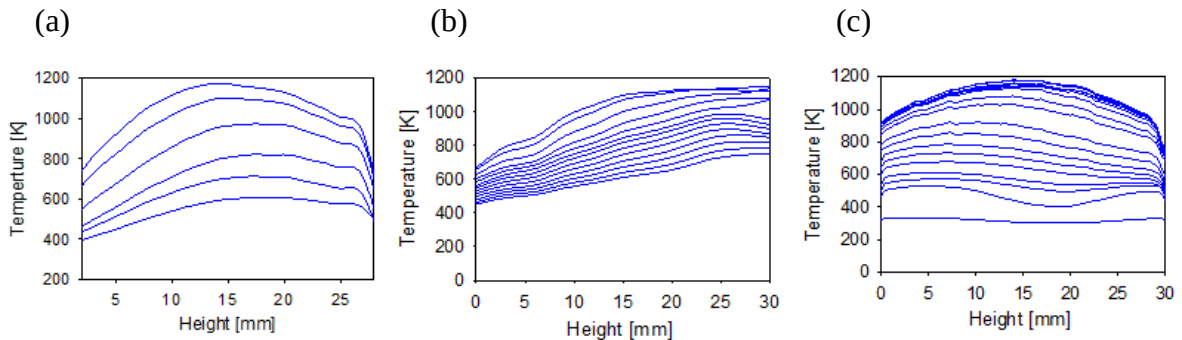


Figure 3.6: The spatial temperature evolution along the side (top to bottom) surface of the basalt cube: (a) Curves in time interval of 60 s corresponding to the no electrode case; (b) curves in time interval of 20 s corresponding to the one electrode case; (c) curves in time interval of 20 s corresponding to the ten electrodes case.

3.1.4. Reflection coefficient measurements (S_{11})

Reflection coefficient measurement, taken by HOMER impedance tuner (S-Team Ltd), shows a relatively large drop in the microwave reflection, associated with the thermal runaway effect and rapid changes in material properties. When the core reaches a critical mass of molten material (of high electric conductivity), the microwave energy absorbance of the core increases, leading to the drop in the reflection coefficient.

Figure 3.7 shows the measured reflection coefficient and the surface temperature. The drop of the reflection coefficient is accompanied with a slight drop in the surface temperature, implying energy concentration in the molten core due to thermal-runaway effect. Hence, by measuring the reflected power we can assume the temperature, phase and state of the irradiated rock core.

Reflection coefficient measurement in the multiple-electrode experiments offers ambiguous results due to local hotspots near the electrodes and arbitrary plasma ejections from the electrode tips. Figure 3.8 shows a reflection coefficient measurement of a multiple electrodes experiment.

3.2. Plasma ejection

Plasma ejection from basalt is initiated in this study by a further irradiation of the molten basalt. In this process, the microwave localization effect continues to evolve up to a point in which plasma is ejected from the melt (shown in the thermal image in Fig. 3.11a). As shown in Fig. 3.10, starting from a molten spot (a), the plasma is first emitted (b), and evolves as a stable fire-column (c). Finally, in some cases, another plasmoid in a form of a fireball is also evolved, either alongside the fire-column (d) or detached from it, as shown in Fig. 3.11d. The detached fireball is less stable than the fireball attached to a fire-column. It tends to move towards the microwave source magnetron, and to disappear within a few seconds. Other modes of plasma ejection from basalts observed in this study are a fire-column eruption from a dome made of a thin layer of obsidian (blown up by the lava eruption), and a simultaneous ejection of two plasma columns from two different hotspots (Fig. 3.11).

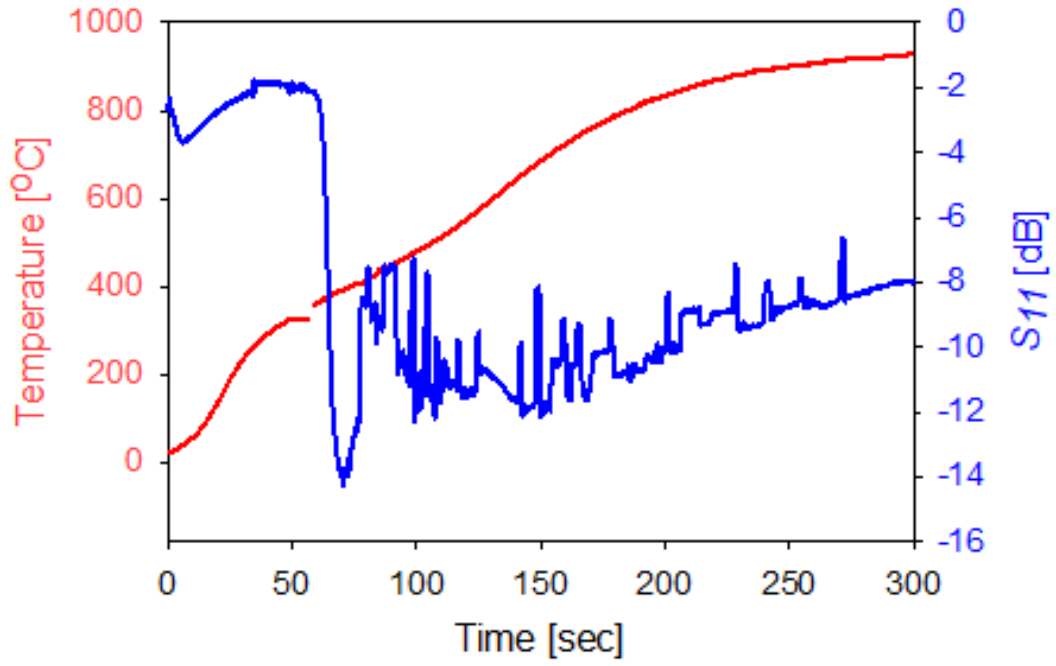


Figure 3.7: The reflection coefficient and center-face temperature during heating process.

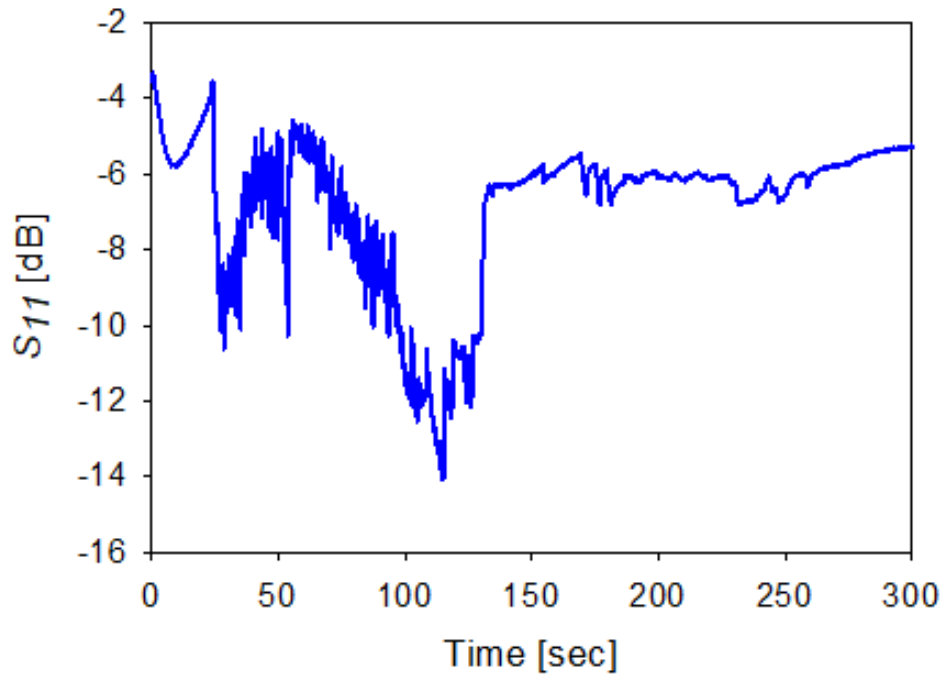


Figure 3.8: Reflection coefficient measurement in a multi-electrode heating experiment.
Multiple hotspots and plasma ejection cause an unstable reflection coefficient.

3.2.1. Microwave scattering, reflection and transmission analyses

The microwave scattered from the plasmoid is analyzed in order to find its microwave impedance, and hence to estimate its dielectric properties (discussed in Section 4.5). First, the reflections of the 2.45-GHz input power are measured during the various stages presented in Figs. 3.10a-d. The results for the complex reflection coefficient Γ are shown in Fig. 3.9a, in time-dependent presentations, and the magnitude with respect to the hotspot temperature in Fig. 3.9b.

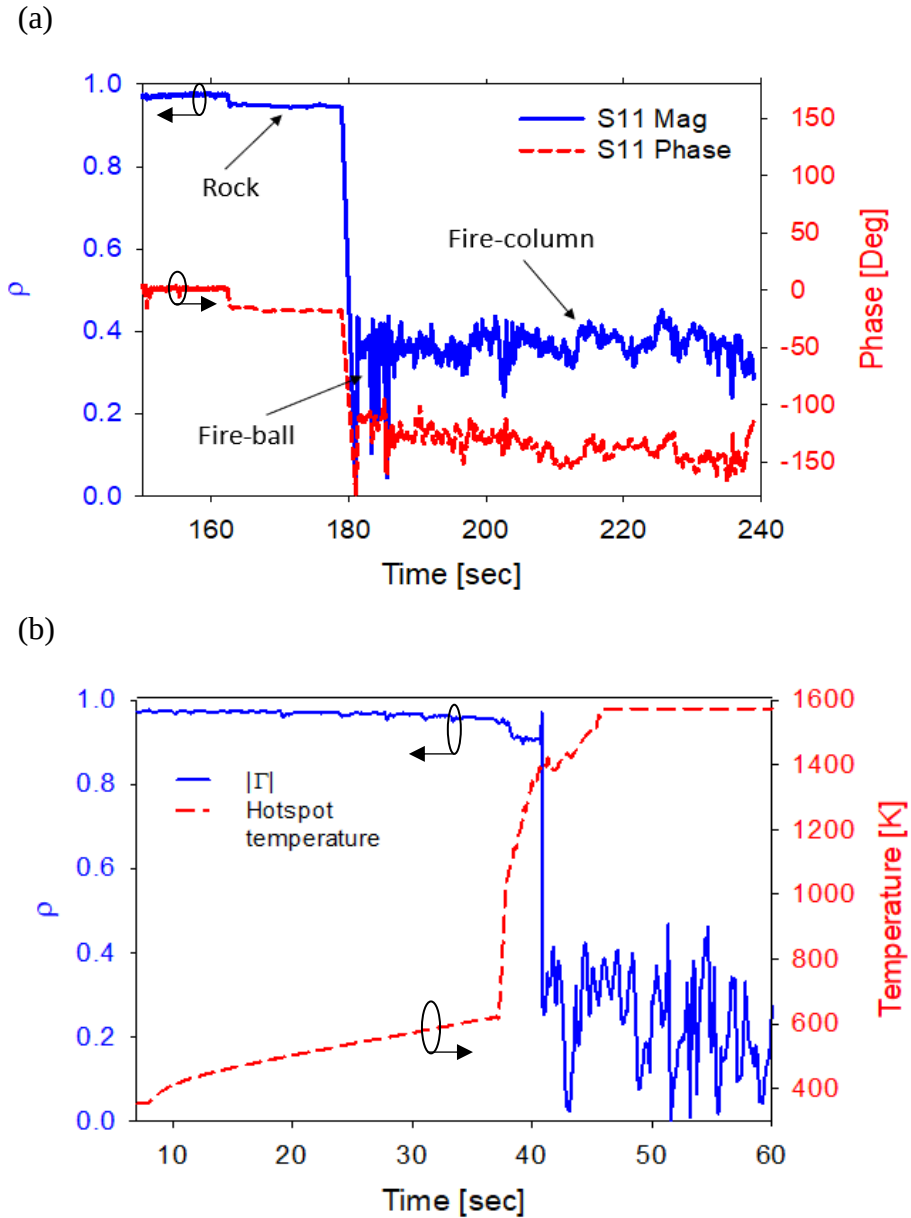


Figure 3.9: Microwave reflections during the various stages of the plasmoid evolution: (a) Complex reflection coefficient (b) The reflection coefficient $\rho = |\Gamma|$, and the hotspot temperature measured vs. time.

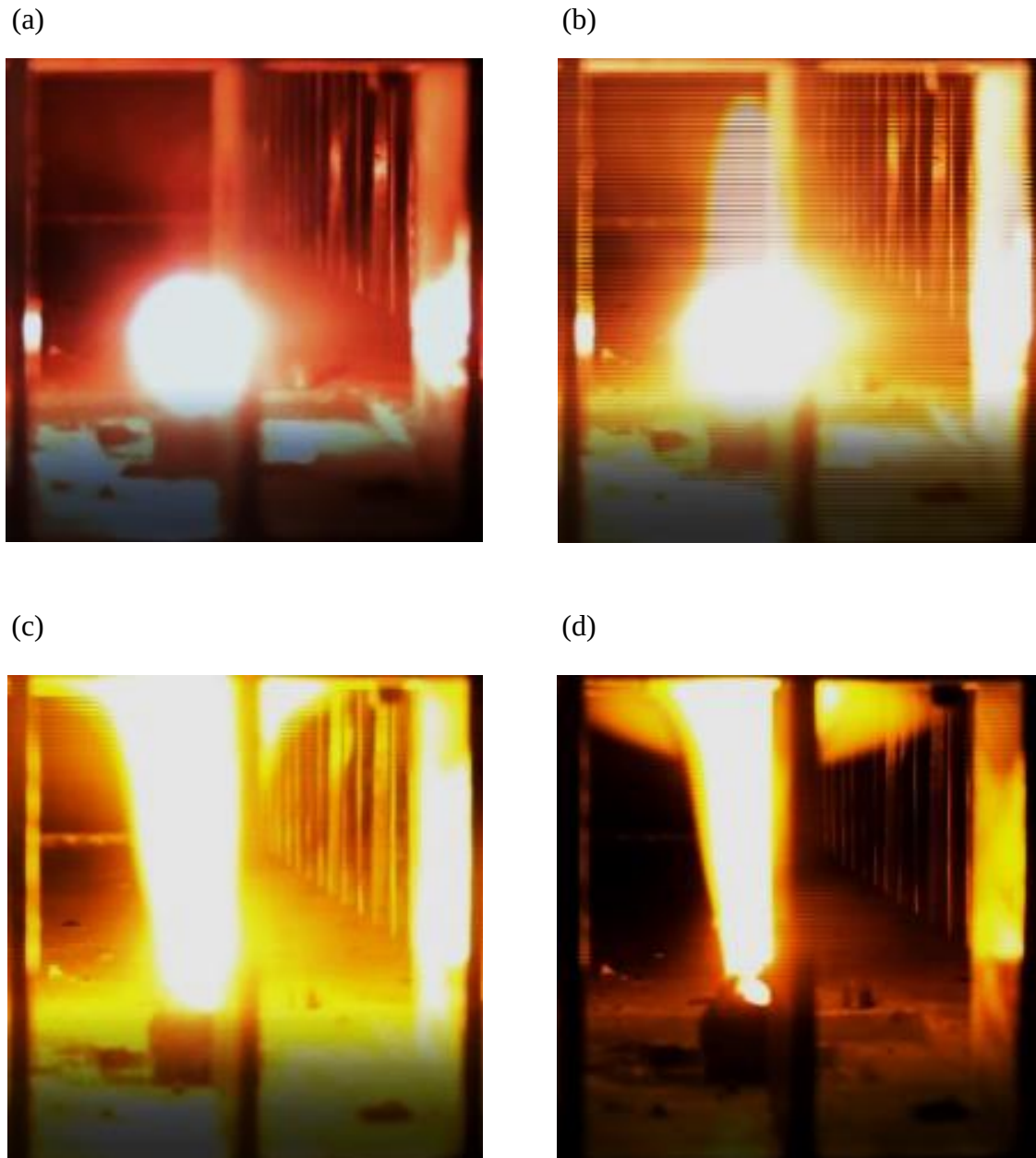


Figure 3.10: The plasma evolution from a molten hotspot in basalt: (a) The hotspot is formed and further evolved; (b) a plasma plume is initially ejected from the hotspot; (c) a stable fire-column is fed by the hotspot emission; and (d) a fireball is separated from the fire-column (beyond it), and hovers in the air atmosphere.

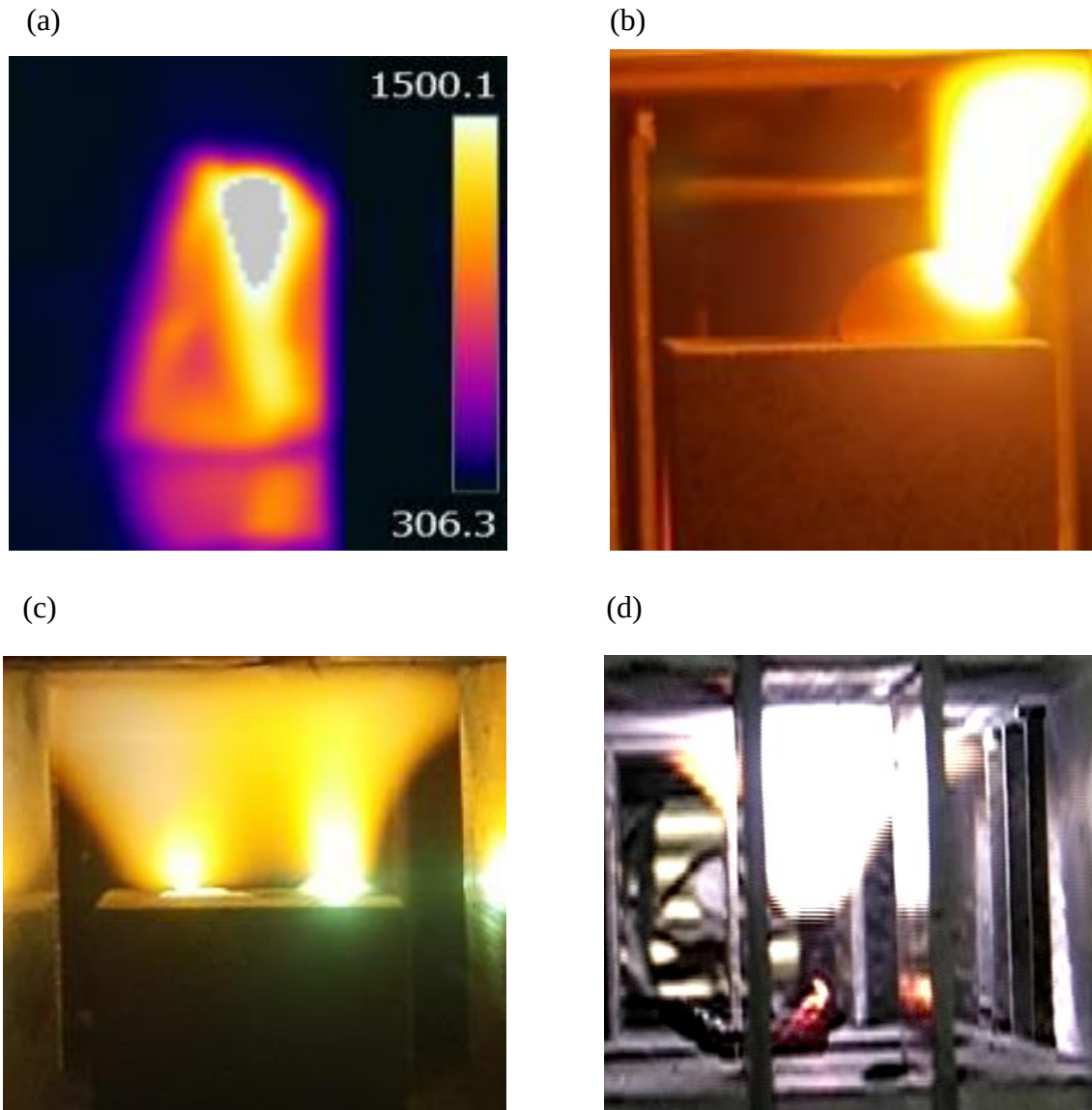


Figure 3.11: Various observations of plasma ejection from molten basalts: (a) A thermal image of a hotspot with a temperature exceeding 1,500K, which emits plasma; (b) a plasma column ejected from a dome of molten basalt, blown up by the lava eruption from the core; (c) a pair of plasma columns ejected from two adjacent hotspots evolved on the basalt surface; and (d) a stand-alone fireball floating in the air atmosphere within the cavity.

The results show that the microwave reflection tends to decrease after the plasma ejection (from $|\Gamma| \sim 1$ to ~ 0.4), and to further decrease when a fireball emerges (to $|\Gamma| \sim 0$). At the steady fire-column stage, the reflection coefficient is bouncing around ~ 0.4 . The corresponding power reflection is smaller than 20% at this stage, hence the plasmoid interacts with its microwave source as an adaptively matched load. At the fireball stage, this self-tuning mechanism tends to adaptively maximize the microwave power absorbed by the fireball, by optimizing its intensity, position and size. The reflection measurements enable us to find the impedance and dielectric properties of the fire-column by simulating an equivalent load, with the same shape, position and size (discussed in Section 4.5.1.4). Figure 3.9b shows an abrupt drop of the microwave reflections ($\rho = |\Gamma|$) during plasma ejection and a slight drop in temperature, indicating plasma ejection and energy concentration by the plasmoid.

Microwave scattering measurements are also performed at other frequency ranges, by irradiating the plasmoid by another probing signal (as discussed in Section 2.3.1). Scattering results for the microwave transmission coefficients at a lower frequency band (< 2 GHz) are shown in Fig. 3.12. The measurement shows the single frequency evolution in time with respect to the hotspot temperature (Fig. 3.12a) and the full spectrum S_{21} evolution (Fig. 3.12b). The results show that around the anti-node frequency of the coupling between Ports 1 and 2 (at ~ 1.25 GHz, which corresponds to $L \sim \lambda/2$ coupler arm length), the transmission coefficient S_{21} is increased by > 20 dB in the presence of plasma.

Indeed, below the plasma frequency, the fire-column behaves like a conductor, which shorts the coupler's arm length by half ($\sim L/2 \sim \lambda/4$) and hence significantly increases the transmission. The transmission increase follows the temperature rise, as shown in Fig. 3.12a for the anti-node frequency, indicating changes of the dielectric and electromagnetic properties in the heated basalt, until the point in which the plasma actually switches between the decoupled and coupled states of Ports 1 and 2. The drop in the hotspot temperature after ejection is attributed to the plasma absorbing the microwave energy.

High frequency measurements did not show significant observable changes, it could be attributed to the sensitivity of the measurement equipment and setup, cavity structure, plasma location inside the cavity, etc. Measurement and simulation results are shown in Section 4.5.2. Some coupling changes are visible with some ambiguity.

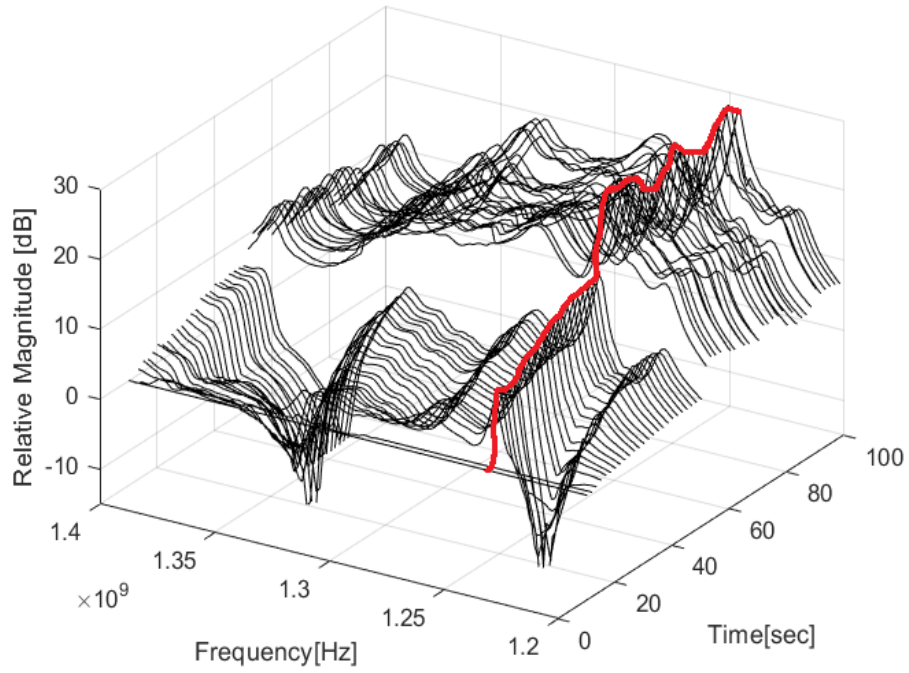
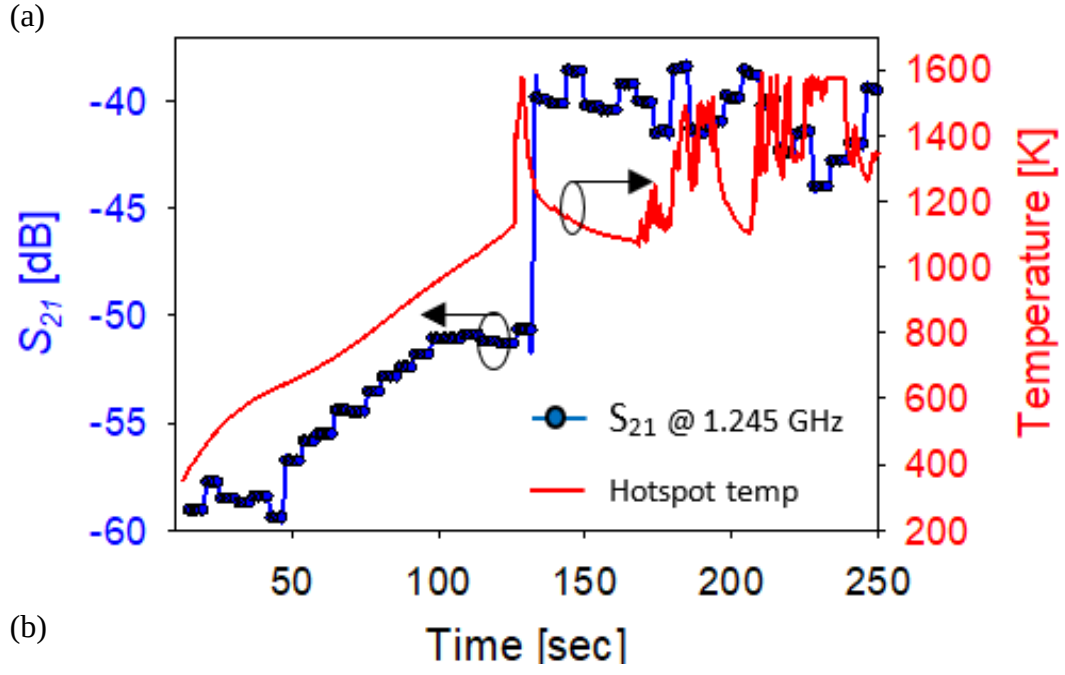


Figure 3.12: Scattering analyses of the plasma by an additional probing signal: (a) The S_{21} transmission variation vs. time at the anti-node frequency (~ 1.245 GHz) with respect to the hotspot temperature variation, and (b) the S_{21} spectral variation vs. time during the plasma evolution, with the anti-node frequency (~ 1.245 GHz) highlighted in red.

3.2.2. Plasma composition detected by optical emission spectroscopy (OES)

The optical spectrum of the light emitted by the basalt plasma-column presents spectral lines near 222, 328, 402, 567, 601 and 670 nm, identified as Fe, P, C, O, Si, Al and Ca lines respectively, as shown in Fig. 3.13a. Due to the limited sampling resolution of the spectrometer (0.3 nm), some of these lines are overlapping or weak (for instance, the continuum seen at the 750-800 nm range, may also indicate the presence of Mg). Nevertheless, all the elements detected by the plasma optical spectroscopy are known as components of basalt rocks.

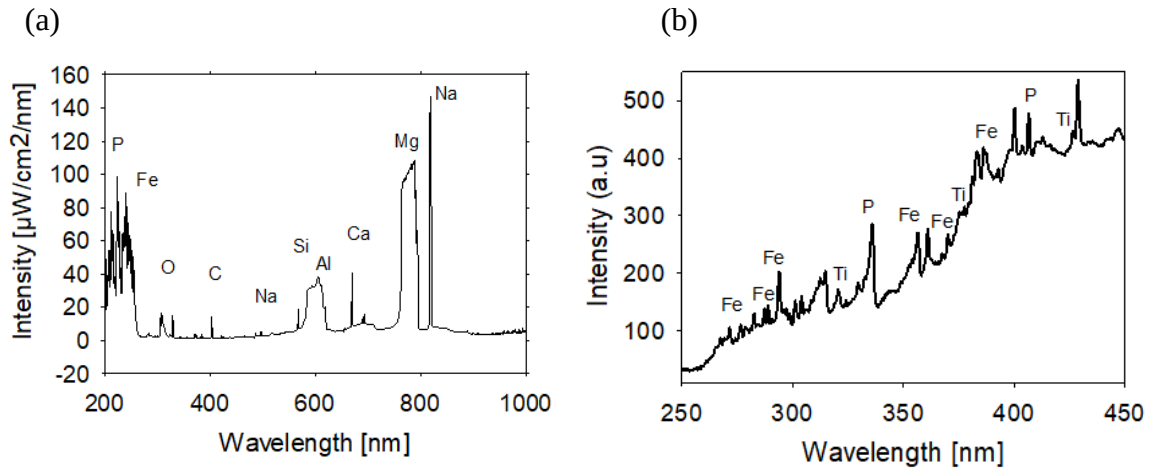


Figure 3.13: Spectrums of atomic emission spectral lines displayed in two ranges and different Intensity scales.

3.2.3. Excitation, rotational, and electron temperatures

The Boltzmann plot technique is implemented on the Fe and Ti lines (Fig 3.13b), as shown in Fig. 3.14a. These result in estimates of the excitation temperature T_{exc} around ~ 0.3 and ~ 0.6 eV, respectively. The spectral emission of OH radicals is identified and compared by fitting to LifBaseTM database [71]. As shown in Fig. 3.14b, the rotational temperature T_{rot} is also roughly estimated in the same range of ~ 0.3 - 0.6 eV.

The Langmuir-like probe measurements in Fig. 3.15a shows the resulting current drawn by the plasma. Such results are only observed when the melt exists (as seen in Figs. 3.15b, c), and closes the electric circuits to enable the current flow

(otherwise much smaller currents are erroneously measured). A presumable reason is the isolating powder oxide accumulated on the metal surfaces, which adds a significant resistance to the plasma current loop. The hysteresis observed in the I - V curve may also indicate a contamination of the electrode, which leads to a capacitive effect. The rough estimates of the electron temperature by the curves' slopes results in ~ 0.3 and ~ 0.6 eV for the voltage rise and fall (arrowed), respectively, in Fig. 3.15a. This range of electron temperatures resembles the estimated optical results above.

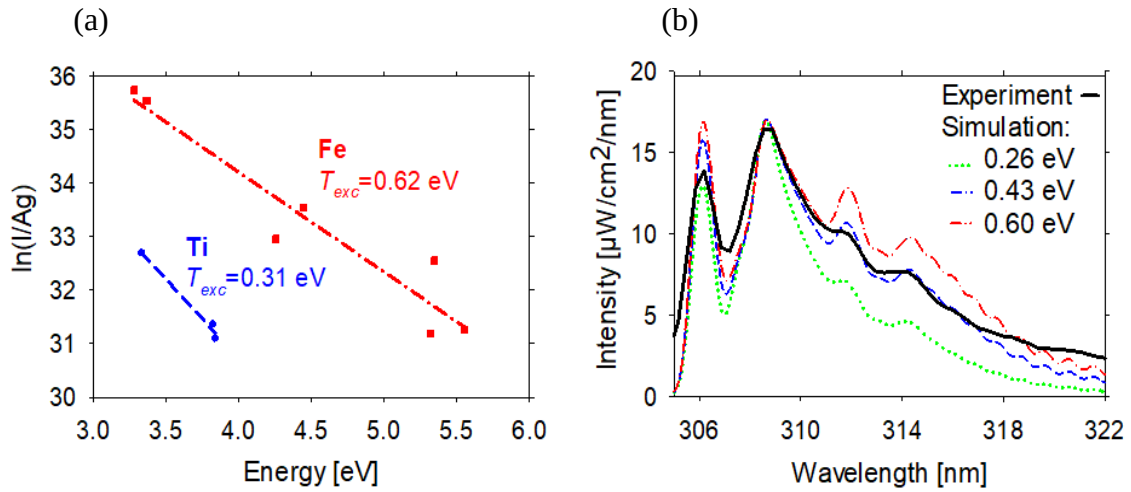


Figure 3.14: (a) A Boltzmann plot of the Fe and Ti lines, and (b) the OH-radical emission compared (by fitting) to LifBase™ simulation.

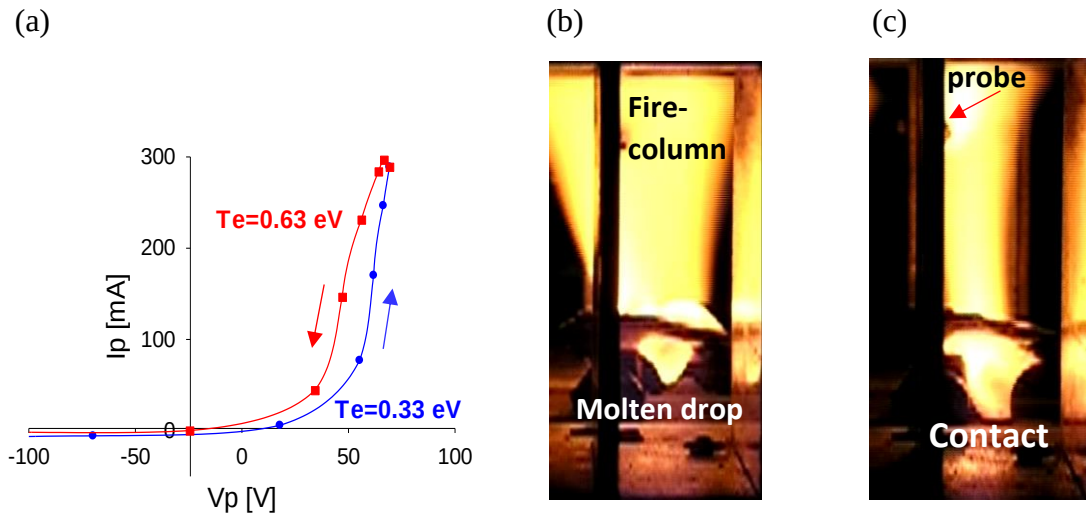


Figure 3.15: Characteristic I - V curve measurements: (a) A typical I - V curve measured. Such I - V curves are only obtained when the melt exists as illustrated in (b), (c), and closes the electric current loop.

3.3. Ex-situ analyses, SEM and XRD observations

The basalt rock, molten glass (Obsidian) and powder (deposited on the collector) are characterized by a FEI Quanta 200FEG environmental SEM. The chemical (elemental) composition is analyzed using energy dispersive spectroscopy (EDS) with a Si (Li) liquid-nitrogen cooled Oxford INCA X-ray detector.

3.3.1. Irradiated rock

First, an observation of a microwave treated basalt sample was performed. The inner core of the sample is melted, and the diagnostics focused on the non-melted surface of the sample. It can be seen from Fig. 3.16 that the material inside the porous region has been slightly crystalized. An EDS analysis of the two regions (porous and flat) was performed. According to Fig. 3.17, the chemical contents of the two regions are nearly the same except a slight increase of Mg, Fe and O in the porous region. An EDS analyses was also performed on a non-treated rock sample face, with a similar porous region (Fig 3.18), a comparison can be seen in Table 3.1. No significant changes in chemical content can be deduced.

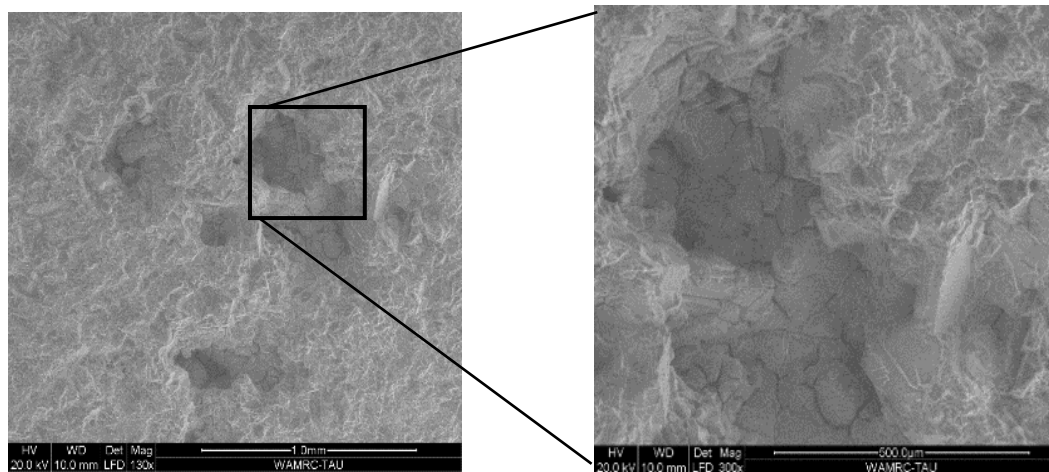


Figure 3.16: SEM image of the non-melted surface of the microwave treated basalt (the inset shows the porous region).

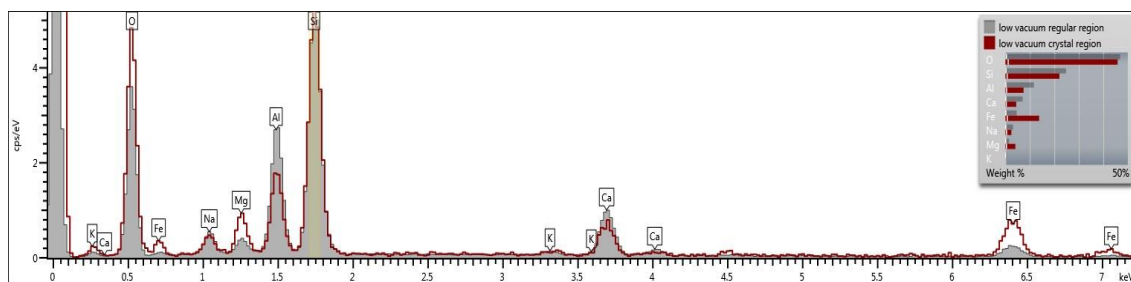


Figure 3.17: EDS analyses on the microwave treated basalt non-melted surface for the slightly crystallized and non-crystallized surfaces.

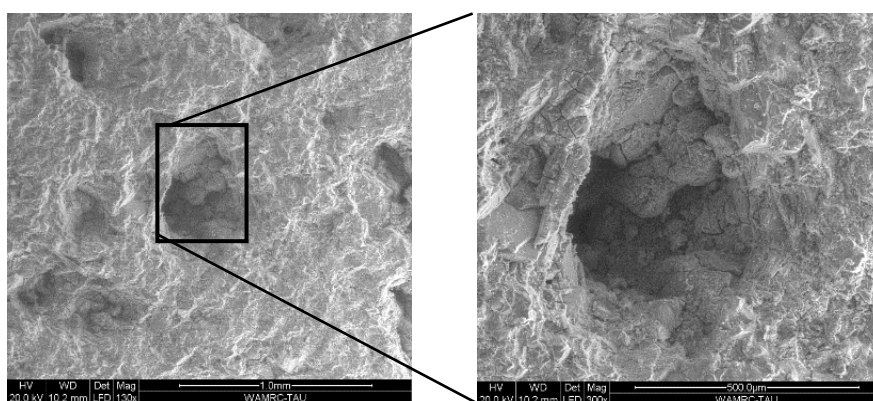


Figure 3.18: SEM image of non-treated basalt sample (inset shows the porous region).

Table 3.1: A comparison between EDS analyses of treated and untreated solid basalts.

Element	MW treated sample (WT%)	Non-treated sample (WT%)
O	45.82	46.53
Na	2.43	2.68
Mg	4.05	2.5
Al	7.42	8.19
Si	22.07	23.84
K	-	0.67
Ca	4.41	6.1
Ti	-	0.43
Fe	13.8	9.07

3.3.2. Obsidian

A molten region that has been solidified to black volcanic glass (Obsidian), was analysed by SEM (Fig 3.19) and EDS. The sample exhibits white grain-like structures (could be as a result of dirt contamination) surrounded by a black region (bright indicates elevation). A closer look on the black region, shows stream-like darker structures in the solidified obsidian, with brighter island-like areas. EDS analyses of the white and black regions shows similar chemical contents and a slight increase in Fe in the darker region (Table 3.2).

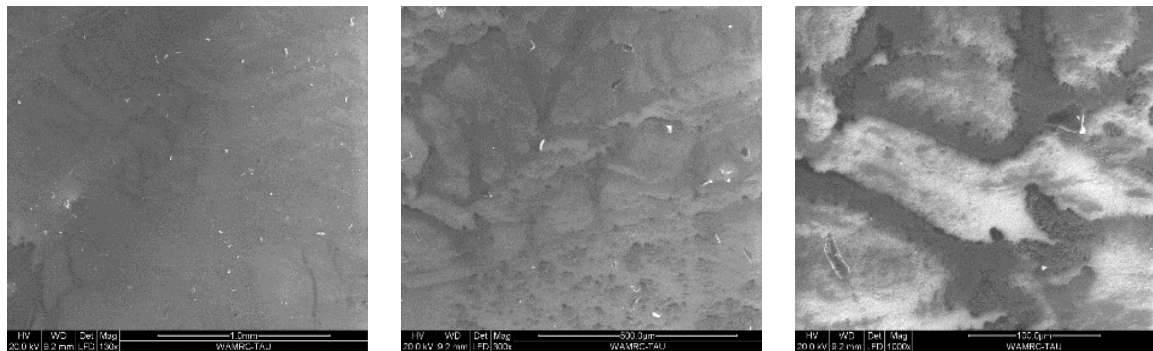


Figure 3.19: SEM images of the volcanic glass obtained shows stream-like darker structures in the vitrified obsidian, with brighter island-like areas.

Table 3.2: EDS analyses of the dark and bright areas of the obsidian sample.

Element	Dark region (WT%)	Bright region (WT%)
O	51.78	57.94
Si	20.08	21.29
Ca	7.93	8.57
Ti	1.81	1.4
Fe	18.39	12.92

3.3.3. Plasma dust accumulated on the collector

Powder accumulation measurements were performed in several ejection methods and different rock samples (cut stone and different natural shaped stones). Here we present the SEM-SE and EDS analyses for different experiments.

First, analyses of the aluminum collector (Fig 3.20) on which the powder is accumulated were performed. The analyses indicate that the substrate consists mostly of Al and some Mg as expected (Table 3.3).

Table 3.3: EDS analyses of the aluminum collector.

Element	Collector (WT%)
O	5.65
Mg	1.3
Al	91.52
Si	0.79

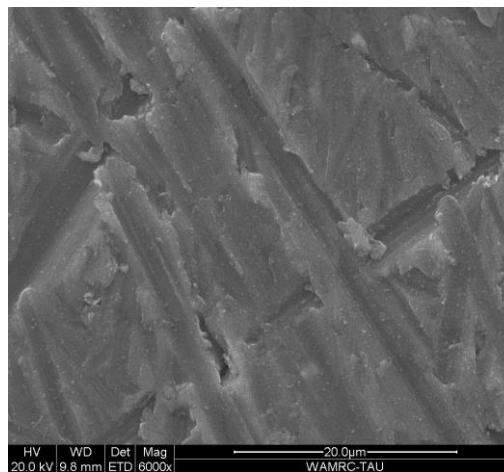


Figure 3.20: A SEM image of the aluminum substrate.

For powder accumulated by plasma ejection from a natural shaped stone (with no electrode involved in the process) a relatively unified distribution of particles is accumulated on the substrate (Fig. 3.21). The particles observed are shaped in a flower-like form of a $\sim 5 \mu m$ size.

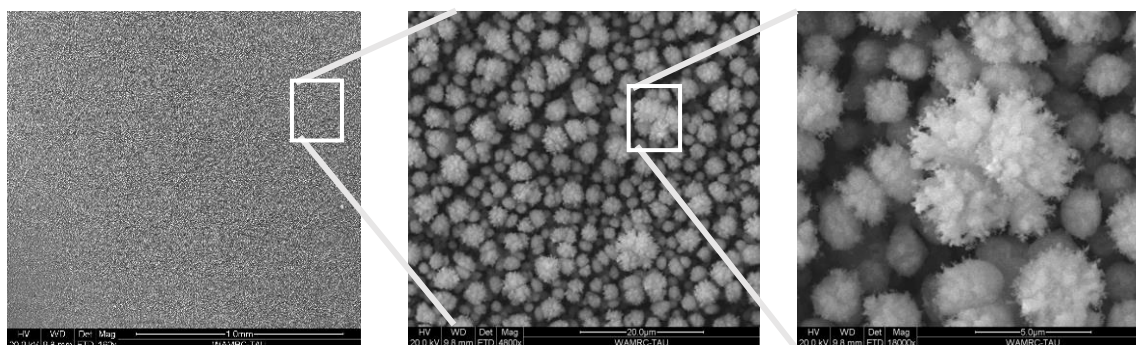


Figure 3.21: SEM observations of the collector show aggregates of flower-like shapes uniformly spread over the collector surface.

Table 3.4: EDS analyses of the powder region and a single flower-like particle.

Element	Powder region (WT%)	"Flower" particle region (WT%)
O	41.37	45.47
Na	21.74	25.18
Al	5.1	-
Si	4.02	4.42
P	16.83	15.17
S	0.51	1.05
Cl	0.39	0.29
K	7.39	6.47
Fe	2.66	1.64

EDS analyses (Table 3.4) show a relatively high concentration of P in the accumulated powder as opposed to the rock and obsidian samples. The only source of P could be in the basalt rock. It is not observed in the analyses of the rock samples because of its low content relative to the other elements.

Similar experiments resulted in particle distribution with random accumulation (Fig. 3.22). Analyses of the two regions (nanoparticles and larger particles) are presented in Table 3.5.

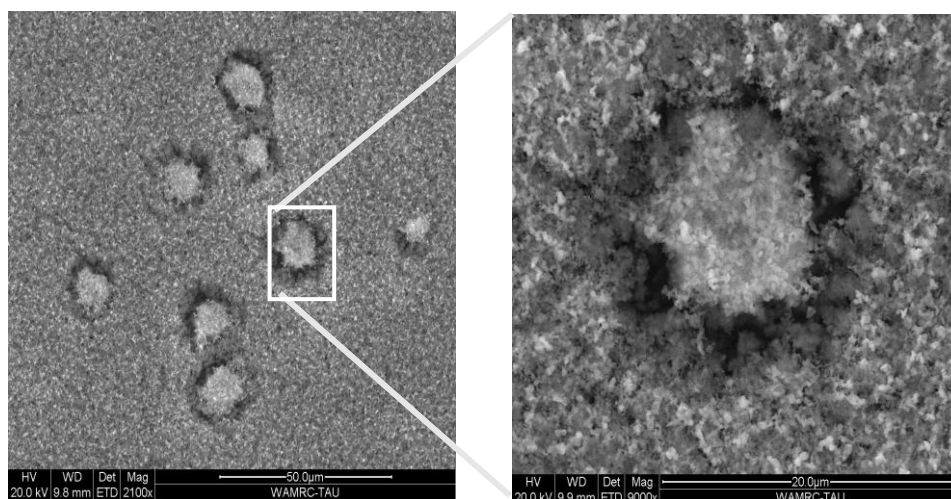


Figure 3.22: SEM image of aggregates of flower-like shapes spread over the collector with random accumulations of larger ($\sim 20\text{-}\mu\text{m}$ sized) structures.

Table 3.5: EDS analyses of the powder region and the large particle region.

Element	Powder region (WT%)	Large particle region (WT%)
O	40.12	41.09
Na	20.87	22.29
Al	10.07	3.16
Si	4.38	3.77
P	14.65	17.52
S	1.24	0.61
Cl	0.84	0.23
K	6.07	8.22
Fe	1.76	2.24

The powder produced by plasma ejected from a cut stone sample by a tungsten electrode was also examined. SEM observations revealed diamond-shaped tungsten particles in addition to the powder particles (Fig. 3.23). EDS analyses of these particles confirm that they mostly consist of W, with its source being in the tungsten electrode.

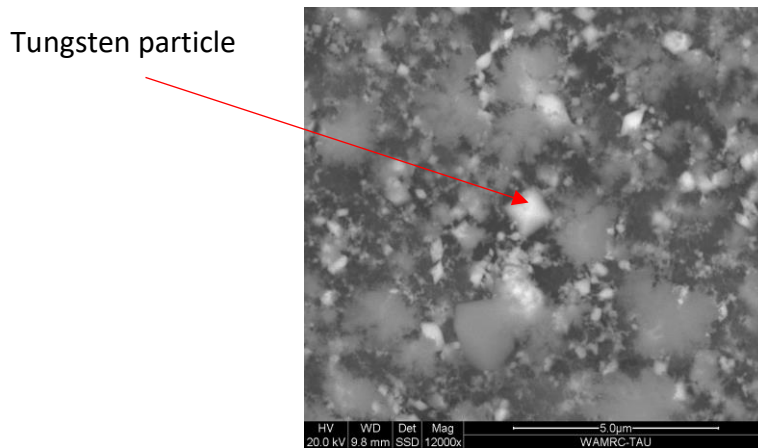


Figure 3.23: A SEM image of collected powder with a diamond shaped tungsten particle from the tungsten electrode.

As discussed above, the use of an electrode to eject plasma for powder accumulation is contaminating the process. Further plasma ejection experiments were conducted with natural shaped rocks in a triangular form.

In additional powder-accumulation experiments, the plasma generated larger spherical particles, in sizes of tens of microns, as seen on the collector (Figs 3.24a, b). Also noticed on the collector surface, are holes of similar sizes (Fig. 3.24c) and spherical particles reside in craters (Fig. 3.24d), which could be attributed either to particle bombardment or to a plasma localization effect.

In order to estimate the thickness of the powder layer accumulated on the collector, a fissure was scratched in the powder layer (Fig 3.25). Height measurements at different positions inside the fissure and on the powdered substrate were taken. The powder thickness was estimated to be around 10-40 μm in a deposition rate of $\sim 0.2 \text{ m}^3/\text{s}$.

Finally, a comparison between the compositions of the natural stone, vitrified product (obsidian) and the accumulated powder was performed. Table 3.6 shows the relative concentrations of the constitutive elements in the various phases of this process. The weight ratios differ between the various phases. However, it seems that the deposited powder consists of elements provided solely by the basalt (e.g. Si and Fe). These ingredients of the powder were most likely extracted from the basalt, evolved and transported by the plasma to the collector. Accordingly, these elements were also detected by optical spectrometry in the plasma (section 3.2.2).

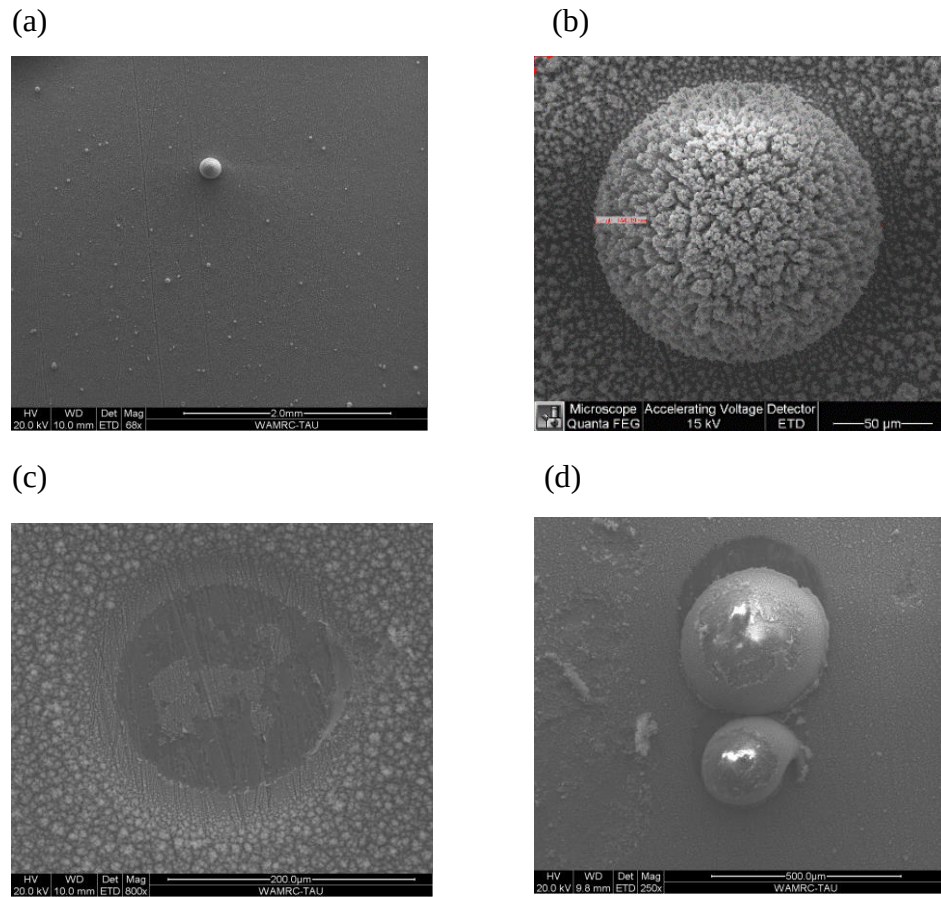


Figure 3.24: SEM observations of the collector: (a, b) An example of a larger sphere observed, of a ~ 0.2 -mm diameter; (c) craters and holes of comparable sizes of ~ 0.1 -mm diameter; (d) spherical particles reside in craters.

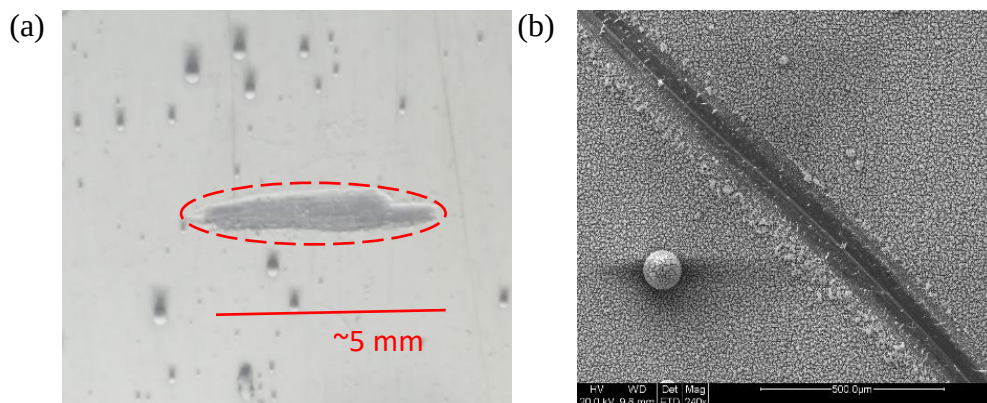


Figure 3.25: (a) An optical image of the white powder accumulated on the collector, with relatively large spherical particles observed. (the scratch marked in Red was made in order to evaluate the layer thickness). (b) A SEM image of a thinner fissure manually carved in the powder substrate in order to measure powder width.

Table 3.6: The elemental composition of the original basalt stone and the two solid phases produced, the vitrified melt (obsidian) and the powder accumulated on the collector (typical values in Wt %).

	O	Si	Al	Fe	Ca	Na	Mg	K	Ti	P
Basalt	48	23	8	7	6	3	2	1	1	1
Obsidian	52	21	6	6	6	2	5	<1	1	<1
Powder	42	4	5	3	-	22	-	7	-	17

3.3.4. XRD measurements

(a) Obsidian

The diffraction pattern of a massive sample has been collected and analyzed (Fig 3.26). The sample show low intensity of diffraction peaks, suggesting that the great part of the sample is a glass. From the position of the peaks, it is possible to determine that the two major crystalline phases are an oxide of aluminum, magnesium and iron, and a silicate of magnesium and iron. No further analyses were possible for the low quantity of crystalline phases in the sample.

(b) Plasma dust

The diffraction pattern of the white deposition on the aluminum collector is shown in Fig 3.27. The signal was low, but sufficient to determine the presence of some crystalline phases. From the analyses, there is the presence of three silicate phases (Enstatite $\text{Mg,Fe}_2\text{Si}_2\text{O}_6$, Augite $\text{Ca,Mg,Fe}_2\text{Si}_2\text{O}_6$ and Mullite $3\text{Al}_2\text{O}_3 \bullet 2\text{SiO}_2$) and two oxides (Periclase MgO and Wustite $\text{Fe}_{0.9}\text{O}$).

Any peak can be related with a phase with phosphor, it is possible that these phases are in the amorphous phase. The presence of amorphous phase is confirmed by the presence of a modulated background. The two big peaks at 35° and 45° are due to Al substrate.

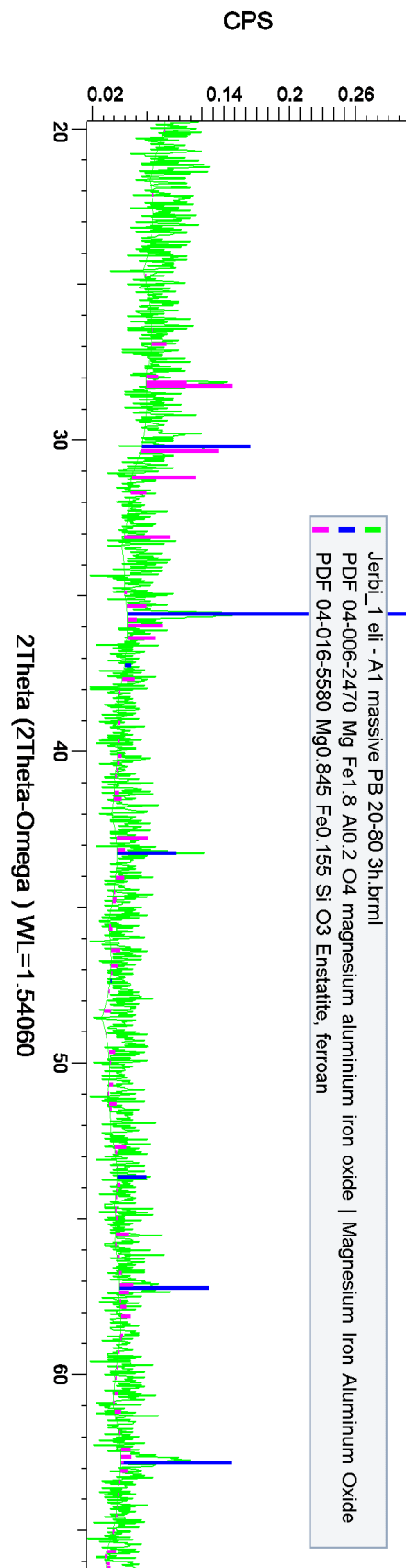


Figure 3.26: XRD analyses of molten basalt (obsidian) sample.

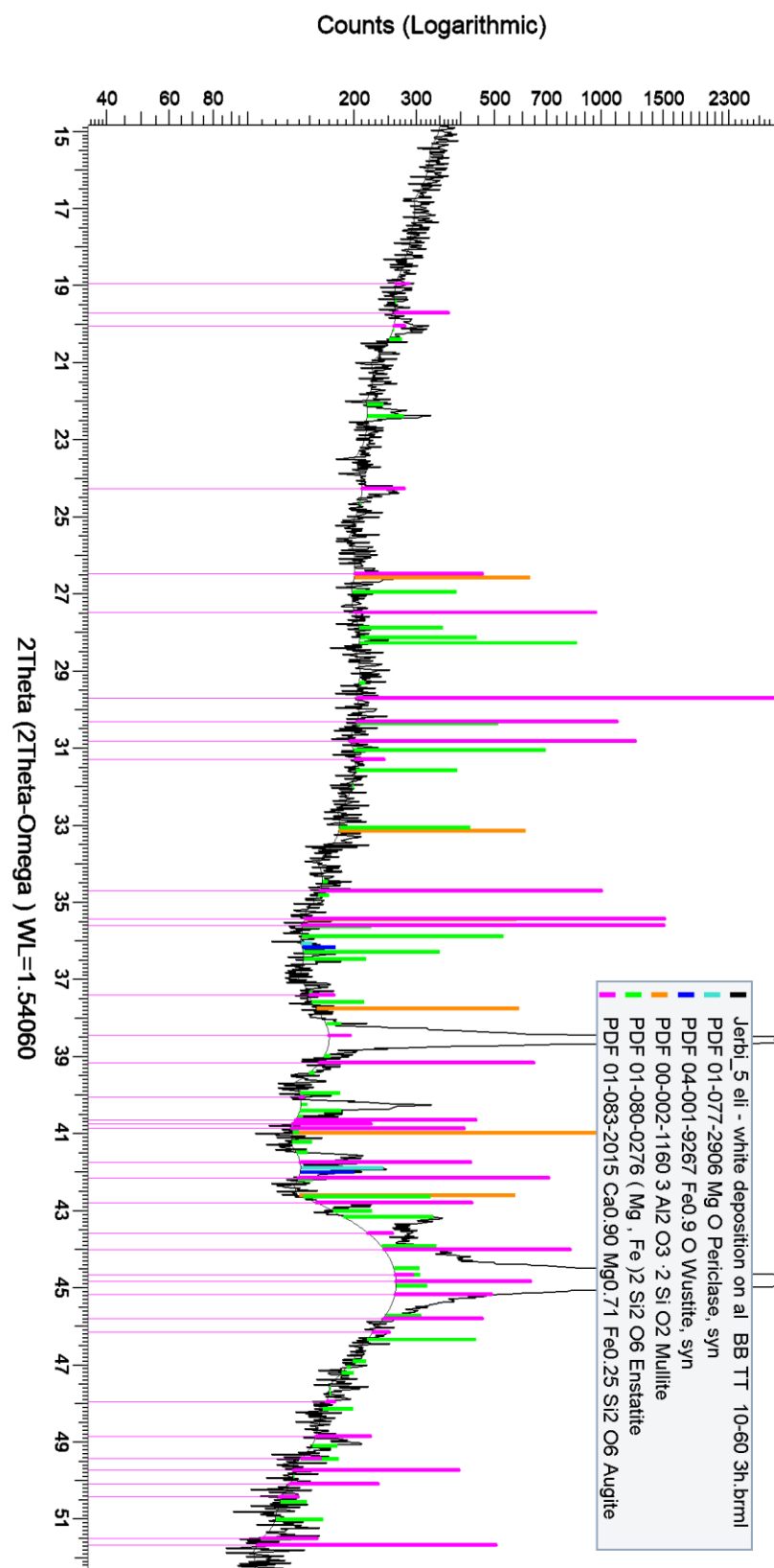


Figure 3.27: XRD analyses of the plasma generated powder accumulated on the aluminum collector.

4. Theoretical analyses

4.1. Localized microwave-heating (LMH) model

A theoretical analysis of the LMH effect and the thermal-runaway instability is performed in this thesis, based on [32]. The model presented here includes dielectric, convective and magnetic loss mechanisms. In the case of basalt stone, the dielectric and convective loss are the major factors in the heating process. The analyses also include melting and phase changes.

4.1.1. Material properties

The LMH process is attainable due to the temperature dependence of the irradiated material properties, such as thermal properties (i.e. thermal conductivity and thermal capacitance), mechanical properties (density), and electromagnetic properties (permittivity, permeability and electric conductivity). In this case, a small non-uniformity in the field pattern causes local variation of the material properties due to local temperature changes. Examples for temperature dependent properties, such as the dielectric loss factor $\varepsilon_r''(T)$ of NylonTM (Ertalon[®]-6), mullite, and soda-lime glass, are found in literature [72-75]. Such non-linear parametric dependencies on temperature introduce a positive-feedback response to the heating process, causing greater energy dissipation in a local region where the temperature increases. The local temperature increase, and the further rise in the loss factor there, lead to a rapid temperature's rising, and eventually to the initiation of a local thermal-runaway instability. The temperature at the hotspot region exceeds a certain threshold above which the temperature increase accelerates [32]. This very unstable stage (in some cases a destructive stage) is ceased by melting, evaporation or plasma generation from the hotspot.

The thermal-runaway instability is known in conventional processes for the accidental damages that it may cause to processed materials in volumetric microwave applications, thus efforts are being made there to stabilize the process. In our case, this instability is intentionally initiated, either for melting, cracking and weakening of the rock, or for hotspot formation from which plasma is ejected. Ways to initiate, intensify and maintain the LMH effect for basalt rocks is further discussed in this thesis.

4.1.2. Coupled Maxwell-heat equations

The heating process is modelled by a coupled EM-thermal model. The model consists of Maxwell equations and the heat equation, both for an inhomogeneous lossy media, with the suitable boundary conditions (e.g. of the cavity is presented in Fig. 4.4). The temperature-dependent properties of the heated material are incorporated in the loss mechanisms and the heat equation.

The EM equations used for the calculation of the fields are presented in the frequency domain as follows

$$\nabla \times E = -j\omega\mu H, \quad (4.1a)$$

$$\nabla \times H = J + j\omega\varepsilon E, \quad (4.1b)$$

where E , H and J are the electric, magnetic and current density vector components, respectively in a phasor form. The complex, temperature-dependent permittivity and permeability are expressed as

$$\varepsilon = \varepsilon_0(\varepsilon'_r - j\varepsilon''_r), \quad (4.2a)$$

$$\mu = \mu_0(\mu'_r - j\mu''_r), \quad (4.2b)$$

where ε_0 and μ_0 are the permittivity and permeability of vacuum, respectively. The current density is related to the electric field by

$$J = \sigma E, \quad (4.3)$$

where σ is the temperature-dependent electrical conductivity. By combining Equations 2.1a,b and 2.2a,b,c we get the EM-wave equation:

$$\nabla \times \left[(\mu'_r - j\mu''_r)^{-1} \nabla \times E \right] - \left[\varepsilon'_r - j \left(\varepsilon''_r + \frac{\sigma}{\omega\varepsilon_0} \right) \right] k_0^2 E = 0. \quad (4.4)$$

The heat equation is presented in the time domain as follows:

$$\rho C_p \frac{\partial T}{\partial t} = Q + \nabla \cdot (k_{th} \nabla T), \quad (4.5)$$

where ρ is the local density of the material, C_p and k_{th} are its heat capacity and thermal conductivity, respectively, T is the temperature, and Q is the power density absorbed by the material during the microwave irradiation.

The EM and heat equations might be coupled together by the three local EM heating mechanisms (namely the ohmic, dielectric, and magnetic loss components), as follows:

$$Q = Q_{ohm} + Q_{die} + Q_{mag} = \frac{1}{2} \sigma |E|^2 + \frac{1}{2} \omega \epsilon_0 \epsilon_r'' |E|^2 + \frac{1}{2} \omega \mu_0 \mu_r'' |H|^2. \quad (4.6)$$

We assume here that the temperature is slowly varying with respect to the EM wave, hence the distinction between the typical time scales of the EM wave propagation (~ 1 ns) and the much slower thermal evolution (> 1 ms) allows the two-time scale approximation [30].

Furthermore, the EM bandwidth is sufficiently narrow to neglect parametric variation with frequency. Hence, the wave equation (4.4) is solved here in the frequency domain whereas the heat equation (4.5) is computed in the slowly-varying time domain.

4.1.3. The microwave-drill effect

The present work utilizes the microwave-drill effect. As discussed above, in most cases microwave heating is volumetrically performed, in ovens, applicators and furnaces, with a typical heat-affected zone comparable to the microwave wavelength. The microwave-drill concept [16] intentionally utilizes the thermal runaway effect in order to drill into hard nonmetallic materials. A principle scheme of the microwave-drill applicator is illustrated in Fig. 4.1. The microwave energy propagates through the coaxial waveguide in which the inner conductor tip is shaped as a sharp electrode and has a physical contact with the projected substrate. The electric field is concentrated on the inner conductor tip due to its sharp shape, and generates a small hotspot there, much smaller than the microwave wavelength itself.

The enhancement ratio between the electric field in the contact area and the incident field propagating in the waveguide depends on the electrode shape and the waveguide impedance with respect to the load impedance. A proper design of the applicator can lead to improved performance and efficiency.

In a thermal-runaway process, the material becomes soften or molten inside the bound hotspot. The center electrode is then inserted into the soft hotspot to form the hole.

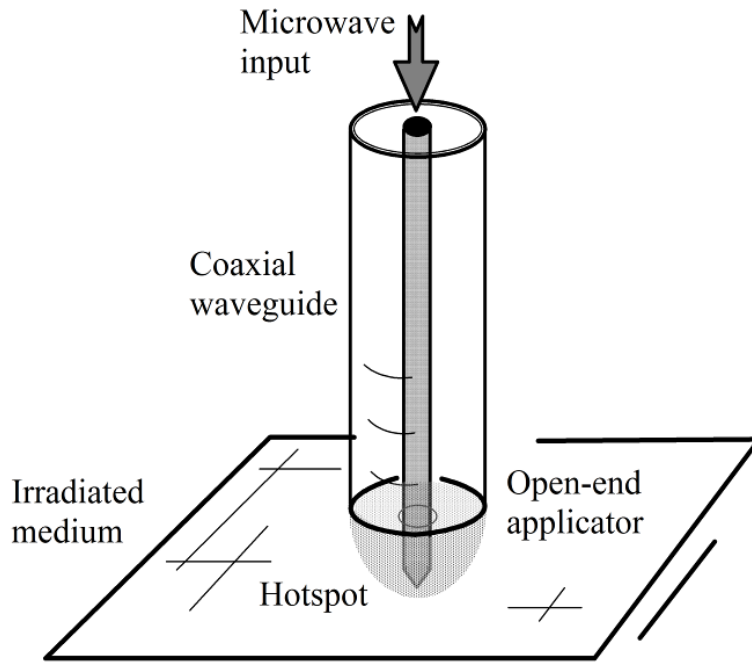


Figure 4.1: The open-end coaxial applicator for microwave concentration [16].

A similar effect can be achieved by introducing an electrode inside a cavity [42]. As in the microwave drill process, the sharpened edge electrode is in contact with the irradiated sample inside the cavity, concentrating the electric field to the region of the electrode tip forming a hotspot in the material substrate.

4.2. Numerical simulations

The heating process is modelled by solving the coupled electromagnetic-thermal equations in the 3D domain, using COMSOL-MultiphysicsTM software and database. The basalt thermal and EM properties, in particular the dielectric permittivity and electrical conductivity, are imported to the model with their temperature dependence, as listed in Table 4.1. The waveguide walls are modelled as perfect conductors (PEC) in the boundary conditions for the EM equations. The heat generated within the cubical basalt stone is transferred to the environment by a blackbody radiation, air convection (to the surrounding atmosphere), and heat transfer to the cavity floor (at room temperature).

4.2.1. Finite-element method (FEM)

The physics of space- and time-dependent problems are usually expressed in terms of partial differential equations (PDE's). For the vast majority of geometries and problems, these PDE's cannot be solved by analytical methods. In these cases, an approximation of the equations can be constructed. The equations can be discretised and solved by using numerical methods. The finite-element method (FEM) can be used to compute such approximations. Dividing the space and geometry of the problem into finite elements, solving the numerical equations for each element, and updating its boundary conditions, is the core of FEM. The size and shape of each element has significant impact on the accuracy of the solution, and on the complexity and computation time as well. Dividing time variable into time steps allows solving the equations for each time step, updating the time dependent variables, and advancing to the next time step.

4.2.2. The two-time scale approximation

The distinction between the typical time scale of the EM wave propagation (~ 1 ns) and the few orders of magnitude slower temperature variation (> 1 ms), enables one to apply the two-time scale approximation [30]. An indication for the validity of this approximation can be provided by the condition $\rho C_p d_{hs}^2 / k_{th} \gg T$, where d_{hs} is the hotspot width, and $T = 2\pi / \omega$ is the EM-wave period. Due to the distinct two-time scales of the LMH problem, the EM and thermal equations can be solved by two separate solvers; the electromagnetic fields in the frequency domain, and the temperature in the time domain [30]. In COMSOL-MultiphysicsTM [76], in order to solve the problem in the time domain, more than 10^5 RF cycles (iterations) need to be computed by the solver before the problem evolves to the periodic steady-state solution. By solving the fields equations in the frequency domain, the change in the field over a single RF cycle does not need to be resolved, and thus the periodic steady state solution is reached much more rapidly. In this case, the transient thermal response of the object of interest is computed, but small fluctuations (e.g. in temperature) over any RF cycle are neglected.

4.2.3. Material properties

As previously discussed, the material properties and their temperature dependencies have a great influence on the heating process. The simulation consists of a basalt rock surrounded by air inside a metal cavity. The cavity walls are modelled as perfect electric conductors (PEC) and are contributing only to the boundary conditions (discussed in Section 4.2.5). The properties of air were taken from COMSOL-Multiphysics™ library. The basalt rock material properties and its references are presented in Table 4.1.

Table 4.1: The temperature-dependent basalt's properties [77-83] used in the numerical simulation. The melting temperature is $T_m = 1360$ K and the N -th order polynomial is denoted by $P_N(a_0, a_1, \dots, a_N) = \sum_{n=0}^N a_n T^n$.

Property	Symbol	Unit	Value / Expression		Ref.
			$T < T_m$	$T_m < T < 1600^\circ\text{K}$	
Relative Permittivity	$\varepsilon_r = \varepsilon' - j\varepsilon''$		$8.18 - j0.708$	$5.95 - j0.208$	[77, 78]
Mass density	ρ	$\text{kg} \cdot \text{m}^{-3}$	2670	2650	[79]
Specific heat	C_p	$\text{J} \cdot \text{kg}^{-1}\text{K}^{-1}$	$P_4(-217.4, 6.4, -0.012, 10^{-5}, -3.1 \cdot 10^{-9})$	$P_3(250.22, 2.56, -0.003, 1.26 \cdot 10^{-6})$	[*]
Thermal conductivity	k_{th}	$\text{W} \cdot \text{m}^{-1}\text{K}^{-1}$	$P_3(0.4145, 0.0018, -9.2315 \cdot 10^{-7}, -1.1836 \cdot 10^{-10})$		[80]
Electric conductivity	$\sigma(T)$	$\Omega^{-1}\text{m}^{-1}$	$110 \cdot \exp(-12,093T^{-1})$	$3.8 \cdot 10^5 \exp(-16,977T^{-1})$	[81]
Emissivity	ε_s		0.74		[82]
Relative permeability	μ_r		1		[*]
Latent heat	H	$\text{J} \cdot \text{kg}^{-1}$	40,000		[83]

* COMSOL-Multiphysics™ database.

Figure 4.2 describes the parameters k_{th}, σ_c , namely the thermal conductivity and electrical conductivity, respectively, and their temperature dependencies. The abrupt ascent in the electric conductivity curve is attributed to the basalt phase variation from solid to liquid.

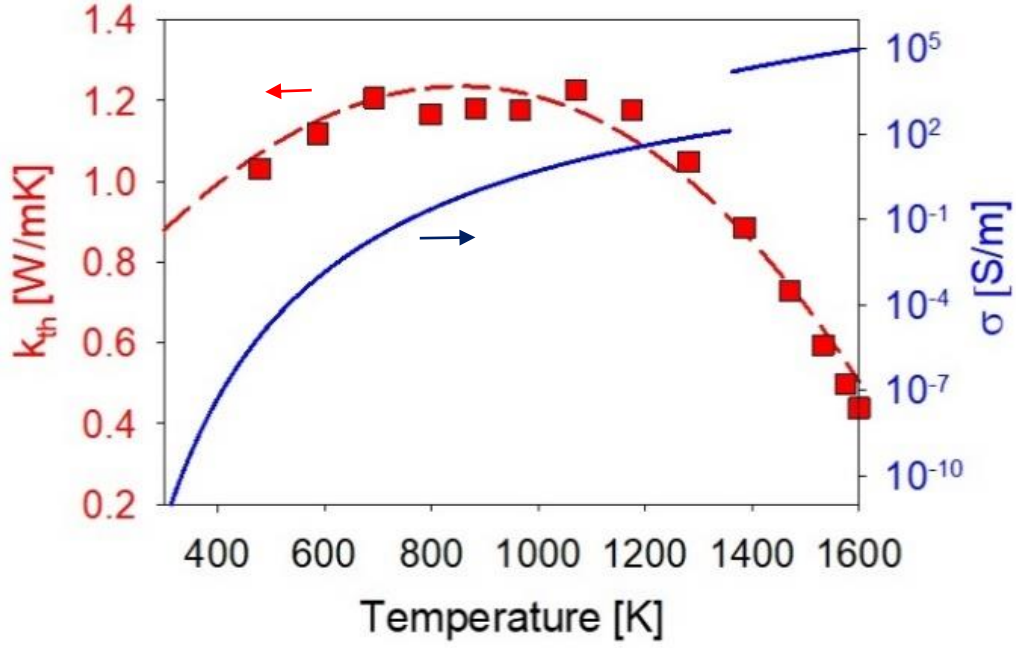


Figure 4.2: The temperature dependencies of the thermal and electric conductivities, k_{th} [80] and σ_c [81], respectively.

With respect to microwave heating, the parameters concerning microwave energy conversion to heat energy can be incorporated as the effective relative dielectric loss coefficient, in the form:

$$\varepsilon''_{eff}(T, \omega) = \varepsilon''(T, \omega) + \frac{1}{\omega \varepsilon_0} \sigma_c(T, \omega) , \quad (4.7a)$$

Or alternatively by the effective electric conductivity defined as

$$\sigma_{eff} = \omega \varepsilon_0 \varepsilon''(T, \omega) + \sigma_c(T, \omega) . \quad (4.7b)$$

Figure 4.3 describes an approximation of the effective electric conductivity (Eq. 4.7b).

The rise of two orders of magnitude in the effective electric conductivity between solid basalt and molten basalt (lava) in the region of the melting temperature (>1200 K) leads to heat accumulation in the molten areas as a result of excessive microwave energy absorption in those areas as compared to the adjacent areas which have not been melted. Regarding these differences, the thermal-runaway instability is generated and simulated.

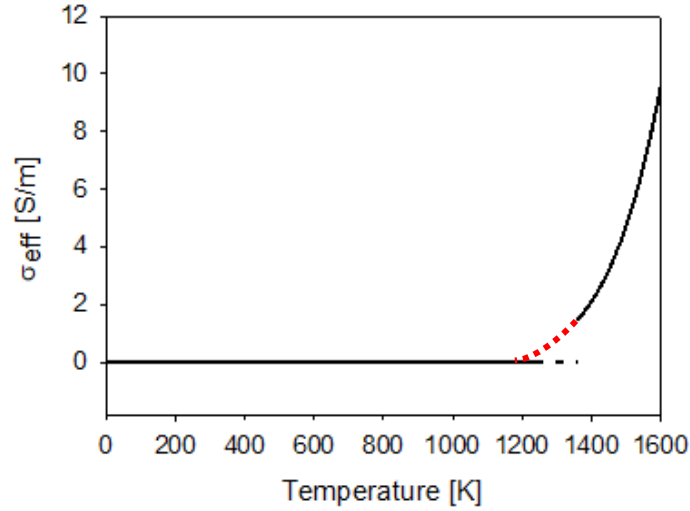


Figure 4.3: An approximation of the effective electrical conductivity σ_{eff} of basalt.

Additional aspects of basalt's LMH are also taken into consideration, as follows:

- The physical properties taken from literature are valid up to temperatures of 1600 K (Table 4.1). Beyond 1600 K the software code performs an extrapolation, which is not accurate compared to experiments.
- In some cases, as seen in experiments, the rock is heated to a point where lava erupts from cracks in the rock surface. These effects are not simulated and causes dissimilarities between experiments and simulation.
- In some cases, as seen in experiments, plasma is ejected from the irradiated basalt rock. This effect causes changes in microwave power scattering and absorption, and thermal and electromagnetic boundary conditions. This effect is not simulated hence it may cause dissimilarities between experiment and simulation in these cases.
- The emissivity of blackbody radiation becomes significant above temperatures of 350 K. According to [82], also the emissivity is temperature dependent. According to some references, it may vary from 0.74-0.995, but no explicit temperature dependency was found in the literature.
- Basalt rock is not a well-defined material and its components, such as the different minerals and water content, may variate between different rocks. Hence, the electromagnetic and thermal properties of each rock may differ by the origin and type of the rock.

4.2.4. Geometry

The simulation consists of three main geometrical objects, namely the waveguide, the basalt rock, and the electrode(s). The basalt rock is represented by a cube with a side length of 30 mm. The waveguide used in the experiment, is a periodic waveguide as shown in Fig. 4.4. The 22 mm opening between each two vanes prevents microwave leakage at the 2.45-GHz frequency, and yet provides a line-of-sight to the heated object.

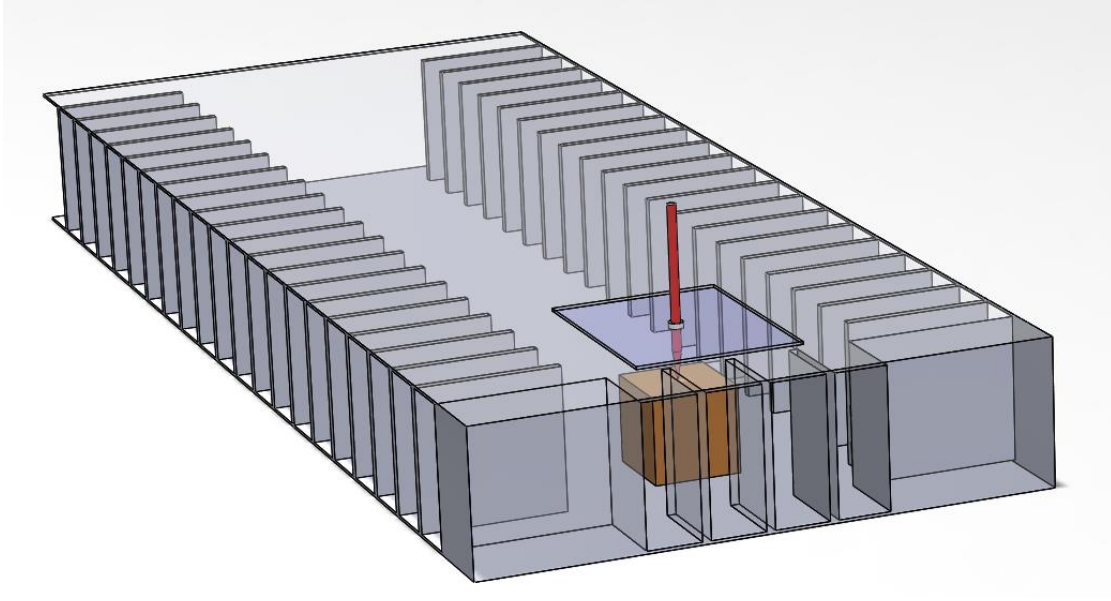


Figure 4.4: The simulated experimental setup (presented by a SolidWorks™ model).

In order to simplify the model and accelerate computation time, the slotted waveguide was simulated as a continuous metallic rectangular waveguide shortened at one of its ends. The open side was pointed as an electromagnetic wave source at 2.45-GHz. The port was placed on the XY plane at $z = 0$, and the cavity dimension were taken to be the inner dimension of the slotted waveguide designed for fundamental mode (TE01) propagation of 2.45-GHz (see Table 4.2).

Table 4.2: The wave guide dimensions

Width	(x axis)	100 mm
Length	(z axis)	426 mm
Height	(y axis)	50 mm

A simulation of the unloaded cavity was conducted in order to find the spot where the electric-field norm attains its peak (later shown in Fig. 4.11a). The cube was then located at that spot (in the simulation as well as the experiment). This spot was measured to be 42 mm from the waveguide shortened wall, on the waveguide longitudinal axis. Finally, a box of air was built surrounding the basalt cube, its goal and dimension are further discussed below.

(a) No-electrode case

The basic simulation includes the basalt cube radiated inside the cavity in order to simulate the inner hotspot formation and core melting to the point where lava erupts (Fig. 4.5).

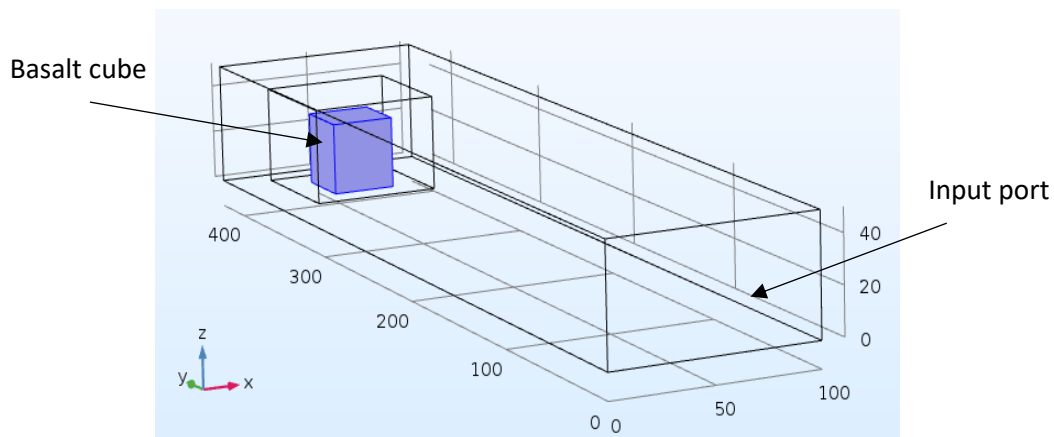


Figure 4.5: A model of the no-electrode simulation (COMSOL Multiphysics™).

(b) Single electrode case

In order to simulate the setup consisting of a single electrode, two sequential simulations may be conducted. As seen in the experiments (Fig. 3.3c), the electrode concentrates the electromagnetic energy by its tip, resulting in hotspot formation in the vicinity of contact with the rock. The material near the surface is melted, and lava erupts from the hotspot, leading to the formation of a lava dome outside the cube.

The simulation is divided into two parts: (a) Simulating up to the melting of a certain volume beneath the electrode (Fig. 4.6a). (b) A sequential simulation starting with a crater (of the same molten volume removed) under the electrode (Fig. 4.6b) with

the initial temperature taken to be the average temperature resulted in the previous simulation.

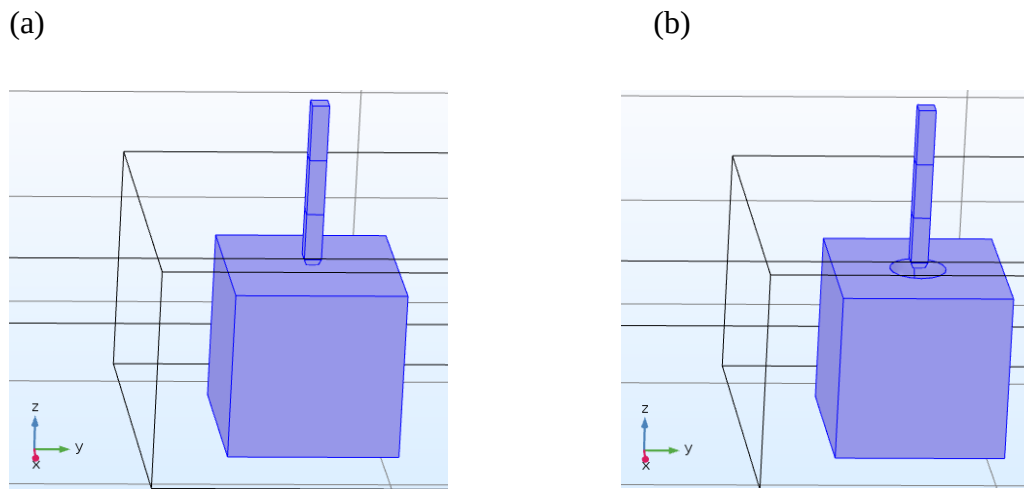


Figure 4.6: A model of the basalt cube with one electrode: (a) At first the electrode is touching the rock surface creating a hotspot at the rock surface. (b) The molten area (in reality burst as lava upwards) is taken out, in order to continue the simulation without simulating lava flow (COMSOL-Multiphysics™ models).

(c) 10-electrode arrays

Arrays of 9 and 18 electrodes were simulated by multiplying the fundamental electrode and positioning the new electrode in arrays as in the experimental setup (Fig. 4.7).

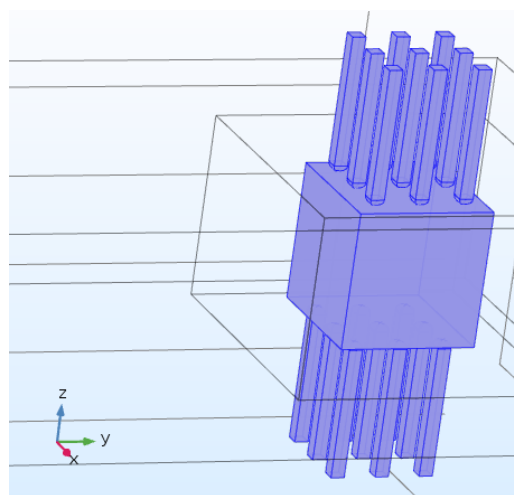


Figure 4.7: A model of the 18-electrode array simulation (COMSOL-Multiphysics™).

4.2.5. Boundary conditions

The region of interest is within the basalt cube, hence the heat equation is solved inside the basalt domain and not in the other domains such as the surrounding air and cavity walls. Therefore, the area near the basalt cube defines the thermal boundary conditions in this simulation.

The EM model consist of the irradiated material and the space between the cavity walls, thus Maxwell equations are solved in all the relevant space. Therefore, the electromagnetic boundary conditions are determined on the cavity walls.

(a) Thermal boundary conditions

During the LMH process, the heat generated in the core of the basalt cube is dissipated by (a) convection through the surface boundaries, (b) radiation from the surface, and (c) conduction by the bottom surface in contact with the aluminium cavity (Fig. 4.8). The latter is taken into account as a constant temperature boundary of 300 K, namely assuming the aluminium cavity is a perfect thermal conductor. This is a reasonable assumption though aluminium is not a perfect thermal conductor in practice, and the basalt-aluminium contact is not an ideal contact. However, the temperature rise at the contact region is less significant, compared to other regions of the basalt cube.

The heat flux dissipated by convection from the surface of the basalt cube to the surrounding air is given by

$$q_{conv} = h_f (T - T_a), \quad (4.8a)$$

where h_f is the convective heat-transfer coefficient of air and T_a is the ambient temperature.

The blackbody heat radiation is given by

$$q_{rad} = \sigma \varepsilon_s (T^4 - T_a^4), \quad (4.8b)$$

where σ is the Stephen-Boltzmann constant and ε_s is the surface emissivity.

(b) Electro-magnetic boundary conditions

All metal parts in this LMH structure (including the waveguide walls and electrodes as shown in Fig. 4.8) are assumed to be perfect electric conductors (PEC), thus the field component at the metallic boundaries obey the following conditions,

$$n \times E = 0, \quad n \cdot H = 0. \quad (4.8c)$$

Hence, the transverse electric field and the perpendicular magnetic field vanish on the metallic boundaries. Being a perfect electric conductor, no energy loss is dissipated on the waveguide walls.

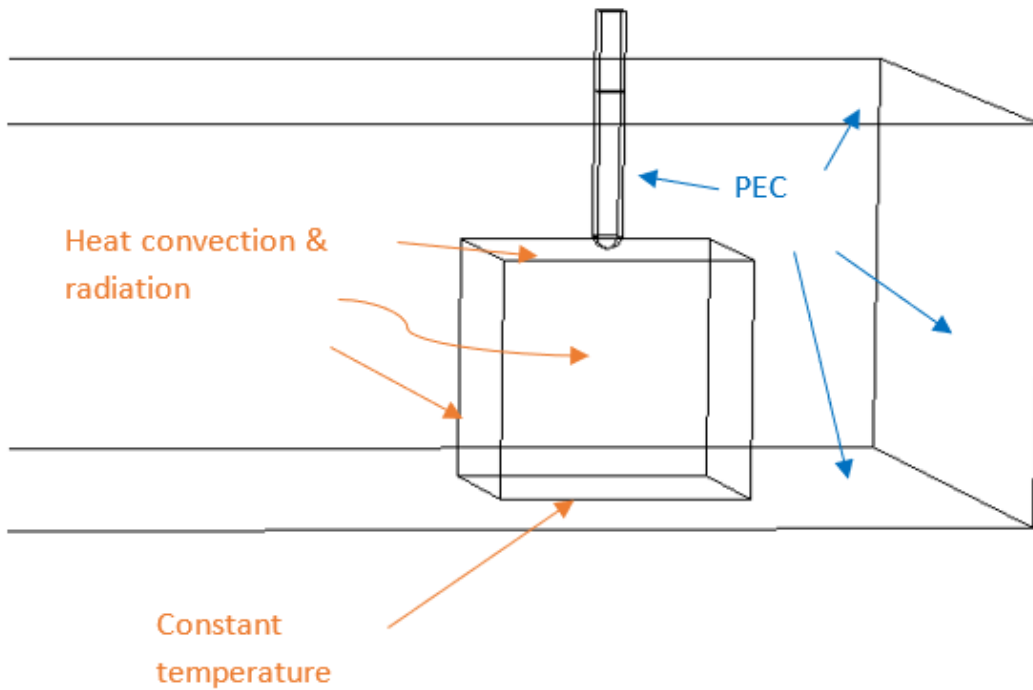


Figure 4.8: Illustration of thermal and electromagnetic boundary conditions.

4.2.6. Simulation mesh

As discussed in Section 4.2.1, the basic concept of FEM is the subdivision of the mathematical model into non-overlapping components called elements. The response of each element is expressed by known functions, and the response of the whole model is then considered to be approximated by assembling the collection of all elements. Therefore, the finite-element technique requires discretization of the domain.

This is done by meshing, and by nodal representations of the geometry and its functional relations. The guidelines for meshing used in our simulation are as follows:

- The sub-geometry of the mesh in the interface region of the basalt cube and electrodes will be highly dense (defined by COMSOL-MultiphysicsTM in physics-controlled-mesh as an extremely-fine mesh), in order to improve accuracy.
- The sub-geometry density of the mesh will be highly dense (extremely-fine) in the basalt cube and electrodes surrounding region, intended for accurate EM-field computations encircling the cube. This can be achieved by building a box of air around the cube, whereas the mesh within the box will be set to be finer than the other air geometry in the simulation (Table 4.3).
- The same applies to the excitation port region in order to get accurate reflection coefficient calculation.
- The sub-geometry sizes in all other regions of the simulation should not exceed a quarter-of-wavelength of the EM-wave propagating in the waveguide.

Table 4.3: The air-box dimensions.

Width	(x axis)	60 mm
Length	(z axis)	60 mm
Height	(y axis)	40 mm

The mesh parameters of the basalt cube and the air box were chosen to be a tetrahedral mesh (the default element type for most physics [76]), as given in Table 4.4. The excitation port was chosen to be a free triangular mesh with size defined as extremely fine. An illustration of the mesh in the simulation is presented in Fig. 4.9.

Table 4.4: Mesh parameters of the simulation.

Maximum element size	3.52 mm
Minimum element size	0.0352 mm

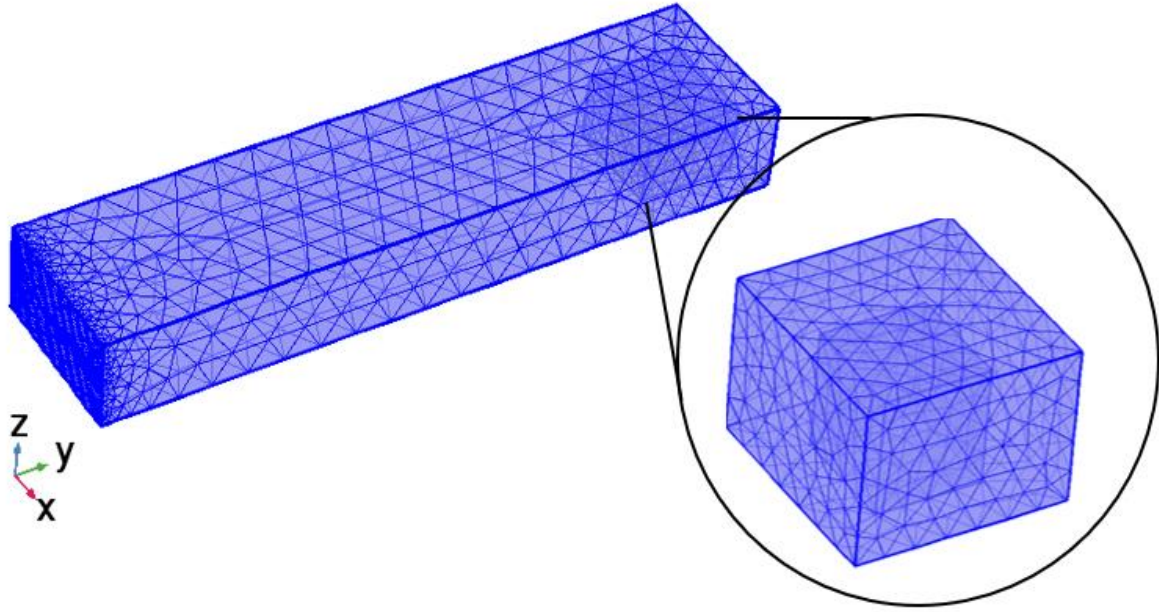


Figure 4.9: Illustration of the modeled mesh of the entire domain, and the irradiated cube of basalt and air in the inset. The mesh is extremely fine in the port and basalt-cube area.

4.2.7. Latent heat

The latent heat, q_{latent} , is the amount of energy needed for a phase transition of a material at its melting temperature. The LMH model is extended beyond the point of phase transition (melting), to further heating of the molten rock, and therefore the latent heat should be taken in to account in the simulation.

There are two methods to include the latent heat in a COMSOL-Multiphysics™ simulation:

(1) The apparent heat capacity method:

By applying a tailored interface for modelling the phase variation, the latent heat is included as an additional term in the heat capacity. This built-in function, $\alpha(T)$, considers a smooth transition between the phases within an interval of $\Delta T_{1 \rightarrow 2}$ around the phase-change temperature $T_{pc,1 \rightarrow 2}$. A new state of mixed material properties exists within this interval. The smaller the interval, the sharper is the transition.

The material properties for the solid and liquid phases are specified separately.

Additionally, the phase change operator receives three inputs:

- $T_{pc,1 \rightarrow 2}$: The phase-change temperature between Phase 1 and Phase 2, in Kelvin.
- $\Delta T_{1 \rightarrow 2}$: The transition interval between the two phases, in Kelvin.
- $L_{1 \rightarrow 2}$: The latent heat, in J/kg

The relevant material properties are expressed then as

$$X = X_{solid} \cdot (1 - \alpha(T)) + X_{liquid} \cdot \alpha(T), \quad (4.9a)$$

with X as the specific material property defined for each phase. For a pure solid $\alpha(T) = 0$ whereas for a pure liquid $\alpha(T) = 1$.

The latent heat is presented by an additional term in the heat capacity, and hence the method uses the following expression for the heat capacity [76]:

$$C_p = C_{p,solid} \cdot (1 - \alpha(T)) + C_{p,liquid} \cdot \alpha(T) + L_{1 \rightarrow 2} \frac{d\alpha}{dT}. \quad (4.9b)$$

The disadvantages of this method are the computation complexity and much longer calculation time. Moreover, the main disadvantage is that this complexity considerably affects the computation of the scattering parameters.

(2) The continuous function method

The second approach to compute the latent heat is to set the material properties either as continuous functions or piecewise functions of temperature, regardless to its phase. The material properties define the actual material phase. Therefore, each sub-geometry is treated according to its phase (solid, liquid or gas) determined by the actual values of its material properties. As mentioned above, the latent heat is included in the heat capacity as a piecewise function, in the form of the latent heat value at the phase transition temperature. The transition interval is determined by the smoothing function applied in the piecewise characterization (see e.g. in Fig 4.10).

In both methods, the transition interval is determined either by literature or by trial and error. In the case of basalts, no specific transition interval was found in literature. Considering the diversity of basalt rocks and their ingredients (and the variation of environmental conditions), the trial and error approach was considered by altering the smoothing parameters with respect to the experimental results. The latent heat of basalt used in the simulation is $L_{1 \rightarrow 2} = 40,000 \text{ J/kg}$ [83].

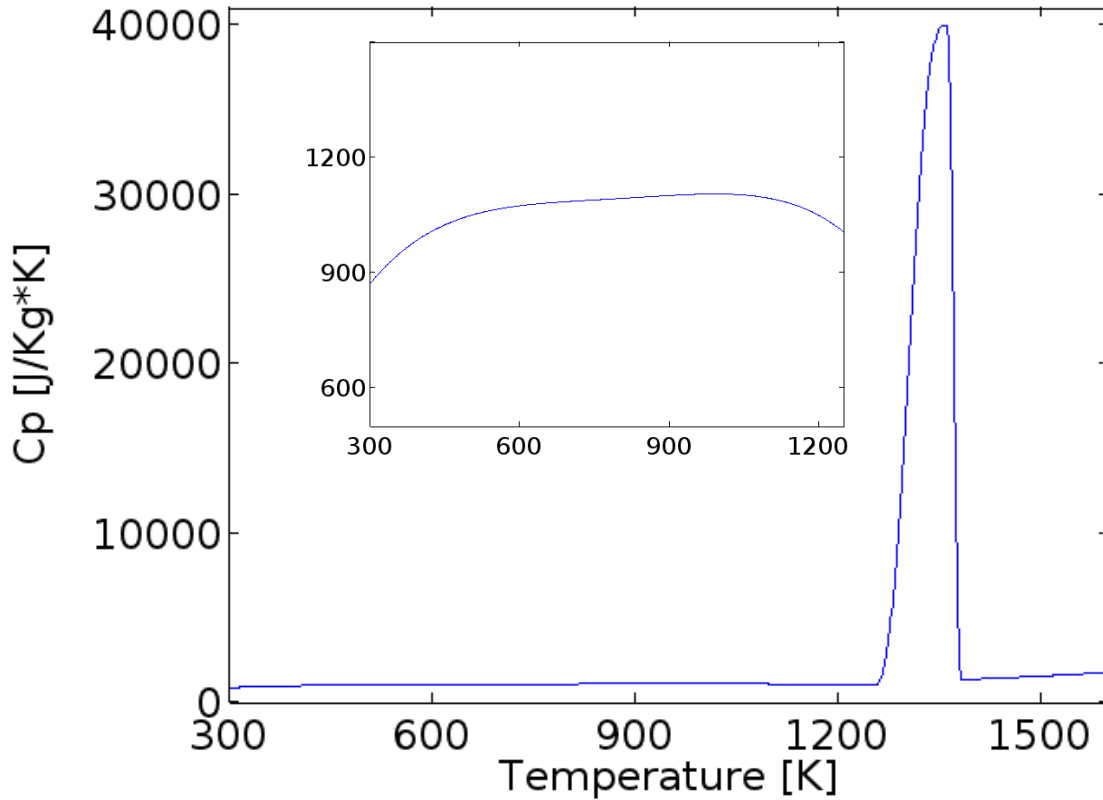


Figure 4.10: The heat capacity vs. temperature with the latent heat incorporated in it. The inset shows the heat capacity at lower temperatures.

4.2.8. Adaptive tuning

The experimental setup comprises tuning capabilities that channel most of the transmitted power to the load with minimized reflections. This capability is achieved by combining an adaptive tuning device, in our case an impedance auto-tuner (Homer, S-Team Ltd., Bratislava, Slovak republic), which adaptively matches the source impedance to the varying load impedance of the heated rock and to the plasma ejected in special cases. Introducing this capability to the COMSOL-Multiphysics™ simulation can be done in two methods:

- In port definitions, a check-box named *Specify Deposited Power* is applicable. When checking this box, a feedback condition is added to the model, and the applied power is adjusted so that the desired power is deposited within the load. The power level set in the input power tab will be the power absorbed by the load, namely simulating an auto-tuning device in series to the source. The main problem with using this COMSOL-Multiphysics™ capability is that no *S*-parameter calculation is performed. This feature is satisfying for heating simulation, and is quick and easy to use.
- In order to simulate the adaptive tuning capability and still calculate *S*-parameters as conducted in the experimental setup, a bit more complicated method is implemented. The *S*-parameters measurement is utilized to convey the desired amount of power P_{eff} to the material. We define the input power in the input power tab as $P_{IN} = P_{eff} / (1 - \Gamma^2)$, where Γ is the measured reflection coefficient, named comp1.Sparam1, and calculate it in each iteration. In the first step, an arbitrary amount of power is launched towards the load, in order to detect the reflection coefficient. In the second step the power is adjusted as above. Although the desired amount of power is absorbed by the material, this procedure does not minimize the reflections, unlike real impedance-matching mechanisms. It only adjusts the source input power to a level that will satisfy the desired absorbed power. This enables the *S*-parameter calculation, such as de-embedded features in real impedance matching systems, and hence it can be compared to experimental results.

4.3. Numerical results and comparison with experiments

The LMH process simulation can be observed in two points of view. One is the electromagnetic simulation, including the EM-wave spatial distribution, power delivery and energy concentration. The other is the heat-evolution simulation, including temperature evolution in time, spatial distribution and phase changes. The analyses employ the material temperature-dependent properties as shown in Fig. 4.2. The other parameters are taken from Table 4.1.

The simulation in COMSOL-MultiphysicsTM combines several physical models into a single coupled model. In our case, we use the microwave-heating module, which couples between the electromagnetic and the thermodynamic effect. The coupling is based on Equations 4.4, 4.5, and 4.6, and is tied by the material temperature-dependent properties. In each iteration the temperature is updated, leading to an update of the material properties, and hence modifying the electromagnetic properties, and vice versa.

4.3.1. **No-electrode case**

The operation in a cavity with no electrode inserted is the most controllable process. The simulation tracks the experimental process to the point of cracking and lava eruption. The cracking itself may affect the heat conduction inside the rock, and hence it is not simulated. The lava eruption has a great effect on the heating process. As the material geometry is changed, the lava flows out of the molten core to the open air and hence another cooling mechanism (not simulated) is taking place. Therefore, only experiments with no lava eruption can be compared to the simulation (or predicted). In all other cases the simulation is only valid before the lava eruption time begins.

First, a simulation of the unloaded cavity was conducted in order to determine the preferred position for the basalt cube in the experimental setup. As can be seen in Figs. 4.11a, b, the position was selected to be at the electrical field maximum, since the heating mechanism is mostly electrical.

A parameter comparable with the experiments is the reflection coefficient, as shown in Fig. 4.11c. The reflection coefficient measurement can depict physical processes within the core (e.g. phase change, temperature, and the amount of molten material), which can also be seen in the simulation. From the thermal simulation we can learn about the heating rate and heating profile, and also about the core temperature (not measured in the experiment).

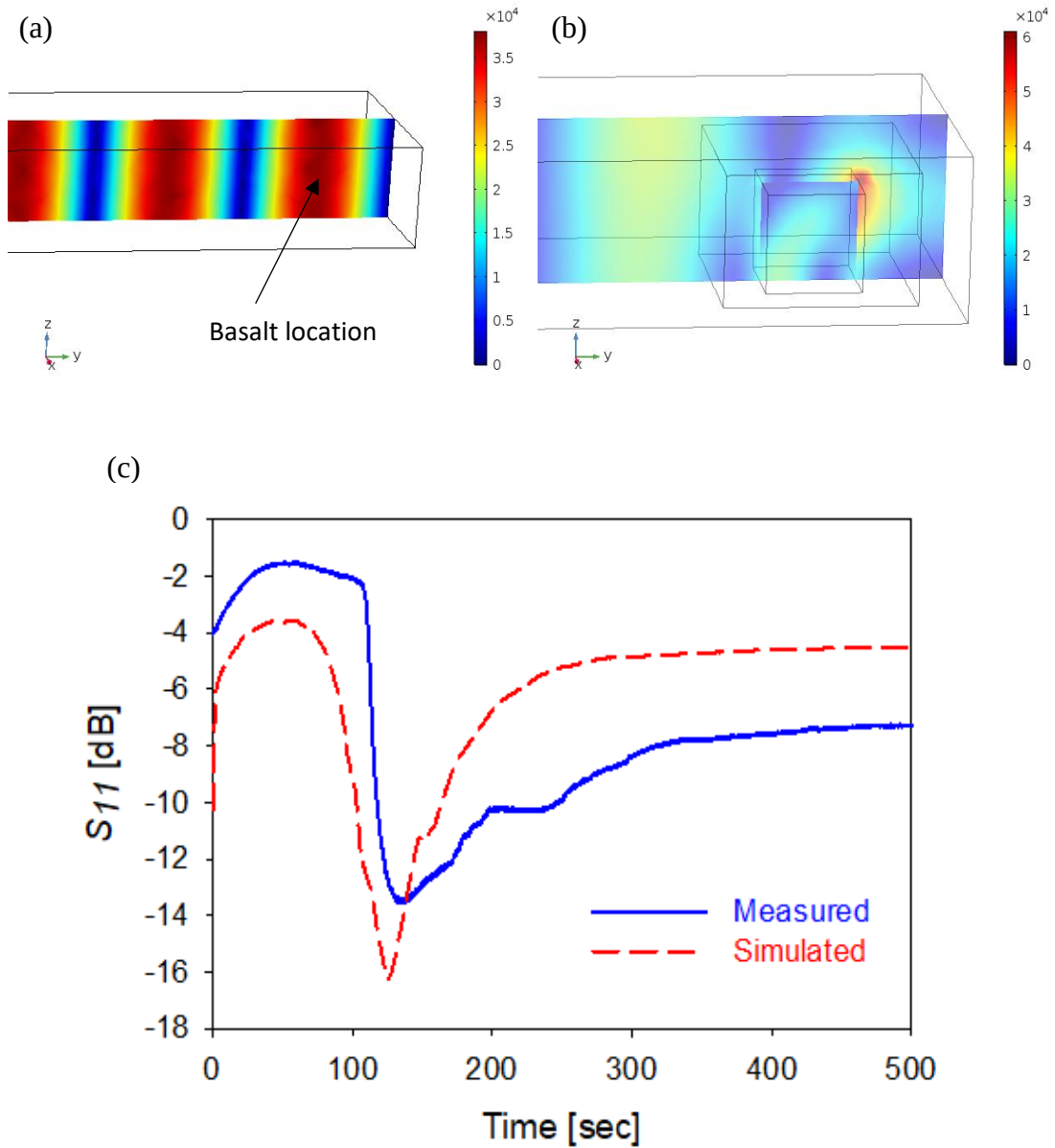


Figure 4.11: The electric-field norm distribution in the unloaded (a) and loaded (b) cavities; and the simulated and measured reflection coefficients from the irradiated basalt (c).

The surface temperature evolution is compared with thermal camera measurements. Figure 4.13a shows a comparison of the temperature evolution on a specific point on the rock surface, and Figure 12.a shows a comparison of the temperature evolution on the central line of the rock surface. The core temperature is important hence the concept of LMH and microwave inner heating depicts that the centre of the rock (assuming symmetry) will be rapidly heated up, as compared to the rest of the rock. The simulation agrees with this hypothesis, as seen in Fig 4.12b. The rapid temperature rise is resulted by the phase change occurring in the rock core.

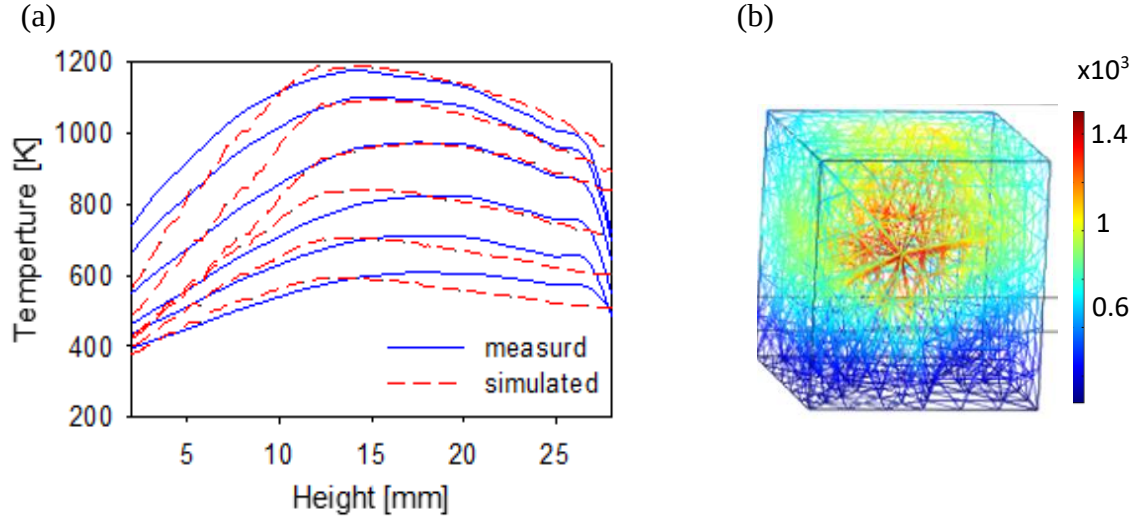


Figure 4.12: Theoretical simulations of basalt LMH with no electrodes: (a) A comparison between numerical and experimental results of the spatial evolution of the temperature profiles on the basalt cube surface (successive curves in time intervals of 60 s). (b) A numerical simulation of the basalt LMH process shows the molten core inside the irradiated basalt cube, reaching 1400 K after 70 s, with no electrode.

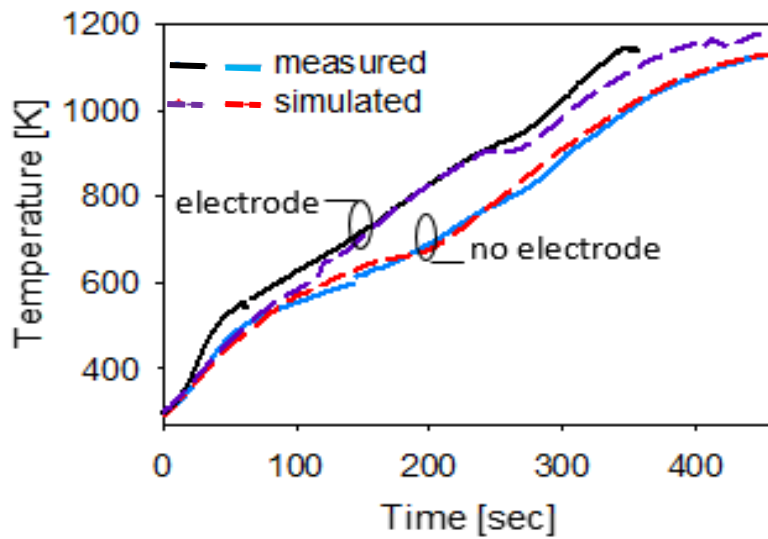
4.3.2. Single-electrode case

The same experiment conducted with a single electrode is a more sensitive process, and is more difficult to simulate and to predict, mainly due to plasma ejection. The electrode creates a hotspot on the surfaces which in some cases ejects plasma to the air atmosphere. This plasma tends to attract the electromagnetic energy, thus once a plasma is ejected, the comparison between simulation and experimental process is not valid. In order to overcome lava eruption from the electrode contact, the simulation is divided to two parts as further clarified in Section 4.2.4.

For experiments with no plasma ejection, the simulation depicts the thermal process, and can also find the temperature evolution of a hotspot near the electrode (Fig. 4.14b). The electrical-field focusing can be seen in Fig 4.13b, depicting a hotspot formation on the rock surface in contact with the electrode.

The temperature evolution at a certain point on the rock surface (in experiments and simulations) is compared to the no-electrode simulation (in Fig. 4.13a). It can be seen that despite the hotspot formation on the top surface (in addition to the core hotspot), and the lava eruption, the rise of the surface temperature is accelerated by the electrode.

(a)



(b)

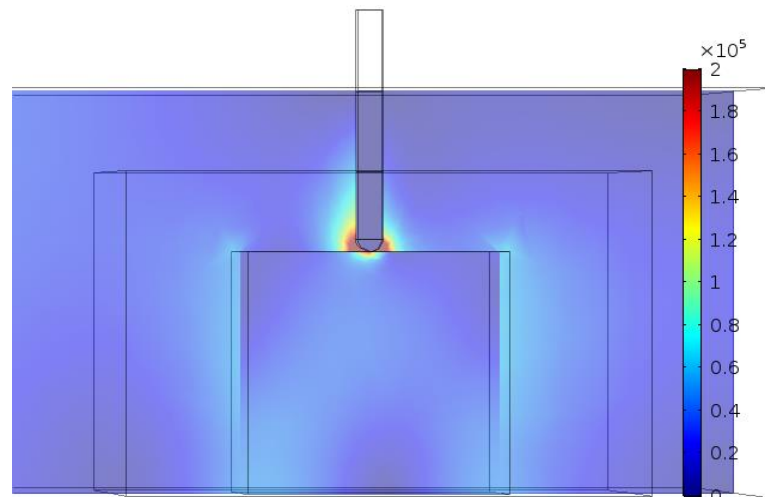


Figure 4.13: (a) A comparison between simulation results of the temporal evolution of the temperature on the side center, with and without electrode; (b) The electric-field norm distribution around the electrode tip.

4.3.3. Electrode array

Experiments with arrays of 10 and 18 electrodes are even more complicated random processes. The plasma ejection and lava eruption occur rapidly, simultaneously and randomly, during irradiation. Therefore simulating the heating process with respect to experiments is not feasible in the means employed in this study. The simulation is therefore limited up to the point of plasma ejection. The temperature profile evolution (as in Fig. 14c) may provide an estimate for the time-to-melt (TTM) prediction, and for the hotspot formation and its location.

4.3.4. Pyramid-like sharp stone

In order to enhance the plasma ejection process from basalt rocks, without using electrodes, a different means of concentration is needed. The rock can be shaped in a way that concentrates the electromagnetic energy to a hotspot near its surface, from which plasma is later ejected. A simulation of such a shape was conducted. As shown in Fig. 4.14a, the hotspot is evolved at the rock tip.

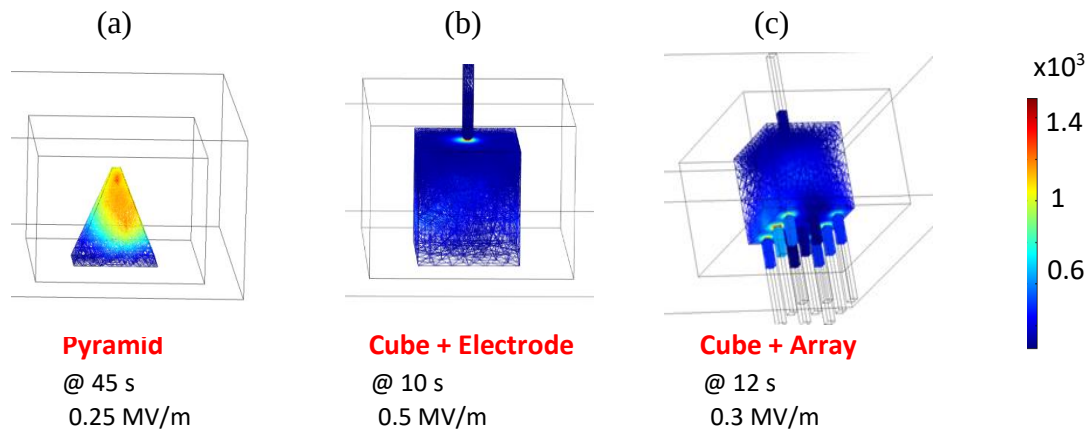


Figure 4.14: Numerical simulations of various shapes intended to intensify the LMH process as compared to the no-electrode scheme simulated Fig. 4.12b: (a) A pyramid shape; (b) a single electrode, and (c) an electrode array. The most rapid surface melting is obtained with a single electrode (0.5 MV/m in 10 s) whereas the no-electrode scheme provides an inner core melting (Fig 4.12b).

4.4. Dusty-plasma interaction with microwaves

The effective dielectric permittivity ε_r , which consists of the complex plasma permittivity $\varepsilon_p' - j\varepsilon_p''$ and of the dust conductivity σ_{ed} [84], is given by

$$\varepsilon_r = \varepsilon_p' - j\varepsilon_p'' - j\sigma_{ed}/\varepsilon_0\omega, \quad (4.10)$$

similarly to Eq. (4.7a). The dielectric permittivity of the plasma is given by its approximated real and imaginary components,

$$\varepsilon_p' = 1 - \frac{\omega_p^2}{\omega^2 + \nu^2}, \quad \varepsilon_p'' = \frac{\omega_p^2\nu}{\omega(\omega^2 + \nu^2)}, \quad (4.11a, b)$$

respectively, where $\omega_p = \sqrt{e^2 n_e / m_e \varepsilon_0}$ and ν are the plasma and the collision frequencies, respectively, and e , n_e and m_e are the electron charge, density and mass, respectively. The dusty plasma conductivity is given in similar conditions [84], as

$$\sigma_{ed} \cong \eta_{ed} \frac{\omega}{k} \left[\frac{\omega^2 - \nu\nu_{ch}}{(\omega^2 + \nu_{ch}^2)(\omega^2 + \nu^2)} + j\omega \frac{\nu + \nu_{ch}}{(\omega^2 + \nu_{ch}^2)(\omega^2 + \nu^2)} \right], \quad (4.12)$$

where $\eta_{ed} = \omega_p^2 \varepsilon_0 / l_d$ is the charging factor, $l_d = (n_d \pi r_d^2)^{-1}$ is the dust collision length factor, n_d and r_d are the dust grain density and the average particle radius, respectively, and k is the spatial angular frequency (the wave wavenumber for a uniform plasma).

The electron effective collision frequency is given by $\nu = V_{Te} \sigma_n N_n$, where

$V_{Te} = \sqrt{k_B T_e / m_e}$ is the electron thermal velocity, σ_n and N_n are the neutrals cross-section and density, respectively, T_e is the electron temperature, and ν_{ch} is the dust charging frequency. Assuming $\nu\nu_{ch} \ll \omega^2$ and $\nu_{ch} \ll \omega^2$, and substituting Equation (4.12) into Equation (4.10), the latter is further reduced to [47,48]

$$\varepsilon_r \cong 1 + \frac{\omega_p^2}{\omega^2 + \nu^2} \left[-1 + \frac{\nu}{kl_d} - j \left(\frac{\nu}{\omega} + \frac{1}{kl_d} \right) \right] g, \quad (4.13)$$

where g is a spatial profile function for the plasmoid in the waveguide.

The Boltzmann-plot analysis of the spectra measured in Section 3.2.3 results in electron temperatures in the range of 0.3–0.6 eV. The corresponding electron thermal velocity is $V_{Te} = 2.7 \times 10^5 \text{ ms}^{-1}$ hence the collision frequency is $\nu = 1.2 \times 10^{10} \text{ s}^{-1}$ (assuming $\sigma_n \sim 4.4 \times 10^{-20} \text{ m}^2$ and $N_n \sim 10^{24} \text{ m}^{-3}$ for a weakly ionized dusty plasma in air atmosphere [84]). The dust size and number densities ($n_d \sim 10^{15} \text{ m}^{-3}$ and $r_d \sim 60 \text{ nm}$) are assumed by the SAXS analyses of similar dusty plasmas [43, 44, 47, 48]. The mean free path for electron-dust collisions is then $l_d \sim 10^{-2} \text{ m}$. In these conditions, for

$$\omega \sim 1.6 \times 10^{10} \text{ s}^{-1}, \text{ the absolute value of the real part in Equation (4.13), } \left| -1 + \frac{\nu}{kl_d \omega} \right|,$$

is significantly smaller than the imaginary part, which plays the dominant role in the microwave absorption by the plasmoid.

4.5. Parametric plasma estimates

4.5.1. Scattering S_{11} simulation

The microwave scattering parameters of the plasmoid are analysed in order to estimate its microwave impedance, and hence its dielectric properties. The S_{11} parameter is measured during the various stages of the plasmoid lifecycle (no plasma, ejection, steady state, and fireball stages). The simulation models the steady state of the plasmoid in its fire-column form. The experimental results are then compared with the different simulation results, in order to estimate the plasma characteristics.

The physical model employed in the simulation assumes microwave load in steady state. The plasma is ejected from a hotspot located on the rock surface. Once ejected, the microwave energy is focused almost entirely into the plasmoid. Thus, the basalt rock is not further heated to its melting temperature, except the surficial hotspot. A precise calibration is needed in order to compare the reflection coefficient (magnitude and phase) between the simulated and measured results.

(a) Materials

The physical models in this simulation is entirely electromagnetic, thus the only equations solved in the cavity and in the plasma are the EM-equations. The plasma is characterized as a bulk of homogenous material with given electrical permittivity, conductivity, and magnetic permeability. These are taken as constant parameters, with a unified spatial distribution and no temperature dependence. The magnetic permeability is assumed to be unity, and the dielectric permittivity is represented by the real and imaginary part of ϵ_r , namely $\epsilon_r = \epsilon' - j\epsilon''$ (Eq 4.7a). A sweep is implemented on the relative real and imaginary parts of the dielectric coefficient.

The rock is also characterized by its dielectric coefficient. As discussed above, the rock temperature is well under its melting temperature. It is measured by the thermal camera to be around 600 K, thus the dielectric coefficient (taken from Table 4.1) is of the solid rock. As seen from Table 4.1 the electrical conductivity is negligible at these temperatures hence it is taken for simplicity to be zero.

(b) Geometry

The plasma geometry is set similar to the experiment. The fire-column is modelled as a cylinder with a diameter of 15 mm and height of 50 mm. The cavity dimensions are the same as used in the experimental setup and in the thermal simulation. The excitation port is set to be the waveguide opening. The location of the port (namely the waveguide length) is hard to be determined from the experimental setup, since the experimental setup is composed of a waveguide exited by a magnetron, isolator and an impedance tuner. The waveguide length is important for phase measurements and for the comparison to the experiment. Therefore, a calibration process is conducted in order to determine waveguide length and port location.

(c) Dummy-load calibration

In order to determine the port position for phase measurement purposes, a calibration process is carried out. A glass tube filled with tap water is placed at a specific location in the microwave cavity. The water dielectric properties at room temperature are taken from [85]. The water dielectric coefficient is temperature dependent. Therefore, the reflection coefficient was measured in the first few seconds after the power excitation (so the water tube is still near room temperature). The glass is assumed transparent to the microwave radiation hence a simulation of a water cylinder (at the same position and size as the fire-column) models the experimental setup (Fig. 4.15a). At the first simulation run, the measured and simulated magnitudes of the reflection coefficient are identical. A sweep on the waveguide length and the port location changes the simulated reflection coefficient phase. The waveguide length is determined by comparing the simulated and measured results of the water reflection coefficient. The simulated waveguide can be shorter than the real experimental waveguide by an integer number of wavelengths. Figure 4.15b shows the simulated (Red) and measured (Blue) reflection coefficient from the water dummy load, for various waveguide lengths.

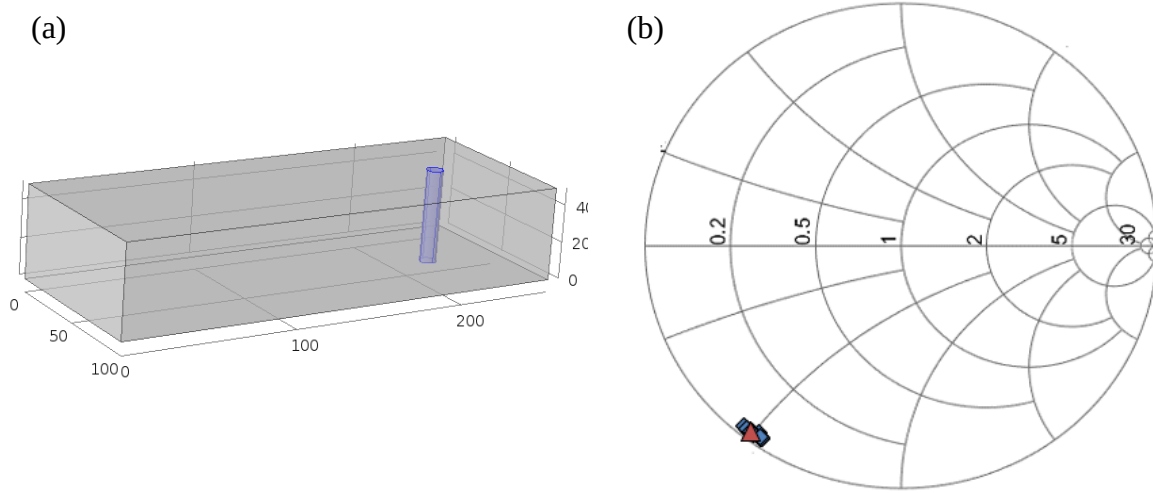


Figure 4.15: The water tube calibration measurements and simulation: (a) The COMSOL-Multiphysics™ model simulated in order to determine the phase calibration. (b) Measured and simulated reflection coefficient from the water tube (in Blue and Red respectively).

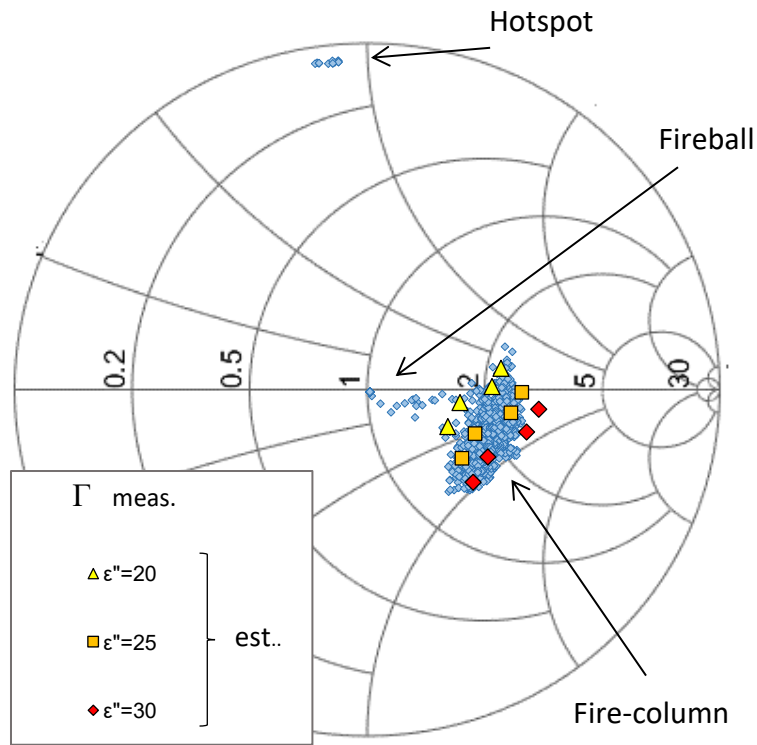


Figure 4.16: A Smith-chart presentation of the complex reflection coefficient Γ , starting from a mismatch ($|\Gamma| \sim 1$) at the hotspot stage, then reduced to $|\Gamma| \sim 0.4$ at the fire-column stage, and finally matched ($|\Gamma| \sim 0$) by the self-adapted fireball. Simulation results of dielectric dummy loads (of $\epsilon_r'' = 20, 25, 30$) with the same column shape in various positions, are shown as benchmarks for the effective plasma impedance.

4.5.1.1 Benchmark results and comparison with experimental results

After the calibration and dimension setting, the dielectric properties of the plasma can be estimated. As seen in Fig. 4.16, the scattering of the reflection coefficient measurements on the Smith-chart can be attributed to the fire-column movement, and to small changes of dielectric loss in time.

4.5.2. Scattering S_{21} simulations at other frequency bands

For additional plasma characterization, scattering analyses for a wider frequency band has been carried out by irradiating the plasmoid by another probing signal (in addition to the main signal exited by the magnetron). In these experiments, as opposed to previous scattering measurements, the scattering parameter under examination is the transmission coefficient S_{21} .

As discussed above, the simulation of the process is crucial for characterization of the plasma. The main challenge in scattering measurements is protecting the measurement equipment from the high power (1kW) running in the microwave cavity. Hence, as discussed, two different measurement setups were applied, one for frequencies below the magnetron frequency and another for frequencies above. Therefore, two simulation projects were conducted. One for two-electrode excitation ports (at lower frequencies) and one for the two waveguide ports (with a cutoff frequency higher than the magnetron frequency).

The simulation was built in CST Microwave StudioTM suite. As in the S_{11} measurement simulation, the plasma is modelled as a bulk of unified distributed material with electromagnetic properties. First, a reference simulation has to be performed and compared to experimental reference. The reference is measured with the rock inside the cavity. The rock is simulated as a bulk cube of electromagnetic properties of basalt taken from Table 4.1. A sweep is performed on the dielectric coefficient's real and imaginary parts and compared to the measurement results.

In order to get accurate results, the waveguide cavity geometry is modelled as the experimental one. No assumptions are made in order to ease the computation process. The electrode probes and SMA ports are modelled as well. The cavity vanes and their dimensions are all taken from the experimental setup.

For a reference simulation a cube of 2 cm^3 (representing the equivalent load of a natural-shaped rock) is modelled and simulated (Fig 4.17).

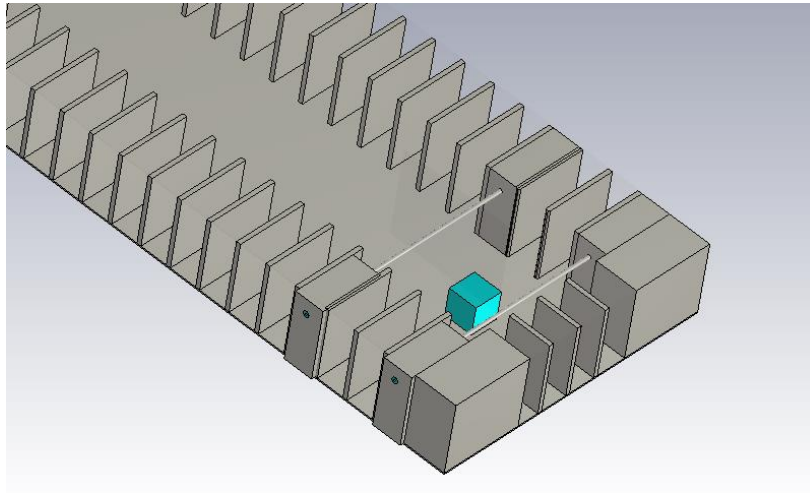


Figure 4.17: A model for the basalt rock scattering simulation.

The simulation is extended in a frequency range of 0.1 to 2 GHz. First, a simulation run of a cubical basalt brick was performed in order to simulate the reference measurement. Different plasma dielectric coefficients (ϵ_r) result in different transmission-coefficient calculations. These were compared to the measured transmission coefficient in order to derive the estimated dielectric properties of the plasma. Figure 4.18 shows a comparison between the measured S_{21} magnitude and its simulated magnitude in the frequency range 0.8-1.5 GHz for a plasma with a dielectric permittivity of $\epsilon_r = 0.3 - j50$.

For transmission measurements at higher frequency (3-6 GHz), two WR-187 waveguides were used as excitation ports for the probing signal. A reference simulation was performed as well, similarly to the reference simulation discussed above. As in the low frequency simulation, a CST Microwave-Studio™ suite model was developed. The plasma, the cavity, and the waveguide ports (including the SMA excitation ports) were modelled according to the experimental setup dimensions (Fig. 2.3b).

The simulation results (Fig. 4.19), as well as observed in the experimental results, do not show significant variations in the magnitude and phase of S_{21} . This could imply that the setup is not proper for the higher frequency band, or that the plasma is not significantly interacting with this frequency band.

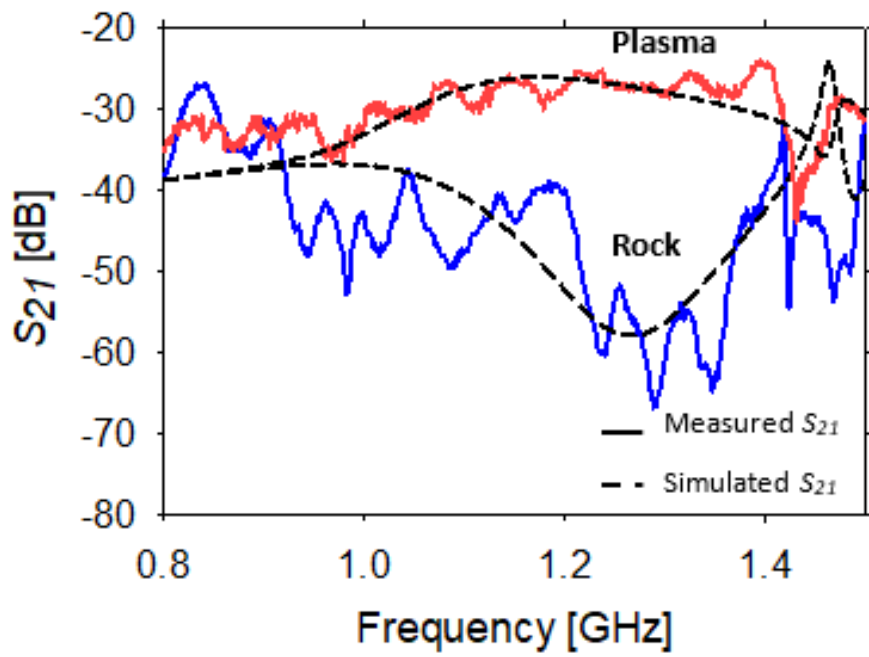


Figure 4.18: Scattering analyses (measured and simulated) of the plasma by an additional probing signal in the frequency range 0.8-1.5 GHz.

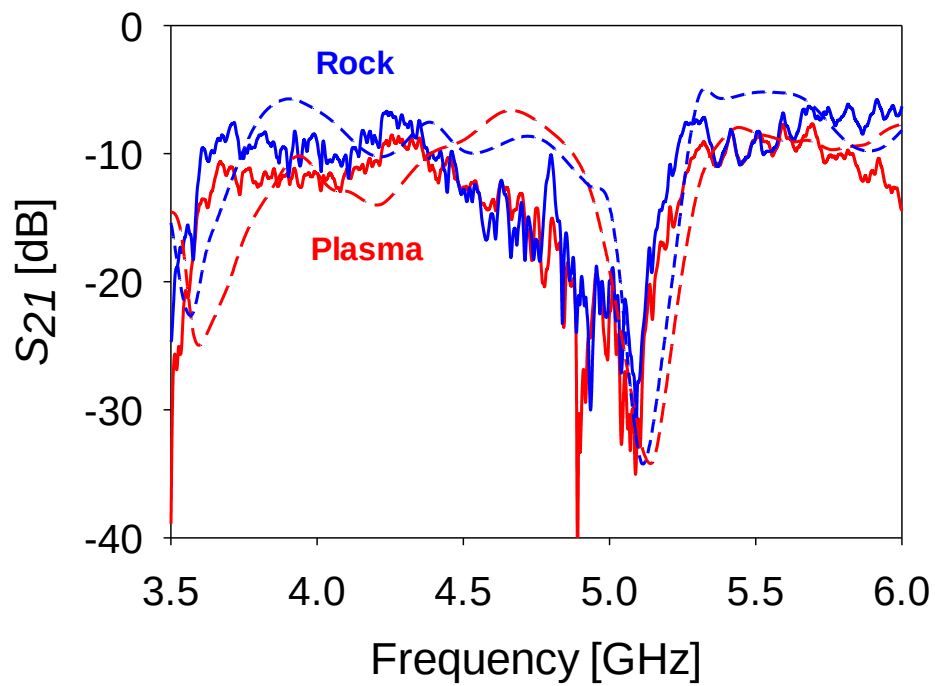


Figure 4.19: without the plasma does not show significant changes in the coupling between the two ports.

The simulated results and the experimental observations are presented in the Smith chart shown in Fig. 4.16. The simulated results give an estimate for the effective dissipation factor (including conductivity) as $\epsilon_r'' \sim 25$, where the average experimental reflection is ~ 0.3 . The corresponding electron density in this case is estimated by

$$n_e \sim \frac{m_e \epsilon_0 \epsilon_r''}{e^2} \frac{\omega^2 + \nu^2}{\nu/\omega + 1/\hat{k}l_d}, \quad (4.14)$$

where k is approximated by $\sim \pi/d_{PC}$ for the finite transverse profile of the plasma column. Equation (4.14) yields $n_e \sim 10^{17} - 10^{18} \text{ m}^{-3}$, similarly to other plasma forms and flames at atmospheric pressure in air.

5. Discussion and conclusions

This thesis presents an experimental and theoretical study of a localized microwave-heating (LMH) interaction with basalt rocks, including effects of hotspot formation, melting, cracking and plasma ejection. The feasibility of basalt melting by LMH is demonstrated here. The thermal-runaway instability is most significant at the initial melting moment, associated with a relatively large drop in the microwave reflection as observed in Fig. 3.7. This effect is attributed to the difference between the effective electrical conductivity of basalts in solid and liquid/lava states (this difference exceed almost two orders of magnitude). This transition leads to the sudden rise in the core temperature, due to the thermal-runaway instability in the "seed" location (where the initial phase transition occurs). This may explain the significant difference in temperatures between the inner core of the stone and its outer surface. This temperature gradient might lead to thermal stresses which eventually lead to crack formation and lava eruptions, as observed here.

In this study we have shown that LMH of basalt may also cause ejection of intense plasma from the molten stone [86,91]. This plasma in a form of a fire-column (and possibly also of a fireball), is found to be rich of nano- and micro-particles, and hence regarded as a dusty plasma.

A simplified model of microwave interaction with dusty plasma (Section 4.4) yields an estimate for the electron density in the basalt-originated plasma as $n_e \sim 10^{17} - 10^{18} \text{ m}^{-3}$ assuming an electron temperature of $T_e \sim 0.3 - 0.6 \text{ eV}$ (obtained by the various measurements above), assuming a dust number density of $n_d \sim 10^{16} \text{ m}^{-3}$ (as measured by small-angle X-ray scattering, SAXS, of similar elemental dusty plasmas [43-48]). Therefore, this plasma is characterized as a weakly-ionized, low-density dusty plasma, which contains components of the original basalt stone (e.g. silicon and iron oxides), deposited in a powder form on the collector installed at the chamber ceiling. The remaining stone is cracked and partly melted to form a volcanic glass (obsidian).

Ball-lightning and fireballs effects have been long associated with volcanic eruptions and volcanic air pollution [62]. The atomic spectra measurements obtained on the plasma plume and fireball reveals components of basalt. It also resembles spectrograph measurements of natural ball-lightning such as Si, Fe, Ca and Al [63], as

in soil, in line with the theory suggesting that ball lightning is generated by lightning striking the soil [64].

The phenomena of lightning within eruptions of volcanic ash plumes has also been recorded and studied [87]. This high-energy lightning chemically modifies and melts the volcanic ash [88], generating spherical shapes similar to those observed in the powder accumulated in this study. The composition of the volcanic ash sampled in Mount St. Helens eruption on May 18, 1980 [89] includes SiO_2 (65%), Al_2O_3 (18%), Fe_2O_3 (5%), and smaller amounts of MgO , CaO , Na_2O , K_2O , TiO_2 and P_2O_5 . The powder accumulated from the basalt-generated plasma consists of similar elements, as listed in Table 3.6. The relatively high concentration of phosphorous in the powder can also be associated with volcanic ash [90] (noting its environmental significance).

Major challenges have remained for further research, including in-depth investigations and modeling of (a) the LMH interaction with basalt in phase transitions; (b) the plasma ejection mechanism and effects of thermionic emission; (c) microwave interactions with dusty plasma and particle agglomeration therein; and (d) plasma-surface interactions and plasma-chemical reactions occurring in this process.

In an interdisciplinary approach, a further progress in this LMH-plasma study is also oriented to volcanology and geosciences. The controllable basalt core-melting and lava-flow effects observed here in a miniature scale (and their resemblance to giant natural volcanic phenomena) can be utilized for experimental small-scale simulations of various volcanic phenomena. The natural ball-lightning phenomena occasionally associated with volcanoes can also be experimentally modeled by the LMH-generated fire-column and fireball-like plasmas.

In a more technology-oriented approach, this study of LMH in basalts could be relevant to various potential applications, including for instance (a) microwave drilling in basalt; (b) cracking and crushing of basalts, for mining and construction operations; (c) basalt vitrification and glass production; (d) joining of basalt bricks; (e) direct extraction of minerals in a powder form from basalts; and (f) analyses of rock contents in the field (e.g. by portable microwave-induced breakdown spectroscopy (MIBS) systems [45]).

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Appendix A: Computer-controlled experimental setup

In order to control, monitor and synchronize the various measurement and diagnostic means, the use of a computer software (LabViewTM) is necessary. A LabView script was written for each of the setup components. They were later combined to a unified LabView script operating simultaneously in order to control the entire setup and to synchronize between operating times and stored data.

The LabView monitor (Fig. A1) is divided into four main parts: (1) Homer control and monitor, (2) MagDrive control and monitor, (3) Langmuir probe voltage monitors, and (4) video camera control and monitor.

(1) The Homer control performs the following:

- The “running” and “sending” buttons control the data display on the monitors.
- The “autotune” button switches HOMER auto-tune mechanism on and off.
- “Polar” and “Mag” plots at the bottom shows the measured S_{11} in a polar presentation and Log-Mag presentation.
- The stubs position graph shows the three stubs real time position. They can be manually controlled by clicking on the graph.
- Other real-time measurements such as frequency, incident and reflected power, etc. are shown on the monitor.
- The communication port chosen is the one connected to the HOMER auto-tuner.
- The measured data is stored to a txt file every 50 ms. The data contains timestamp, relative timestamp (in 1 ms from beginning of run), measured S_{11} (magnitude and phase), de-embedded S_{11} (magnitude and phase), incident power, reflected power, input power and autotune (on/off).

- (2) The MagDrive control module governs the transmitted power level. It can be controlled by writing the wanted transmitted power, or by clicking the down and up arrows, as monitored by the scale bar. Note that the communication port chosen is the one connected to the MagDrive module.
- (3) The Langmuir-probe module monitors the measured voltages. The data is presented in real-time on the voltage-vs-time plots, and stored in a txt file for later analyses. The data contains the relative timestamp, V_{sup} and V_{meas} .
- (4) The video-camera module controls and monitors the video camera. The live feed is displayed in the white box. A check box states the video recording to a file. A display of timestamp and incident power is present on the recorded video.

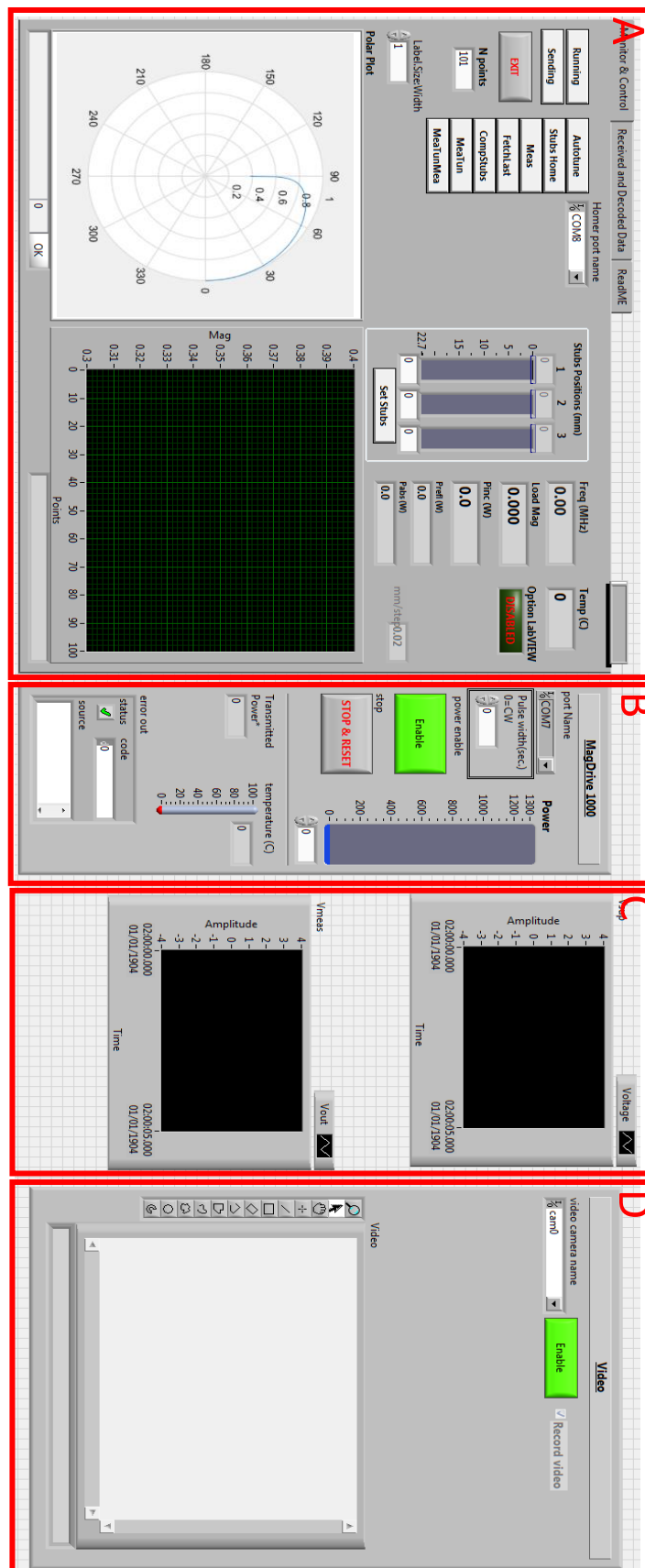


Figure A1: A screenshot of the LabView operating screen divided to the various modules.

Appendix B: Tektronix TTR 500 continues sampling

In order to sample and store the transmission coefficient measurements, carried out continuously by the Tektronix TTR500 VNA, a MATLAB™ script was written. The software connects to the VNA, samples and stores the measured data in a controlled rate. The data is saved as an S2P format, later processed also by MATLAB. The measurement definitions are set on the VNA.

```
clear all;
clc;
% Find a tcpip object.
obj1 = instrfind('Type', 'tcpip', 'RemoteHost', '127.0.0.1',
'RemotePort', 5026, 'Tag', '');

% Create the tcpip object if it does not exist
% otherwise use the object that was found.
if isempty(obj1)
    obj1 = tcpip('127.0.0.1', 5026);

else

    fclose(obj1);
    obj1 = obj1(1)
end
% Connect to instrument object, obj1.
fopen(obj1);

% Communicating with instrument object, obj1.
i=0;
fprintf(obj1, 'MMEMory:STORe:SNP:DATA C:\Users\Eli
Jerby\Desktop\YN%d.slp');
tic
data43 = str2double(query(obj1, 'SENSe1:SWEep:TIME:DATA?'));
while i<500
    format longg
    c = num2str(clock);
    k = regexp(c, ' ', '');
    format1 = 'MMEMory:STORe:SNP:DATA %s%s.s2p';
    filepath = 'C:\Users\Jerby\Documents\MATLAB\tektronix\';
    file = sprintf(format1,filepath,k);
    fprintf(obj1, file);
    pause(data43)
    i=i+1;
end
toc
% Disconnect from instrument object, obj1.

fclose(obj1);
```